# Non-Abelian Matrix-Condensate Geometrogenesis

Edward Schipani

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# Preface

This book develops Non-Abelian Matrix-Condensate Geometrogenesis (NAMCG) as a single derivation chain: from a microscopic matrix order parameter to an emergent Lorentzian metric, massless spin-2 dynamics, covariantly conserved entanglement backreaction, and a set of astrophysical corollaries.

The text is written for any reader who will follow an argument in order. The working language is the Einstein equation, geodesic deviation, and the idea of a Lagrangian. Quantum field theory is reviewed where it is used. Entanglement entropy is developed from scratch.

Two reading rules are in force throughout.

First, a composite metric is not general relativity until the infrared spectrum contains two massless transverse-traceless polarizations, the Einstein–Hilbert prefactor is computed from microscopic data, and every extra stress tensor satisfies $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$. Analog-gravity metrics and phenomenological drag terms are treated as insufficient.

Second, numerical laboratories in later chapters are reduced models. They illustrate conservation and dissipation after the covariant theory has been stated. They do not replace that theory.

Chapter 1 states the derivation problem. Chapter 2 installs only the geometry and field theory consumed by later proofs. Chapters 3–11 are the core. Chapters 12–15 are applications. Chapter 16 restates the chain. Chapter 17 lists limitations and falsifiers.

*Lake Ontario, 2026-09-19*
Edward Schipani, Independent Researcher

# Notation

Throughout, spacetime indices $\mu, \nu, \ldots$ are coordinate indices, $A, B, \ldots \in \{0, 1, 2, 3\}$ are tangent-space Lorentz indices, and $I, J, \ldots$ are internal gauge indices of the matrix condensate. The metric signature is $(-, +, +, +)$. Repeated indices are summed.

| Symbol | Meaning | Introduced |
|---|---|---|
| $\Phi^I{}_A$ | matrix order parameter | Ch. 3 |
| $v$ | condensate vacuum expectation value | Ch. 3 |
| $\hat{X}^\mu$ | matrix coordinates | Ch. 3 |
| $\Theta^{\mu\nu}$ | non-commutativity tensor | Ch. 3 |
| $\Lambda_{\mathrm{UV}}$ | ultraviolet cutoff $\sim \Theta^{-1/2}$ | Ch. 3 |
| $e^A_\mu$ | emergent vierbein $\langle \partial_\mu \Phi^{IA} \rangle$ | Ch. 4 |
| $g_{\mu\nu}$ | composite metric $\eta_{AB} e^A_\mu e^B_\nu$ | Ch. 4 |
| $T_{\mu\nu}$ | symmetric trace-free shear mode | Ch. 5 |
| $S_D$ | entanglement entropy of causal diamond $D$ | Ch. 6 |
| $H_D$ | modular Hamiltonian $-\ln \rho_D$ | Ch. 6 |
| $U_{\mathrm{ent}}$ | mutual-information potential | Ch. 6 |
| $G_{\mathrm{eff}}$ | emergent Newton coupling | Ch. 8 |
| $T^{\mathrm{ent}}_{\mu\nu}$ | entropic stress tensor | Ch. 11 |
| $\mathbf{a}_{\mathrm{irrev}}$ | irreversible drag acceleration | Ch. 11 |
| $S_{\mathrm{HBEV}}$ | holographic boundary-entropy action | Ch. 12 |
| $\Lambda_{\mathrm{eff}}$ | regulated cosmological constant | Ch. 12 |
| $r_0$ | core-regularization scale in $f(r)$ | Ch. 13 |

Where a later chapter needs a second meaning for a symbol, the local definition is restated at the start of that section. The table above is the default.

# Chapter 1

# What Geometrogenesis Must Derive

**Learning outcomes.** After this chapter the reader should be able to:

1. state the geometrogenesis problem as a derivation task, not a metaphor;
2. explain why analog-gravity metrics are not general relativity;
3. identify the representation content that a microscopic order parameter must carry if massless spin-2 modes are to exist;
4. list the consistency conditions treated as non-negotiable in this book: diffeomorphism invariance, $\nabla^\mu T^{\rm total}_{\mu\nu} = 0$, a bounded infrared Hamiltonian, and an explicit map from microscopic couplings to $G_{\rm eff}$;
5. see why Chapter 2 is required before any later proof can be stated.

## 1.1 What this book is trying to derive

General relativity is an infrared theory of a dynamical Lorentzian metric $g_{\mu\nu}$ whose vacuum excitations are two massless, transverse-traceless spin-2 polarizations, and whose coupling to matter is fixed by local energy-momentum conservation and the equivalence principle. In standard presentations the metric is primitive. One writes an Einstein–Hilbert action

$$S_{\rm EH}[g] = \frac{1}{16\pi G}\int \mathrm{d}^4x\,\sqrt{-g}\,R \tag{1.1}$$

and proceeds.

Geometrogenesis asks a more primitive question:

> Can $g_{\mu\nu}$, the Einstein–Hilbert dynamics, and the two graviton polarizations be obtained as the long-wavelength collective description of a microscopic quantum system that does not contain a metric as a fundamental field?

That question is the subject of this book. The framework developed here is Non-Abelian Matrix-Condensate Geometrogenesis (NAMCG). The microscopic object is not a single complex scalar. It is a matrix-valued order parameter $\Phi^I{}_A(x)$ carrying an internal gauge index $I$ and an emergent tangent-space index $A \in \{0, 1, 2, 3\}$. The metric is a composite,

$$e^A_\mu(x) \equiv \langle \partial_\mu \Phi^{IA}(x) \rangle, \qquad g_{\mu\nu}(x) = \eta_{AB}\, e^A_\mu(x)\, e^B_\nu(x). \tag{1.2}$$

The later chapters argue that entanglement equilibrium on causal diamonds forces the infrared mass of the symmetric trace-free shear mode to vanish, that a heat-kernel expansion supplies an explicit formula for $G_{\rm eff}$, and that the entanglement backreaction stress tensor remains covariantly conserved.

Those are theorems *inside the framework*. This first chapter does not prove them. It makes precise what would count as a proof, and why the obvious scalar shortcuts fail.

## 1.2 Analog gravity, and the ceiling it cannot break

A large and useful literature constructs effective metrics for perturbations on condensed-matter backgrounds. The standard example is a weakly interacting Bose gas with order parameter $\Psi = \sqrt{\rho}\, e^{i\theta}$. Phase fluctuations $\theta$ obey, after linearization, a wave equation

$$\frac{1}{\sqrt{-g_{\rm ac}}} \partial_\mu \Big( \sqrt{-g_{\rm ac}}\, g^{\mu\nu}_{\rm ac} \partial_\nu \theta \Big) = 0 \tag{1.3}$$

in an acoustic metric $g^{\mu\nu}_{\rm ac}$ built from the background density and flow. Horizons, ergoregions, and Hawking-like spectra can be engineered in the laboratory.

This is analog gravity. It is not geometrogenesis of general relativity, for four structural reasons.

**Only one scalar mode.** The Goldstone field of a single complex scalar is a scalar. The acoustic metric is a rewriting of that scalar wave equation. It does not create two independent tensor polarizations.

**Screening.** Gross–Pitaevskii interactions introduce a healing length $\xi$. The static potential felt by a localized source is of Yukawa or modified-Bessel type, not a $1/r$ Newtonian potential with radiative $1/r$ tails.

**Wrong infrared gravity.** A relativistic scalar condensate coupled to an auxiliary metric generically yields Nordström gravity (conformal, with no bending of light in the pure-scalar sector) or scalar–tensor gravity. The Einstein tensor is not selected as the unique linearized kinetic operator for a massless spin-2 field.

**Dissipation inserted by hand.** Binary inspiral in general relativity follows from quadrupole radiation in a theory with massless spin-2 waves. Replacing that derivation by a Peters formula or a drag term in Newton's equation may fit a curve. It does not derive the curve from the microscopic action.

NAMCG is written to climb over that ceiling, not to decorate it.

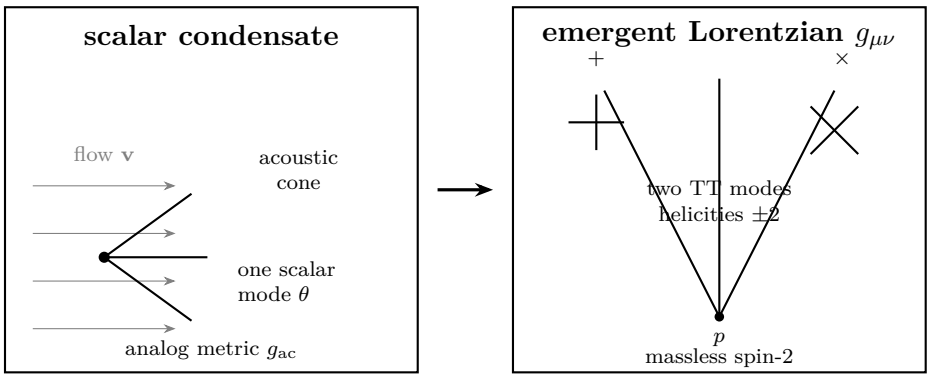



Figure 1.1: A scalar order parameter supplies an acoustic metric. It does not supply the two massless spin-2 polarizations of general relativity.

## 1.3 Representation theory in one paragraph

Local Lorentz invariance acts on tangent-space indices. A graviton is a symmetric, traceless, divergenceless tensor $h_{\mu\nu}$ with two on-shell degrees of freedom in four dimensions. A single scalar $\phi$ can produce a conformal mode proportional to $g_{\mu\nu}\delta\phi$, and perhaps a vector current if a global symmetry is gauged, but it cannot produce an irreducible shear tensor as an independent propagating excitation of the order parameter itself.

That is the representation-theoretic reason the microscopic field is promoted to $\Phi^I{}_A$. The tangent index $A$ is not cosmetic. It is the handle on which local $\mathrm{SO}(3,1)$ will later act. Chapter 5 decomposes

$$\delta\Phi^I{}_A = \frac{1}{4}\eta_{\mu\nu}\delta g^{\mu\nu}\, I + A_\mu \gamma^\mu + T_{\mu\nu} \tag{1.4}$$

into conformal, vector, and symmetric trace-free pieces. Chapter 7 is the argument that $T_{\mu\nu}$ becomes massless.

If that argument fails, NAMCG collapses back to analog gravity. The book is organized so the reader can see exactly where it would fail.

## 1.4 What must be delivered before one may say "this is GR"

A derivation of infrared general relativity from a microscopic model is not complete unless all of the following are obtained, not assumed.

1. **Composite metric.** $g_{\mu\nu}$ is defined from microscopic expectation values. It is not put into the fundamental action as an independent Einstein metric.

2. **Local Lorentz frames.** The vacuum must lock internal indices to tangent indices so that $e_\mu^A$ transforms as a vierbein and no physical Goldstone scalars remain. That is Chapter 10.

3. **Two massless TT modes.** The infrared kinetic operator of $T_{\mu\nu}$ must be the Lichnerowicz / Einstein operator, with $m_T^2 = 0$. That is Chapters 6–7.

4. **Coefficient map.** The prefactor of $R$ must be a computed function of microscopic parameters $(v, \Lambda_{\rm UV}, g_c, N)$. That is Chapter 8:

$$\frac{1}{16\pi G_{\rm eff}} = \frac{\mathrm{Tr}(I)\,\Lambda_{\rm UV}^2}{16\pi^2} + \frac{v^2}{12}\ln\Big(\frac{\Lambda_{\rm UV}^2}{\mu^2}\Big). \tag{1.5}$$

5. **Bianchi consistency.** Whatever extra stress is attributed to entanglement or condensate backreaction must satisfy

$$\nabla^\mu\big(T_{\mu\nu}^{\rm matter} + T_{\mu\nu}^{\rm tensor} + T_{\mu\nu}^{\rm ent}\big) = 0. \tag{1.6}$$

   That is Chapter 11. Without it, one has a non-conservative force, not geometry.

6. **Ultraviolet control.** Higher-derivative terms generated by integrating out the condensate cannot leave propagating Ostrogradsky ghosts in the infrared. That is Chapter 9.

7. **Vacuum energy.** The same heat-kernel expansion produces a term $c_0\Lambda_{\rm UV}^4$. A framework that claims to recover Einstein's equation must say why this is not a Planck-scale cosmological constant. That is Chapter 12.

8. **Phenomenology as corollary.** Rotation curves, regular cores, inspiral drag, and tidal disruption are allowed only after items 1–7. They are not evidence that items 1–7 have been proved.

This checklist is the spine of Parts II–IV. Parts V–VI apply it.

## 1.5 The NAMCG derivation in one view

The logical order used from Chapter 3 onward is

$$\begin{aligned}\Phi^I{}_A \;&\longrightarrow\; \langle\Phi\rangle = v\,\delta^I{}_A \;\longrightarrow\; e^A_\mu = \langle\partial_\mu \Phi^{IA}\rangle \\ &\longrightarrow\; g_{\mu\nu} = \eta_{AB} e^A_\mu e^B_\nu \;\longrightarrow\; \delta\Phi = \text{conformal} + \text{vector} + T_{\mu\nu} \\ &\longrightarrow\; \delta S_D = 0 \;\longrightarrow\; m_T \to 0 \;\longrightarrow\; S_{\rm EH}[g] + S^{(\text{backreaction})}_{\rm NAMCG}.\end{aligned} \tag{1.7}$$

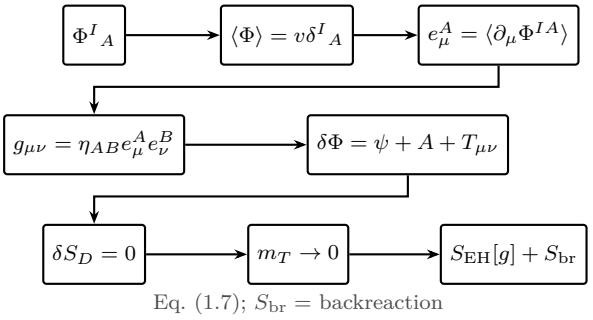



Figure 1.2: The NAMCG derivation spine used in this book. A single scalar phase cannot lock internal indices to local Lorentz frames.

Two comments are essential already.

First, the entanglement step is doing real work. In a generic spontaneously broken matrix model, $T_{\mu\nu}$ may be massive. The central physical claim of the book is that Jacobson's stationarity condition on dyadic causal diamonds,

$$\delta S_D = \delta\langle H_D\rangle = 0, \tag{1.8}$$

is not an extra thermodynamic slogan. It is a Ward identity of the microscopic theory, and that identity forbids $m_T^2 \neq 0$ in the infrared.

Second, conservation of the entropic drag is not optional. The inspiral integrator of Chapter 15,

$$\mathbf{F}_{\rm diss} = -\eta\,\frac{\mu}{d^4}\,\mathbf{v}, \tag{1.9}$$

is a reduced Newtonian laboratory. It is legitimate only because Chapter 11 has already argued that the corresponding covariant object $T^{\rm ent}_{\mu\nu}$ is divergenceless on shell. If the reader treats the Python drag as the theory, the book has failed.

## 1.6 How the later phenomenology sits on the spine

Once the infrared Einstein sector exists, NAMCG adds three controlled deformations, each with a named chapter.

**Holographic Boundary-Entropy Vacuum (HBEV).** A boundary functional involving $K/8\pi G$ and an entropy density $S_{\rm ent}(h_{ij}, \phi)$ is used to regulate $\Lambda_{\rm eff}$. This is how the book addresses the $c_0\Lambda_{\rm UV}^4$ term rather than ignoring it.

**Einsteinian cubic gravity.** A ghost-controlled cubic curvature invariant modifies the core of compact objects,

$$f(r) = 1 - \frac{2Mr^2}{r^3 + r_0^3} + \frac{\Lambda_{\rm eff}}{3} r^2, \tag{1.10}$$

so that $f(r) \to 1 + \Lambda_{\rm eff} r^2/3$ as $r \to 0$.

**Vacuum backreaction in galaxies and compact binaries.** The same conserved extra tensor sources a modification of Poisson's equation and an irreversible but covariantly conserved drag.

None of these is a substitute for Chapters 3–11. They are tests of whether the extra tensors have the right long-distance falloff and the right conservation law.

## 1.7 How to read this book

Chapters 2–5 are definitional. Read them with a pen. If a symbol is used later with two meanings, the fault is in those chapters and should be caught there.

Chapters 6–11 are the proof core. Each theorem is written with hypotheses first. If a hypothesis is an assumption of the framework rather than a lemma, it is labeled as such.

Chapters 12–15 are applications. Numerical sections flag reduced models in the first paragraph.

Chapter 16 restates the whole derivation as a single chain. Chapter 17 lists what would falsify the framework. Read Chapter 17 before deciding that the theory is closed.

**Prerequisites.**

**What you need.** Special relativity, the Euler–Lagrange equation, and the Einstein equation as a field equation for a metric. Linear algebra at the level of commutators and traces. Quantum field theory is helpful but is introduced at the point of use. Entanglement entropy is built in Chapter 6 and Appendix B. If a later proof needs more than that, the extra tool is installed in Chapter 2 or in an appendix, not presupposed as coursework.

## 1.8 Exercises

1. **Acoustic metric.** Starting from the continuity and Bernoulli equations of an irrotational barotropic fluid, derive $g^{\mu\nu}_{\mathrm{ac}}$ and show that linearized phonons see that metric. Where does a healing length appear if the fluid is a Gross–Pitaevskii condensate?

2. **Mode count.** In $3+1$ dimensions, how many on-shell degrees of freedom does a massive symmetric tensor have? A massless one with gauge invariance $\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$? Why does that counting matter for Section 1.3?

3. **Scalar-tensor ceiling.** Consider $S = \int \sqrt{-g}(\phi R - (\partial\phi)^2 - V(\phi))$. Linearize about Minkowski spacetime and a constant $\phi_0$. Show that the propagating excitations are a massless spin-2 field mixed with a scalar. What extra condition would be needed to freeze the scalar?

4. **Conservation as a constraint.** Suppose a two-body equation $\mu \ddot{\mathbf{r}} = -\nabla U + \mathbf{a}_{\mathrm{irrev}}$ is written in Minkowski space. What property must $\mathbf{a}_{\mathrm{irrev}}$ have if there is to exist a stress tensor $T^{\mathrm{ent}}_{\mu\nu}$ with $\partial^\mu (T^{\mathrm{matter}}_{\mu\nu} + T^{\mathrm{ent}}_{\mu\nu}) = 0$? A full curved-space answer is Chapter 11; a Newtonian answer is enough here.

5. **Checklist audit.** Take any analog-gravity paper you know and score it against the eight items of Section 1.4. This exercise is the standard the book applies to itself.

## 1.9 Bridge to Chapter 2

Every later proof is a sentence in a language that has not yet been fixed: manifolds and vierbeine, Einstein–Hilbert variation and Bianchi identities, heat kernels, causal diamonds, and modular Hamiltonians. Chapter 1 stated the derivation problem. Chapter 2 installs the language at the exact thickness later chapters consume—no more, because a reader who must first finish a complete treatise on general relativity will never reach the matrix condensate.

The first object Chapter 2 defines is the one this chapter kept using without ceremony: a Lorentzian manifold with a non-degenerate metric of signature $(-,+,+,+)$, and the difference between a primitive metric and a composite vierbein bilinear.

# Chapter 2

# Mathematical Preliminaries

**Learning outcomes.** After this chapter the reader should be able to:

1. distinguish a primitive metric from a composite vierbein bilinear, and write the conversion between $g_{\mu\nu}$ and $e^A_\mu$;
2. vary the Einstein–Hilbert action to obtain $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ and state the contracted Bianchi identity that forces $\nabla^\mu T_{\mu\nu} = 0$;
3. expand a functional determinant to second Seeley–DeWitt order and recognize which coefficient multiplies $R$;
4. define a causal diamond, its modular Hamiltonian, and the first variation $\delta S_D$;
5. enter Chapter 3 already possessing every geometric sentence used in the matrix action and the later Ward identities.

This chapter is a toolkit, not a treatise. Every section exists because a later proof calls it by name. Results that are used as black boxes are labeled as such. Results that NAMCG must derive, rather than import, are flagged and postponed.

## 2.1 Lorentzian manifolds and index conventions

A spacetime in this book is a smooth four-manifold $M$ equipped with a non-degenerate symmetric tensor $g_{\mu\nu}$ of signature $(-,+,+,+)$. The inverse is $g^{\mu\nu}$, and $g = \det(g_{\mu\nu})$. The volume form is $\sqrt{-g}\,\mathrm{d}^4x$.

Coordinate indices $\mu, \nu, \rho, \sigma$ transform under diffeomorphisms. Tangent-space (Lorentz) indices $A, B, \ldots$ transform under local $\mathrm{SO}(3,1)$. Internal gauge indices $I, J, \ldots$ transform under the microscopic gauge group of the condensate. These three alphabets are never mixed silently.

The Levi-Civita connection compatible with $g_{\mu\nu}$ is

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \tag{2.1}$$

The Riemann tensor, Ricci tensor, and Ricci scalar follow the conventions

$$\begin{aligned} R^{\rho}{}_{\sigma\mu\nu} &= \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}, \\ R_{\mu\nu} &= R^{\rho}{}_{\mu\rho\nu}, \qquad R = g^{\mu\nu} R_{\mu\nu}. \end{aligned} \tag{2.2}$$

The Einstein tensor is

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R. \tag{2.3}$$

*Remark* 2.1 (Mostly-minus versus mostly-plus). Particle-physics texts often use $(+,-,-,-)$. Every energy condition and every sign in a heat-kernel coefficient is convention-dependent. If a formula in a cited paper looks off by a sign, check the signature before inventing new physics.

## 2.2 Vierbeine: primitive geometry versus composite geometry

A vierbein (tetrad) is a local frame field $e^{A}_{\mu}$ satisfying

$$g_{\mu\nu} = \eta_{AB}\, e^{A}_{\mu} e^{B}_{\nu}, \qquad \eta_{AB} = \mathrm{diag}(-1,+1,+1,+1). \tag{2.4}$$

The inverse frame $e^{\mu}_{A}$ obeys $e^{A}_{\mu} e^{\mu}_{B} = \delta^{A}_{B}$ and $e^{A}_{\mu} e^{\nu}_{A} = \delta^{\nu}_{\mu}$. Local Lorentz transformations act as

$$e^{A}_{\mu}(x) \;\longmapsto\; \Lambda^{A}{}_{B}(x)\, e^{B}_{\mu}(x), \qquad \Lambda^{T}\eta\Lambda = \eta. \tag{2.5}$$

The spin connection $\omega_{\mu}{}^{A}{}_{B}$ is the unique metric- compatible connection on the frame bundle (vanishing torsion in the theories used here),

$$\nabla_\mu e^{A}_{\nu} = \partial_\mu e^{A}_{\nu} - \Gamma^{\rho}_{\mu\nu} e^{A}_{\rho} + \omega_{\mu}{}^{A}{}_{B} e^{B}_{\nu} = 0. \tag{2.6}$$

**Primitive metric.** In textbook general relativity one takes $g_{\mu\nu}$ as the fundamental field. A vierbein is an optional rewriting, useful for spinors.

**Composite metric.** In NAMCG the logical order is reversed. A microscopic matrix field $\Phi^{I}{}_{A}$ is fundamental. After condensation one defines

$$e^{A}_{\mu}(x) \equiv \langle \partial_\mu \Phi^{IA}(x) \rangle \tag{2.7}$$

and only then reconstructs $g_{\mu\nu}$ by (2.4). Equation (2.7) is not justified in this chapter. It is the definition that Chapter 4 must earn.

The distinction matters for counting degrees of freedom. A primitive symmetric tensor in four dimensions has ten components, of which diffeomorphisms remove four and local Lorentz transformations are absent because there is no independent frame. A primitive vierbein has sixteen components; local Lorentz invariance removes six, diffeomorphisms remove four, and six remain, matching the metric. When the vierbein is composite, those sixteen numbers are not independent fields. They are expectation values of derivatives of $\Phi$. Gauge redundancies of $\Phi$ and of the emergent geometry must be matched, not double counted. That matching is Chapter 10.

## 2.3 Variation of the Einstein–Hilbert action

The Einstein–Hilbert action with a cosmological term and a matter action is

$$S[g] = \frac{1}{16\pi G}\int_M \mathrm{d}^4x\,\sqrt{-g}\,(R - 2\Lambda) + S_{\mathrm{m}}[g,\Psi]. \tag{2.8}$$

Under a metric variation $g_{\mu\nu} \to g_{\mu\nu} + \delta g_{\mu\nu}$ one uses

$$\delta g^{\mu\nu} = -g^{\mu\alpha}g^{\nu\beta}\delta g_{\alpha\beta}, \qquad \delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}\,g^{\mu\nu}\delta g_{\mu\nu}, \tag{2.9}$$

and the Palatini identity

$$\delta R_{\mu\nu} = \nabla_\rho(\delta\Gamma^\rho_{\mu\nu}) - \nabla_\nu(\delta\Gamma^\rho_{\mu\rho}). \tag{2.10}$$

The covariant-derivative terms become a boundary integral. On a manifold without boundary, or after adding the Gibbons–Hawking–York term

$$S_{\mathrm{GHY}} = \frac{1}{8\pi G}\int_{\partial M}\mathrm{d}^3x\,\sqrt{|h|}\,K, \tag{2.11}$$

the bulk stationarity condition is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\,T_{\mu\nu}, \tag{2.12}$$

where the matter stress tensor is defined by

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{m}}}{\delta g^{\mu\nu}}. \tag{2.13}$$

The same definition will be used in Chapter 11 for the entropic contribution $T^{\mathrm{ent}}_{\mu\nu}$, with $S_{\mathrm{m}}$ replaced by the entanglement functional.

*Remark* 2.2 (Boundary terms are not optional in this book). Chapter 12 promotes a boundary entropy functional to a dynamical object. The combination $K/8\pi G$ appearing in (2.11) is therefore not an afterthought. It is the gravitational piece of the HBEV boundary action.

## 2.4 Bianchi identities and conservation

The Riemann tensor obeys

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0. \tag{2.14}$$

Contracting twice yields the contracted Bianchi identity

$$\nabla^{\mu} G_{\mu\nu} = 0. \tag{2.15}$$

If the field equation is $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$, then necessarily

$$\nabla^{\mu} T_{\mu\nu} = 0. \tag{2.16}$$

This is not a courtesy to matter. It is the geometric content of diffeomorphism invariance.

In an emergent theory the same identity becomes a constraint on the microscopic construction. Suppose the total effective stress tensor is split as

$$T^{\text{total}}_{\mu\nu} = T^{\text{matter}}_{\mu\nu} + T^{\text{tensor}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu}. \tag{2.17}$$

Geometry still requires $\nabla^{\mu} T^{\text{total}}_{\mu\nu} = 0$. It is not enough that ordinary matter be conserved by itself if the extra pieces dump momentum into the world. Chapter 11 exists to prove that the NAMCG split satisfies (2.16). Until that proof, any inspiral "drag" is a phenomenological force and not a geometric backreaction.

Noether's second theorem makes the same point in field language. If an action is invariant under local diffeomorphisms generated by an arbitrary vector field $\xi^{\mu}$, the Euler–Lagrange tensor $\mathcal{E}^{\mu\nu}$ of the metric obeys $\nabla_{\mu}\mathcal{E}^{\mu\nu} = 0$ identically, off shell as an identity of the tensor densities involved, and on shell as conservation of $T^{\mu\nu}$. Mass terms that break $\delta_{\xi} h_{\mu\nu} = \nabla_{\mu}\xi_{\nu} + \nabla_{\nu}\xi_{\mu}$ destroy that identity. That observation is the skeleton of Chapter 7.

## 2.5 Gauge covariant derivatives and a scalar warmup

Let $A_{\mu} = A^{a}_{\mu} T^{a}$ be a connection on a principal bundle with compact gauge group, generators $T^{a}$ anti-Hermitian in the conventions of this book. The covariant derivative

on a field $\Phi$ in a representation $\rho$ is

$$D_\mu \Phi = \partial_\mu \Phi - i A_\mu \Phi \tag{2.18}$$

(up to the representation matrices, restored whenever the index structure requires them). The field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]. \tag{2.19}$$

A complex scalar with Mexican-hat potential,

$$S[\phi] = \int \mathrm{d}^4x \Big( |D_\mu \phi|^2 - \frac{\lambda}{4} \big( |\phi|^2 - v^2 \big)^2 \Big), \tag{2.20}$$

condenses at $|\langle \phi \rangle| = v$. The phase is eaten by $A_\mu$ (Abelian Higgs / Stueckelberg mechanism), and the physical spectrum contains a massive vector and a massive radial scalar. There is no graviton.

This warmup is here for two reasons. First, the NAMCG potential in Chapter 3 is the non-Abelian matrix analogue of (2.20). Second, the failure mode is already visible: a single complex scalar has one Goldstone phase. After it is eaten, nothing remains that can be identified with a symmetric trace-free tensor $T_{\mu\nu}$.

## 2.6 One-loop effective actions and the heat kernel

Let $\mathcal{O}$ be a second-order elliptic (or hyperbolic, after Wick rotation) differential operator on a bundle over $M$, typically of Laplace type,

$$\mathcal{O} = -g^{\mu\nu} D_\mu D_\nu + E, \tag{2.21}$$

with bundle endomorphism $E$. The one-loop effective action obtained by integrating out a fluctuation $\eta$ is

$$W[g] = \frac{1}{2} \mathrm{Tr}\, \ln \mathcal{O} = -\frac{1}{2} \int_0^\infty \frac{\mathrm{d}s}{s}\, \mathrm{Tr}\, e^{-s\mathcal{O}}, \tag{2.22}$$

up to local counterterms and the usual $i\varepsilon$ / Wick-rotation caveats. The operator $e^{-s\mathcal{O}}$ is the heat kernel. Its diagonal admits the Seeley–DeWitt expansion

$$\langle x | e^{-s\mathcal{O}} | x \rangle \sim \frac{1}{(4\pi s)^2} \big( a_0(x) + a_1(x) s + a_2(x) s^2 + \cdots \big) \qquad (s \to 0^+), \tag{2.23}$$

in four dimensions. For a Laplace-type operator the first coefficients are

$$\begin{aligned} a_0 &= 1, \\ a_1 &= \frac{1}{6}R - E, \\ a_2 &= \frac{1}{180}\big(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu} + \tfrac{7}{2}R^2\big) + \frac{1}{2}E^2 - \frac{1}{6}RE + \cdots , \end{aligned} \tag{2.24}$$

where the omitted terms involve derivatives of $E$ and the bundle curvature. The precise tensor combination in $a_2$ will be written in full in Appendix A when it is needed for ghost analysis. What Chapter 8 requires is only this fact:

**Proposition 2.3** (Which coefficient produces Newton's constant)**.** *In the derivative expansion of $W[g]$, the term linear in curvature and quadratic in the ultraviolet cutoff comes from $a_1$. After a cutoff $\Lambda_{\mathrm{UV}}$ is imposed on the proper-time integral* (2.22)*, one obtains a contribution of the schematic form*

$$W[g] \supset \int \mathrm{d}^4x\,\sqrt{-g}\,(c_0\Lambda_{\mathrm{UV}}^4 + c_2\Lambda_{\mathrm{UV}}^2 R + \cdots). \tag{2.25}$$

*The coefficient $c_2$ is the microscopic ancestor of $1/(16\pi G_{\mathrm{eff}})$. The coefficient $c_0$ is the microscopic ancestor of the cosmological-constant problem.*

*Proof sketch.* Insert (2.23) into (2.22) and cut the proper-time integral at $s \sim \Lambda_{\mathrm{UV}}^{-2}$. The $a_0/s^3$ piece produces $\Lambda_{\mathrm{UV}}^4$. The $a_1/s^2$ piece produces $\Lambda_{\mathrm{UV}}^2$ times $\frac{1}{6}R - E$. After the endomorphism $E$ is evaluated on the condensate background, the curvature term remains and is absorbed into the Einstein–Hilbert prefactor. Details and the logarithmic $v^2\ln(\Lambda_{\mathrm{UV}}/\mu)$ piece are Chapter 8. □

Two warnings, both of which the later text will spend chapters respecting.

1. The quartic term $c_0\Lambda_{\mathrm{UV}}^4$ is not small. Declaring $\Lambda_{\mathrm{eff}} \approx 10^{-120}\rho_{\mathrm{Pl}}$ without a mechanism is not a calculation. Chapter 12 is that mechanism, or it is a failure.

2. If the fluctuation operator is fourth order, as in the quadratic action of Chapter 9, the heat-kernel calculus changes and Ostrogradsky poles appear. One cannot quote $a_1$ from a second-order Laplacian and then use a $\Box^2$ kinetic term without further work.

## 2.7 Causal diamonds

Let $p$ and $q$ be events in $M$ with $q$ in the chronological future of $p$. The causal diamond (Alexandrov set)

$$D(p,q) = J^+(p) \cap J^-(q) \tag{2.26}$$

is the intersection of the causal future of $p$ with the causal past of $q$. Its boundary $\partial D$ consists of two null cones joined at a spacelike 2-surface $\mathcal{B}$ (the edge, or bifurcation surface in the small-diamond limit).

A *small* causal diamond is one whose proper size is short compared with the local curvature radius, so that the metric inside $D$ may be written in Riemann normal coordinates

$$g_{\mu\nu}(x) = \eta_{\mu\nu} - \frac{1}{3} R_{\mu\rho\nu\sigma}(0) x^{\rho} x^{\sigma} + O(x^3). \tag{2.27}$$

Jacobson's thermodynamic derivation of Einstein's equation uses precisely such diamonds: the matter flux across $\partial D$ is compared with the variation of entanglement entropy of the diamond interior.

*Dyadic* diamonds $D_{j,n}$ are a nested family obtained by repeated bisection of a reference diamond. They are bookkeeping. The physics is in the statement that first variations of entropy vanish for arbitrary small deformations of $\partial D$, not in the dyadic label.

**Definition 2.4** (Diamond entanglement functional)**.** Let $\rho_D = \mathrm{Tr}_{M\setminus D}\rho$ be the reduced state of the microscopic theory on the diamond interior. The von Neumann entropy

$$S_D = -\mathrm{Tr}(\rho_D \ln \rho_D) \tag{2.28}$$

is the diamond entanglement functional. The modular Hamiltonian is

$$H_D = -\ln \rho_D, \tag{2.29}$$

so that $S_D = \langle H_D \rangle$ up to the additive constant $\mathrm{Tr}(\rho_D \ln \rho_D) + \langle \ln \rho_D \rangle = 0$ written in that form.

For a free field in Rindler space, $H_D$ is proportional to the boost generator (Bisognano–Wichmann). For a small diamond in a curved background the same statement holds approximately, with Unruh temperature

$$T_{\mathrm{Unruh}} = \frac{\kappa}{2\pi}, \tag{2.30}$$

where $\kappa$ is the surface gravity of $\partial D$ as seen by the conformal Killing vector that fixes $\mathcal{B}$.

This book does *not* take Jacobson's equation of state as an axiom imported from thermodynamics of horizons. Chapter 6 derives $\delta S_D = 0$ from the microscopic matrix theory, or states clearly where an assumption remains. The definitions above are only the language in which that derivation is written.

## 2.8 Mutual information

For two regions $A$ and $B$,

$$I(A:B) = S(A) + S(B) - S(A \cup B). \tag{2.31}$$

Mutual information is non-negative, vanishes if and only if $\rho_{AB} = \rho_A \otimes \rho_B$ (for finite-dimensional algebras), and is invariant under local unitaries on $A$ or $B$. In NAMCG the macroscopic potential

$$U_{\text{ent}} = -\lambda\, I(A:B) = -\lambda\big[S(A) + S(B) - S(A \cup B)\big] \tag{2.32}$$

is the interaction energy assigned to two condensed cores $A$ and $B$. The sign is chosen so that positive mutual information lowers the potential: entanglement is attractive in the effective description.

The variation of $U_{\text{ent}}$ with respect to the worldline of a core produces the drag acceleration of Chapter 11. Because $I(A:B)$ is built from diffeomorphism scalars $S(\,\cdot\,)$, that variation is not free to point in an arbitrary non-conservative direction. Making this remark precise is the content of Theorem 2 in the ultraviolet note, rewritten as a textbook proof in Chapter 11.

## 2.9 What this chapter deliberately omits

The following topics are real, and they are not treated here.

- A complete development of spinor bundles and the Dirac operator. NAMCG as formulated in the source notes is bosonic at the microscopic level. Fermionic matter can be coupled to the emergent vierbein later; it is not needed to obtain $G_{\mu\nu}$.
- Full renormalization of quantum gravity as a fundamental metric theory. The metric is not fundamental here. The ultraviolet object is the matrix condensate, with cutoff $\Lambda_{\text{UV}}$.
- A self-contained treatise on Tomita–Takesaki modular theory. Definition 2.4 and the replica formula $S = (1 - n\partial_n) \ln \operatorname{Tr} \rho^n|_{n=1}$ are sufficient for Chapter 6. Appendix B expands them.
- Numerical relativity. The integrators of Chapter 15 are ordinary-differential toy models, announced as such.

If a later proof appears to need a tool that is not in this chapter, the defect is in the later proof or in this chapter, and should be repaired rather than papered over

with a citation.

## 2.10 Exercises

1. **Volume variation.** Derive both identities in (2.9) from $\delta g = g\, g^{\mu\nu}\delta g_{\mu\nu}$ and $g^{\mu\alpha}g_{\alpha\nu} = \delta^{\mu}_{\nu}$.

2. **Einstein tensor from variation.** Starting from (2.8) with $\Lambda = 0$ and no boundary, obtain $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. Keep the Palatini terms long enough to see them become a total derivative.

3. **Contracted Bianchi.** Contract (2.14) twice and obtain $\nabla^{\mu}G_{\mu\nu} = 0$. Do not use the field equations.

4. **Frame versus metric degrees of freedom.** Count local components of $e^{A}_{\mu}$ and of $g_{\mu\nu}$. Subtract local Lorentz transformations and diffeomorphisms. Confirm that both descriptions have six physical components off shell in the usual sense of geometric gravity, before any equations of motion.

5. **Composite-frame warning.** Suppose $e^{A}_{\mu} = \langle \partial_{\mu}\Phi^{IA} \rangle$ and $\Phi$ is subject to a global internal rotation $\Phi \mapsto U\Phi$. What happens to $e^{A}_{\mu}$? What additional structure is required before this rotation can be identified with a local Lorentz transformation of the emergent frame? (The answer is Chapter 10; write the question cleanly now.)

6. **Heat-kernel dimensions.** In four dimensions, assign canonical mass dimensions to $a_0, a_1, a_2$ and to the proper-time variable $s$. Show that $a_0\Lambda^4_{\mathrm{UV}}$ and $a_1\Lambda^2_{\mathrm{UV}}$ are the only two local terms generated by (2.23) that can appear as relevant or marginal couplings of mass dimension four or less, aside from logarithms.

7. **Small diamond.** In Riemann normal coordinates (2.27), compute the spatial 3-volume of a small diamond of radius $\ell$ to order $\ell^5 R$. This is the geometric ingredient in Jacobson's original argument.

8. **Mutual information positivity.** Prove $I(A:B) \geq 0$ from strong subadditivity, or from relative entropy $S(\rho_{AB}\|\rho_A \otimes \rho_B)$. Why does the sign in (2.32) then correspond to an attractive effective potential?

## 2.11 Bridge to Chapter 3

The language is now in place. A metric may be primitive or composite. If it is composite, it is built from a vierbein, and the vierbein must come from a field that carries a tangent index. The Einstein tensor is the Euler–Lagrange tensor of (2.8),

and Bianchi forces conservation. Newton's constant, if it is emergent, lives in the $a_1$ heat-kernel coefficient. Entanglement lives on causal diamonds.

What is missing is the field. Chapter 3 introduces the non-Abelian matrix order parameter $\Phi^I{}_A$, the matrix-coordinate algebra $[\hat{X}^\mu, \hat{X}^\nu] = i\Theta^{\mu\nu}$, and the microscopic action whose condensation will produce the vierbein of (2.7). After that chapter, geometry is no longer an independent ingredient. It is an output.

# Chapter 3

# Matrix Coordinates and the Microscopic Action

**Learning outcomes.** After this chapter the reader should be able to:

1. write the non-commutative coordinate algebra $[\hat{X}^\mu, \hat{X}^\nu] = i\Theta^{\mu\nu}$ and state how $\Theta^{\mu\nu}$ supplies an ultraviolet cutoff;
2. write both microscopic actions used in the NAMCG notes—the pregeometric matrix action and the covariant condensate-field action—and say what each is for;
3. explain why the order parameter must carry an internal index $I$ and a tangent index $A$;
4. expand the Mexican-hat matrix potential about its vacuum and identify the radial and Goldstone directions;
5. enter Chapter 4 knowing which object will condense into a vierbein.

Chapter 2 supplied geometry as language. This chapter supplies the field. After this point the metric is not an independent ingredient. It is something the field must produce.

## 3.1 The scalar failure, restated as a field-content problem

A single complex scalar $\phi$ transforming as a singlet under every tangent-space representation can condense. It can eat one Goldstone phase into a gauge field. It cannot furnish a local Lorentz frame. The infrared excitations of such a condensate are, at best,

- a radial Higgs scalar,

- a massive or massless vector, if a gauge field is present,
- phonons that admit an acoustic metric rewrite.

There is no symmetric trace-free tensor $T_{\mu\nu}$ hiding in $\phi$. Seeking that tensor in a more elaborate potential for the same $\phi$ is a representation-theoretic error, not a calculational one.

NAMCG therefore promotes the order parameter to a rectangular (or square, after a choice of basis) matrix field

$$\Phi^I{}_A(x), \tag{3.1}$$

where $I = 1, \ldots, N$ labels an internal gauge representation and $A = 0, 1, 2, 3$ labels an emergent tangent space. The field is a linear map from the tangent space of an auxiliary frame into the internal space, or the reverse, according to index placement. The placement used in this book is: $I$ up, $A$ down on $\Phi$, unless a formula from the source notes is being quoted verbatim.

The rest of the chapter writes an action for $\Phi$ and for a closely related pregeometric object, the matrix coordinates $\hat{X}^\mu$.

## 3.2 Two presentations, and why both appear

The original notes use two microscopic languages. They are not identical, and treating them as interchangeable by slogan would produce a confused book.

**Presentation A: pregeometric matrix coordinates.** There is, at the outset, no manifold. The dynamical objects are Hermitian operators $\hat{X}^\mu$ acting on a Hilbert space $\mathcal{H}$. Spacetime points, if they appear at all, appear as eigenvalue data of a complete commuting subset of these operators, or as a derived continuum after condensation.

**Presentation B: condensate field on an auxiliary background.** There is an auxiliary manifold with a non-dynamical metric $g^{(0)}_{\mu\nu}$. The dynamical object is the field $\Phi^I{}_A(x)$. The physical metric $g_{\mu\nu}$ is not $g^{(0)}_{\mu\nu}$. The auxiliary metric is a scaffold for writing derivatives and heat kernels. Chapter 8 uses this scaffold.

This book treats Presentation A as the ontological starting point and Presentation B as the background-field language appropriate after a condensate has selected a notion of local coordinates. Chapter 4 is the bridge. Until that bridge is built, the two actions below are written separately and not added together.

**Assumption 3.1** (Auxiliary background is non-dynamical)**.** Whenever Presentation B is in use, $g^{(0)}_{\mu\nu}$ is a fixed scaffold, usually taken flat, $g^{(0)}_{\mu\nu} = \eta_{\mu\nu}$. It is not varied in the Einstein–Hilbert sense. Varying the *emergent* metric $g_{\mu\nu}$ defined in Chapter 4 is a different operation.

## 3.3 Presentation A: matrix coordinates and ultraviolet cutoff

**Definition 3.2** (Matrix coordinates)**.** Let $\mathcal{H}$ be a separable Hilbert space and let $\hat{X}^\mu$, $\mu = 0, 1, 2, 3$, be Hermitian operators on $\mathcal{H}$. The fundamental kinematic relation of Presentation A is

$$[\hat{X}^\mu, \hat{X}^\nu] = i\Theta^{\mu\nu}, \tag{3.2}$$

where $\Theta^{\mu\nu} = -\Theta^{\nu\mu}$ is an antisymmetric tensor with dimensions of length squared. If $\Theta^{\mu\nu}$ is a multiple of the identity on $\mathcal{H}$, it is a $c$-number background. In the fully dynamical theory it may itself be an operator; the notes treat the leading infrared analysis with a $c$-number $\Theta^{\mu\nu}$.

The scale

$$\Lambda_{\mathrm{UV}} \sim \Theta^{-1/2}, \qquad \Theta \equiv \sqrt{\tfrac{1}{2}\Theta_{\mu\nu}\Theta^{\mu\nu}}, \tag{3.3}$$

is the ultraviolet cutoff of the theory. Two operators that fail to commute cannot be simultaneously diagonalized, so the notion of a sharply localized event $(x^0, x^1, x^2, x^3)$ is meaningless below $\Lambda_{\mathrm{UV}}^{-1}$. This is the NAMCG replacement for a hard lattice.

*Remark* 3.3 (What $\Theta^{\mu\nu}$ is not). Equation (3.2) is not yet a solution of the cosmological-constant problem, not yet a black-hole core, and not yet a graviton. It is a kinematic regulator. Claims that begin "because spacetime is non-commutative…" and end at galactic rotation curves are skipping the work of Chapters 4–14.

A Yang–Mills-type matrix action for these coordinates, including a cubic structure-constant coupling to a background condensate tensor $C_{\mu\nu\lambda}$, is

$$\begin{aligned} S_{\mathrm{matrix}} = &-\frac{1}{4g_{\mathrm{YM}}^2}\mathrm{Tr}\big([\hat{X}^\mu, \hat{X}^\nu][\hat{X}_\mu, \hat{X}_\nu]\big) + \frac{1}{3!}\mathrm{Tr}\big(C_{\mu\nu\lambda}[\hat{X}^\mu, \hat{X}^\nu]\hat{X}^\lambda\big) \\ &+ S_{\mathrm{matter}}(\Psi, \hat{X}^\mu). \end{aligned} \tag{3.4}$$

The first term is the IKKT / reduced-Yang–Mills kinetic term. On a background satisfying (3.2) it is proportional to $\mathrm{Tr}(\Theta_{\mu\nu}\Theta^{\mu\nu})$ plus fluctuation pieces. The cubic term selects a preferred vacuum condensation structure; its detailed index symmetries are constrained by the requirement that the vacuum preserve a Lorentzian interpretation after Chapter 4.

Matter fields $\Psi$ in $S_{\mathrm{matter}}$ are functions of the matrices $\hat{X}^\mu$ in the sense of the Weyl map associated with $\Theta^{\mu\nu}$. This book does not need the full star-product calculus until phenomenology with explicit localized sources appears in Chapter 14. The existence of such a map is enough for now.

## 3.4 Presentation B: the covariant condensate action

Once local coordinates $x^\mu$ have been introduced as labels on an auxiliary manifold, the dynamical variable is the matrix field $\Phi^I{}_A(x)$. Let $A_\mu$ be a connection on the internal bundle, with covariant derivative

$$D_\mu \Phi = \partial_\mu \Phi - i A_\mu \Phi \tag{3.5}$$

acting on the $I$ index. The microscopic action used for the heat-kernel analysis is

$$S_{\text{micro}}[\Phi, A] = \int \mathrm{d}^4x \, \sqrt{-g^{(0)}} \, \mathrm{Tr}\Big( g^{(0)\mu\nu} (D_\mu \Phi)^\dagger D_\nu \Phi - V(\Phi^\dagger \Phi) \Big). \tag{3.6}$$

The trace is over internal indices. Tangent indices on $\Phi$ are contracted through the potential and through the identification of $A$ with a frame index after condensation; they are not traced in the same sum.

The potential that triggers condensation is the matrix Mexican hat

$$V(\Phi^\dagger \Phi) = \frac{\lambda}{4} \mathrm{Tr}\big(\Phi^\dagger \Phi - v^2 \mathbb{I}\big)^2. \tag{3.7}$$

Its minima lie on the manifold

$$\Phi^\dagger \Phi = v^2 \mathbb{I} \tag{3.8}$$

in the internal space, up to unitary rotations. A representative vacuum used throughout the book is

$$\langle \Phi^I{}_A \rangle = v \, \delta^I{}_A, \tag{3.9}$$

which is available when the internal dimension is at least four and a basis has been chosen that identifies four internal directions with the tangent directions. If $N > 4$, the remaining internal directions must either decouple or be locked by additional Yukawa-like couplings. The source notes assume that a vacuum of the form (3.9) exists. This book records that as an assumption.

**Assumption 3.4** (Existence of a frame-locking vacuum)**.** There exists a stable vacuum (3.9) (or a locally equivalent vacuum) in which four internal directions are identified with the four tangent directions. Goldstone modes associated with the broken generators are available to be eaten by a connection, as in Chapter 10.

## 3.5 Index structure and the intended infrared spectrum

Under an internal gauge transformation $\Omega(x)$ and a local Lorentz transformation $\Lambda(x)$, the field is required to transform as

$$\Phi^I{}_A(x) \;\longmapsto\; \Omega^I{}_J(x) \, \Phi^J{}_B(x) \, \Lambda^B{}_A(x). \tag{3.10}$$

Equation (3.10) is the transformation law that makes $\Phi$ capable of locking the two groups to a diagonal subgroup. It is a choice of field content. It is not derived from Presentation A alone; it is the additional structure that Presentation B puts in by hand so that a vierbein can exist.

Fluctuations about (3.9) will be decomposed in Chapter 5 into

$$\delta\Phi^{I}{}_{A} = \frac{1}{4}\eta_{\mu\nu}\delta g^{\mu\nu}\, I + A_{\mu}\gamma^{\mu} + T_{\mu\nu}. \tag{3.11}$$

The only fact needed here is that the last piece exists because $\Phi$ has two indices that can be symmetrized and made traceless after the vacuum has identified $I$ with $A$. A scalar has no such piece.

## 3.6 Equations of motion from Presentation B

Varying (3.6) with respect to $\Phi^{\dagger}$ gives

$$D_{\mu}\big(\sqrt{-g^{(0)}}\, g^{(0)\mu\nu} D_{\nu}\Phi\big) + \sqrt{-g^{(0)}}\,\frac{\lambda}{2}\Phi\big(\Phi^{\dagger}\Phi - v^{2}\mathbb{I}\big) = 0. \tag{3.12}$$

Varying with respect to $A_{\mu}$ produces a matrix current conservation law

$$D_{\mu}\big(\Phi(D^{\mu}\Phi)^{\dagger} - (D^{\mu}\Phi)\Phi^{\dagger}\big) = 0, \tag{3.13}$$

up to the conventional factor of $i$ in the current definition. These equations determine the condensate profile on the auxiliary background. They do not yet determine the Einstein tensor of the emergent metric. That requires integrating out fluctuations, which is Chapter 8, after the emergent metric has been defined in Chapter 4 and the tensor mode identified in Chapter 5.

## 3.7 Relating the two presentations

A complete derivation from (3.4) to (3.6) would construct, from the spectra of $\hat{X}^{\mu}$, a commutative manifold, a Weyl map, and a slowly varying matrix-valued field on that manifold whose covariant derivative reproduces the commutator $[\hat{X}^{\mu}, \hat{\Phi}]$. That derivation is not supplied in full in the source notes. The working relation used in this book is the following.

**Assumption 3.5** (Infrared dictionary)**.** In a condensate phase in which the matrices $\hat{X}^{\mu}$ are, to leading order, commutative coordinates $x^{\mu}\mathbb{I}$ plus fluctuations, there exists a slowly varying field $\Phi^{I}{}_{A}(x)$ such that

$$[\hat{X}_{\mu}, \Phi] \;\longleftrightarrow\; i\Theta_{\mu\nu}D^{\nu}\Phi \qquad \text{plus higher-derivative corrections in } \Theta. \tag{3.14}$$

Under this dictionary the quadratic commutator term in (3.4) generates the kinetic term in (3.6), and $\Lambda_{\mathrm{UV}}$ remains the cutoff of the derivative expansion.

Assumption 3.5 is the most important unproved bridge in the microscopic setup. Chapter 17 lists a sharper derivation of this dictionary as an open problem. Every later heat-kernel formula is a calculation *within* Presentation B, contingent on the dictionary.

## 3.8 Energy scales and coupling constants

The dimensionful quantities introduced so far are

$$\Lambda_{\mathrm{UV}}, \qquad v, \qquad \lambda, \qquad g_{\mathrm{YM}}, \tag{3.15}$$

together with the auxiliary-background curvature scale, which is set to zero when $g^{(0)} = \eta$. Their intended roles are:

| Quantity | Role |
|---|---|
| $\Lambda_{\mathrm{UV}}$ | kinematic cutoff from $\Theta^{\mu\nu}$ |
| $v$ | condensate scale; enters $G_{\mathrm{eff}}$ logarithmically |
| $\lambda$ | controls the radial Higgs mass $m_\sigma \sim \sqrt{\lambda}\, v$ |
| $g_{\mathrm{YM}}$ | internal gauge coupling; enters the connection that eats Goldstones |

Chapter 8 will produce

$$\frac{1}{16\pi G_{\mathrm{eff}}} = \frac{\mathrm{Tr}(\mathbb{I})\,\Lambda_{\mathrm{UV}}^2}{16\pi^2} + \frac{v^2}{12}\ln\Big(\frac{\Lambda_{\mathrm{UV}}^2}{\mu^2}\Big). \tag{3.16}$$

Two immediate inequalities follow from insisting that the infrared Newton coupling be finite and that the radial mode be integrable:

$$m_\sigma \ll \Lambda_{\mathrm{UV}}, \qquad v \lesssim \Lambda_{\mathrm{UV}}. \tag{3.17}$$

If $v \sim \Lambda_{\mathrm{UV}}$ the logarithm is order one and the first term in (3.16) dominates unless the internal trace $\mathrm{Tr}(\mathbb{I})$ is arranged to cancel against counterterms. The book does not hide that arrangement. It is part of the coefficient-matching problem, not a detail.

## 3.9 What has been assumed, collected

The microscopic theory, as used from here on, rests on:

1. the coordinate algebra (3.2) with cutoff (3.3);

2. the matrix action (3.4) at the pregeometric level;
3. the condensate action (3.6)–(3.7) after coordinates exist;
4. Assumption 3.1, that $g^{(0)}$ is a scaffold;
5. Assumption 3.4, that a vacuum $\langle \Phi^I{}_A \rangle = v\delta^I{}_A$ exists;
6. Assumption 3.5, the infrared commutator dictionary.

Everything in Chapters 4–11 is a calculation from this list, or an additional assumption announced as such. The list is long enough that the theory is not "derived from nothing." It is derived from a definite microscopic ansatz. That is still a stronger starting point than inserting $g_{\mu\nu}$ by hand, provided the later chapters do the work advertised in Chapter 1.

## 3.10 Exercises

1. **Dimensions.** Assign mass dimensions to $\hat{X}^\mu$, $\Theta^{\mu\nu}$, $\Phi$, $v$, $\lambda$, and $g_{\mathrm{YM}}$ in four dimensions, with $\hbar = c = 1$. Confirm (3.3).
2. **Vacuum of the potential.** Show that (3.7) is bounded below and that its minima satisfy (3.8). Linearize $\Phi = v\,\mathbb{I} + \sigma + i\pi$ in the internal space (ignoring tangent structure) and read off $m_\sigma^2$ and $m_\pi^2$.
3. **Why $N \geq 4$.** If the internal dimension is $N = 2$, can a vacuum of the form (3.9) exist? What infrared representation content is then missing?
4. **Commutator to derivative.** In the Moyal plane with constant $\Theta^{12} = -\Theta^{21} = \theta$ and all other components zero, verify the leading term of (3.14) on a plane-wave matrix $\Phi = e^{ik\cdot\hat{X}}$. Where do higher powers of $\theta$ appear?
5. **IKKT vacuum energy.** Evaluate the first term of (3.4) on the $c$-number background (3.2). Show that it is a constant proportional to $\mathrm{Tr}(\Theta_{\mu\nu}\Theta^{\mu\nu})$. This constant is a preview of the $c_0\Lambda_{\mathrm{UV}}^4$ problem.
6. **Transformation law.** Starting from (3.10) and the representative vacuum (3.9), compute the infinitesimal variation of $\Phi$ under a simultaneous internal rotation and local Lorentz transformation that preserve the vacuum. Which generators are broken? (This is the algebra of Chapter 10, prepared here.)
7. **No graviton in the scalar truncation.** Set $N = 1$ and drop the tangent index, reducing $\Phi$ to a complex scalar. Show that the fluctuation spectrum of (3.6) contains no spin-2 state. This is the Chapter 1 ceiling written as a homework problem.

## 3.11 Bridge to Chapter 4

The field exists, the potential prefers a vacuum $\langle \Phi^I{}_A \rangle = v\delta^I{}_A$, and the coordinate algebra supplies a cutoff. Nothing yet is a metric. Chapter 4 defines the emergent vierbein as the condensate gradient

$$e^A_\mu(x) \equiv \langle \partial_\mu \Phi^{IA}(x) \rangle, \tag{3.18}$$

reconstructs $g_{\mu\nu} = \eta_{AB} e^A_\mu e^B_\nu$, and writes the Seiberg–Witten map that relates the non-commutative background $\Theta^{\mu\nu}$ to the leading correction $\mathcal{G}_{\mu\nu}(\langle \hat{X} \rangle)$. After that chapter, geometry is an output of the action written here.

# Chapter 4

# Condensation, Vierbein Emergence, and the Seiberg–Witten Map

**Learning outcomes.** After this chapter the reader should be able to:

1. explain why a strictly constant vacuum $\langle \Phi^I{}_A \rangle = v\delta^I{}_A$ cannot, by itself, produce a non-degenerate vierbein from $\langle \partial_\mu \Phi \rangle$;
2. write the polar decomposition $\Phi = (v + \sigma)\Lambda$ and construct an emergent frame from the condensate gradient of $\Lambda$;
3. reconstruct $g_{\mu\nu} = \eta_{AB} e^A_\mu e^B_\nu$ and state the non-degeneracy condition;
4. use the leading Seiberg–Witten map to relate $\Theta^{\mu\nu}$ to a correction of the emergent metric;
5. enter Chapter 5 with a definite background about which to expand $\delta\Phi$.

Chapter 3 introduced the field and a preferred vacuum. This chapter turns that vacuum into geometry. No Einstein equation is derived here. The only output is a composite metric, plus the dictionary that connects it to the non-commutative parameter $\Theta^{\mu\nu}$.

## 4.1 The constant-vacuum obstruction

The representative vacuum of Assumption 3.4 is

$$\langle \Phi^I{}_A \rangle = v\, \delta^I{}_A. \tag{4.1}$$

If this expectation value is independent of $x$, then

$$\langle \partial_\mu \Phi^{IA} \rangle = 0. \tag{4.2}$$

The definition advertised in the source notes,

$$e^A_\mu(x) \equiv \langle \partial_\mu \Phi^{IA}(x) \rangle, \tag{4.3}$$

then yields the zero tensor. A zero tensor is not a frame.

This is not a notational quibble. A textbook that passes from (4.1) to a non-degenerate $g_{\mu\nu}$ without an intermediate step is asserting an identity that is false. There are two honest repairs. NAMCG uses both, in different places; they must be kept distinct.

**Repair I: the polar field carries the frame.** Write the field in polar form, as in the ultraviolet note,

$$\Phi^{IA}(x) = \big(v + \sigma(x)\big)\, \Lambda^A{}_I(x), \tag{4.4}$$

where $\sigma$ is a radial singlet and $\Lambda(x)$ takes values in the broken coset that rotates internal indices into tangent indices. Even if $v$ is constant, $\partial_\mu \Phi$ need not vanish:

$$\partial_\mu \Phi^{IA} = (\partial_\mu \sigma)\, \Lambda^A{}_I + (v + \sigma)\, \partial_\mu \Lambda^A{}_I. \tag{4.5}$$

A vacuum with $\langle \sigma \rangle = 0$ and a slowly varying $\langle \Lambda \rangle$ produces a candidate frame from $\langle \partial_\mu \Lambda \rangle$.

**Repair II: identify the vacuum itself with a frame.** Promote the representative vacuum to a spacetime-dependent but still "pure Higgs" configuration

$$\langle \Phi^I{}_A(x) \rangle = v\, e^I{}_A(x), \tag{4.6}$$

and treat $e^I{}_A$ as the fundamental condensate profile. In this repair, equation (4.3) is abandoned as a definition. The vierbein *is* the normalized vacuum,

$$e^A_\mu(x) \equiv \frac{1}{v} \langle \Phi^A{}_\mu(x) \rangle, \tag{4.7}$$

after the internal and tangent bases have been identified.

Repair II is simpler and is the standard Higgs-gravity pattern. Repair I is what the notes write when they insist on (4.3). This chapter adopts Repair I as the official definition, because that is the formula used in the later fluctuation analysis, and records Repair II as an equivalent leading- order gauge choice.

**Definition 4.1** (Official emergent vierbein)**.** In a polar configuration (4.4) with $\langle\sigma\rangle = 0$,

$$e_\mu^A(x) \equiv \frac{1}{v}\,\langle \partial_\mu \Phi^{IA}(x)\rangle\Big|_{\sigma=0} = \langle \partial_\mu \Lambda^A{}_I(x)\rangle. \tag{4.8}$$

The factor $1/v$ restores a dimensionless frame when $\Phi$ has the engineering dimension of $v$. If a given formula in the notes omits $1/v$, it is using units $v = 1$.

*Remark* 4.2 (When the two repairs agree)*.* Choose local coordinates and a gauge in which $\Lambda^A{}_I = \delta^A{}_I + v^{-1}\omega^A{}_{I\mu}x^\mu + \cdots$. Then $\partial_\mu\Lambda$ is constant to first order and Repair I reproduces Repair II at that order. The distinction reappears at quadratic order in fluctuations, which is Chapter 5.

## 4.2 The composite metric

**Definition 4.3** (Composite metric)**.** Given a vierbein (4.8),

$$g_{\mu\nu}(x) \equiv \eta_{AB}\, e_\mu^A(x)\, e_\nu^B(x). \tag{4.9}$$

Non-degeneracy is not automatic.

**Assumption 4.4** (Non-degenerate condensate frame)**.** On the region of spacetime under consideration, $e_\mu^A$ has non-vanishing determinant. Equivalently, the four covectors $e^A{}_\mu \mathrm{d}x^\mu$ are linearly independent.

If Assumption 4.4 fails, $g_{\mu\nu}$ is degenerate and there is no Lorentzian geometry to discuss. Vanishing of $\det e$ is a physical possibility—a condensate defect, a region where the frame-locking vacuum has not formed—not a gauge artifact. This book restricts attention to the condensed phase in which Assumption 4.4 holds.

The inverse metric and inverse frame are defined in the usual way once non-degeneracy is granted,

$$g^{\mu\nu} e_\mu^A e_\nu^B = \eta^{AB}, \qquad e_A^\mu e_\nu^A = \delta_\nu^\mu. \tag{4.10}$$

Local Lorentz transformations act on the frame index,

$$e_\mu^A \;\longmapsto\; \Lambda^A{}_B e_\mu^B, \tag{4.11}$$

and leave $g_{\mu\nu}$ invariant. Diffeomorphisms act on the coordinate index. These two redundancies are the reason a composite metric can have the same geometric content as a primitive metric, provided the Goldstone counting of Chapter 10 works.

## 4.3 What kind of object $e_\mu^A$ is

At this stage $e_\mu^A$ is a condensate profile, not a solution of Einstein's equation. It is closer to a magnetization than to a gravitational field. Three consequences follow.

1. There is no a priori reason for $e_\mu^A$ to satisfy the torsion-free condition (2.6). If torsion is to vanish, that is a dynamical statement about the connection that eats the Goldstones, not a definition.

2. There is no a priori reason for small disturbances of $g_{\mu\nu}$ to propagate as massless spin-2 waves. That is Chapter 7.

3. The auxiliary scaffold $g^{(0)}_{\mu\nu}$ of Presentation B is still present in the microscopic action. It must not be confused with (4.9). Heat-kernel calculations in Chapter 8 are performed on a background that is eventually identified with the emergent metric; making that identification precise is part of the coefficient map.

## 4.4 Seiberg–Witten map and the $\Theta$ correction

Presentation A still has a non-commutative parameter $\Theta^{\mu\nu}$. After a commutative manifold has been introduced, the open-string / non-commutative-gauge-theory dictionary of Seiberg and Witten relates an ordinary metric $g_{\mu\nu}$ and a two-form background to an open-string effective metric. The source notes write the leading relation

$$G_{\mu\nu}(x) = \big((g+\Theta)^{-1}\big)_{\mu\nu} + \mathcal{G}_{\mu\nu}(\langle\hat{X}\rangle). \tag{4.12}$$

Two clarifications are required before (4.12) is used.

**Matrix versus form.** As written, $g+\Theta$ adds a symmetric tensor to an antisymmetric tensor. The honest object is the mixed tensor

$$(g+\Theta)_\mu{}^\nu = g_{\mu\rho}\eta^{\rho\nu} + \Theta_\mu{}^\nu, \tag{4.13}$$

or, equivalently, the linear map $g+\Theta : T^*M \to TM$ after one index of $\Theta$ is raised with the auxiliary metric. The inverse in (4.12) is the inverse of that map. In components, for constant $\Theta$ on a flat auxiliary background,

$$\big((g+\Theta)^{-1}\big)_{\mu\nu} = g_{\mu\nu} - g_{\mu\alpha}\Theta^{\alpha\beta}g_{\beta\nu} + O(\Theta^2). \tag{4.14}$$

The first correction is symmetric if one takes the symmetric part, which is the part that can enter a metric. The antisymmetric remainder is a two-form background, not a correction to $g_{\mu\nu}$.

**The extra term $\mathcal{G}_{\mu\nu}$.** The piece $\mathcal{G}_{\mu\nu}(\langle\hat{X}\rangle)$ in (4.12) is the backreaction of the matrix condensate on the emergent geometry, beyond the kinematic $\Theta$ correction. At this stage of the book it is a name, not a formula. Chapters 8 and 11 compute pieces of it: the Einstein–Hilbert term from the heat kernel, and the entropic stress tensor from mutual information. Until those calculations exist, $\mathcal{G}_{\mu\nu}$ must not be assigned a phenomenology.

**Definition 4.5** (Working infrared metric)**.** Unless a higher-order $\Theta$ correction is being tracked explicitly, "the emergent metric" means (4.9). Equation (4.12) is the next-order kinematic correction plus a placeholder for dynamical backreaction.

## 4.5 A linearised frame around Minkowski spacetime

The fluctuation chapters need a background. Choose coordinates and a local Lorentz gauge so that the condensed frame is the trivial frame plus a small disturbance,

$$e_\mu^A = \delta_\mu^A + \tfrac{1}{2}h^A{}_\mu + \omega^A{}_\mu, \tag{4.15}$$

where $h_{\mu\nu} \equiv \eta_{AB}\delta_\mu^A h^B{}_\nu$ is symmetric and $\omega_{\mu\nu}$ is antisymmetric. Then

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + O(h^2, \omega^2, h\omega). \tag{4.16}$$

The antisymmetric piece $\omega$ is a local Lorentz Goldstone. If Chapter 10 operates as advertised, $\omega$ is eaten and does not propagate. The symmetric piece $h_{\mu\nu}$ is the candidate graviton, still un-decomposed and possibly massive. Chapter 5 splits it; Chapter 7 asks whether its mass vanishes.

This expansion assumes that a Minkowski condensate exists: a vacuum in which $e_\mu^A = \delta_\mu^A$ globally, or in a region large compared with the scale of later experiments. Existence of that vacuum is part of Assumption 4.4, not a new principle.

## 4.6 Signature and the role of $\eta_{AB}$

Equation (4.9) borrows its signature from $\eta_{AB}$. Nothing in the potential (3.7) selected $(-,+,+,+)$ rather than $(+,+,+,+)$. A Euclidean condensate is a priori as natural as a Lorentzian one.

**Assumption 4.6** (Lorentzian condensate)**.** The locked vacuum is Lorentzian: $\eta_{AB} = \mathrm{diag}(-1,+1,+1,+1)$. Dynamical emergence of this signature from a Euclidean matrix model is not proved in the source notes and is not pretended here. Chapter 10 shows how local $\mathrm{SO}(3,1)$ can arise *given* this $\eta_{AB}$.

A reader looking for a derivation of Lorentzian signature from nothing will not find it in this book. What the book does derive, contingent on Assumption 4.6, is the dynamics of the resulting Lorentzian metric.

## 4.7 Worked example: a weak, static frame perturbation

Take Repair I with

$$\Lambda^A{}_I(x) = \delta^A{}_I + \frac{1}{2}\delta^A{}_\alpha\delta^\beta{}_I\partial_\beta\zeta^\alpha(x), \tag{4.17}$$

and $\zeta^\alpha$ small. Then

$$e^A_\mu = \delta^A_\mu + \frac{1}{2}\partial_\mu\partial_\beta\zeta^\alpha\,\delta^A{}_\alpha\delta^\beta{}_., \tag{4.18}$$

which is a pure-gauge linearized frame if $\zeta$ is a diffeomorphism parameter. The associated metric perturbation is

$$h_{\mu\nu} = \partial_\mu\zeta_\nu + \partial_\nu\zeta_\mu \tag{4.19}$$

to first order. This configuration is geometry in name only: it is a coordinate artifact. A physical graviton must be a disturbance that cannot be written as (4.19). Isolating that disturbance is the point of the irreducible decomposition in Chapter 5.

The example is still useful. It shows that Definition 4.1 does produce a metric perturbation of the correct tensor type, and that diffeomorphism redundancy is already visible at the level of $\Lambda(x)$.

## 4.8 What this chapter has not done

The following statements are *not* theorems of Chapter 4.

- $g_{\mu\nu}$ obeys Einstein's equation.
- $h_{\mu\nu}$ contains two massless TT polarizations.
- $G_{\text{eff}}$ has the value (3.16).
- Torsion vanishes.
- The signature is dynamical.
- $\mathcal{G}_{\mu\nu}$ has been computed.

What has been done is narrower and indispensable: a non-degenerate composite metric has been defined from the condensate, the constant-vacuum obstruction has been removed by the polar field $\Lambda(x)$, and the linearized Minkowski background used in every later proof has been fixed.

## 4.9 Exercises

1. **Zero frame.** Prove that a strictly constant $\langle\Phi\rangle$ yields $\langle\partial_\mu\Phi\rangle = 0$. This is the obstruction of Section 4.1 as a two-line exercise.

2. **Polar derivative.** Derive (4.5) from (4.4). Under what condition on $\Lambda$ is $e_\mu^A$ proportional to a Maurer–Cartan form $\Lambda^{-1}\partial_\mu\Lambda$?

3. **Non-degeneracy.** Let $e_\mu^A = \mathrm{diag}(1, 1, 1, \varepsilon)$. Compute $g_{\mu\nu}$ and $\det g$. What fails as $\varepsilon \to 0$? Interpret the limit as a condensate defect.

4. **Symmetric part of the $\Theta$ correction.** From (4.14), extract the symmetric $O(\Theta)$ correction to $g_{\mu\nu}$ and the antisymmetric remainder. Which object can enter geodesic motion at this order?

5. **Linearized metric.** Starting from (4.15), derive (4.16) and show that $\omega_{AB}$ drops out of $g_{\mu\nu}$ at first order. Where does $\omega$ re-enter at second order?

6. **Pure gauge.** Show that (4.19) is the linearized diffeomorphism action on $\eta_{\mu\nu}$. Write one explicit $h_{\mu\nu}$ that cannot be written in this form. (Any TT wave will do.)

7. **Auxiliary versus emergent.** In Presentation B the microscopic action uses $g^{(0)}$. In this chapter the geometry is $g$. Write one sentence stating when it is legitimate to replace $g^{(0)}$ by $g$ inside a heat kernel, and what error term that replacement drops. The full answer is Chapter 8; the sentence is due now.

## 4.10 Bridge to Chapter 5

The background is a condensed frame, taken Minkowski for the linearized theory,

$$\bar{e}_\mu^A = \delta_\mu^A, \qquad \bar{g}_{\mu\nu} = \eta_{\mu\nu}. \tag{4.20}$$

The next object is the fluctuation $\delta\Phi^I{}_A$ about the polar vacuum (4.4). Chapter 5 decomposes that fluctuation under local Lorentz transformations into a conformal piece, a vector piece, and a symmetric trace-free tensor $T_{\mu\nu}$. Only after that decomposition is it meaningful to ask which piece might be a graviton.

# Chapter 5

# Irreducible Lorentz Decomposition of Fluctuations

**Learning outcomes.** After this chapter the reader should be able to:

1. convert a fluctuation $\delta\Phi^I{}_A$ to a spacetime tensor using the background frame;
2. split that tensor into trace, antisymmetric, and symmetric trace-free pieces, and count components;
3. match the split to the schematic formula used in the source notes, $\delta\Phi^I{}_A = \frac{1}{4}\eta_{\mu\nu}\delta g^{\mu\nu} I + A_\mu \gamma^\mu + T_{\mu\nu}$;
4. distinguish algebraic irreducibility from the later dynamical conditions $\nabla^\mu T_{\mu\nu} = 0$ and $m_T^2 = 0$;
5. enter Chapter 6 knowing which piece is the candidate graviton.

Chapter 4 produced a condensed background frame $\bar{e}_\mu^A = \delta_\mu^A$ and a composite Minkowski metric. This chapter is linear algebra on the tangent space. No entanglement condition is used. No mass gap is shown to vanish. The only task is to name the pieces of $\delta\Phi$.

## 5.1 Fluctuations about the polar vacuum

Start from the polar field of Definition 4.1,

$$\Phi^{IA}(x) = \big(v + \sigma(x)\big)\, \Lambda^A{}_I(x), \tag{5.1}$$

and expand about the Minkowski condensate

$$\langle\sigma\rangle = 0, \qquad \langle\Lambda^A{}_I\rangle = \delta^A{}_I, \qquad \bar{e}_\mu^A = \delta_\mu^A. \tag{5.2}$$

Write

$$\Phi^{IA} = v\,\delta^{IA} + \delta\Phi^{IA}, \tag{5.3}$$

with $\delta\Phi$ first order in $\sigma$ and in the deviation of $\Lambda$ from the identity. For the rest of the linearized theory it is convenient to lower the internal index with the background identification $I \leftrightarrow A$ and convert the remaining tangent index to a coordinate index with $\bar{e}$:

$$\delta\Phi_{\mu\nu} \equiv \bar{e}^{A}{}_{\mu}\,\delta\Phi_{IA}\,\bar{e}^{I}{}_{\nu}. \tag{5.4}$$

When $N = 4$ and the vacuum of Assumption 3.4 locks every internal direction to a tangent direction, $\delta\Phi_{\mu\nu}$ is a genuine $4 \times 4$ spacetime tensor. If $N > 4$, (5.4) captures only the locked $4 \times 4$ block. The orthogonal internal block is a collection of extra Higgs fields. They are assumed heavy and integrated out; they are not candidate gravitons.

**Assumption 5.1** (Locked block dominates the infrared)**.** Fluctuations in internal directions orthogonal to the four locked axes have masses of order $v$ or $\Lambda_{\mathrm{UV}}$ and may be discarded from the infrared Lorentz decomposition.

## 5.2 Algebraic split under the Lorentz group

At a fixed event, $\delta\Phi_{\mu\nu}$ is an element of $\mathbb{R}^4 \otimes \mathbb{R}^4$. The Lorentz group $\mathrm{SO}(3,1)$ acts on each factor. The tensor product decomposes into irreducible pieces

$$\mathbf{4} \otimes \mathbf{4} = \mathbf{1} \oplus \mathbf{6} \oplus \mathbf{9}, \tag{5.5}$$

namely a scalar, an antisymmetric tensor, and a symmetric traceless tensor. Explicitly,

$$\delta\Phi_{\mu\nu} = \frac{1}{4}\eta_{\mu\nu}\psi + F_{\mu\nu} + T_{\mu\nu}, \tag{5.6}$$

where

$$\begin{aligned} \psi &= \eta^{\alpha\beta}\delta\Phi_{\alpha\beta}, \\ F_{\mu\nu} &= \frac{1}{2}(\delta\Phi_{\mu\nu} - \delta\Phi_{\nu\mu}), \\ T_{\mu\nu} &= \frac{1}{2}(\delta\Phi_{\mu\nu} + \delta\Phi_{\nu\mu}) - \frac{1}{4}\eta_{\mu\nu}\psi. \end{aligned} \tag{5.7}$$

By construction

$$T_{\mu\nu} = T_{\nu\mu}, \qquad \eta^{\mu\nu}T_{\mu\nu} = 0, \qquad F_{\mu\nu} = -F_{\nu\mu}. \tag{5.8}$$

Component count in four dimensions:

| Piece | Symbol | Components |
|---|---|---|
| trace scalar | $\psi$ | 1 |
| antisymmetric tensor | $F_{\mu\nu}$ | 6 |
| symmetric traceless tensor | $T_{\mu\nu}$ | 9 |
| total | $\delta\Phi_{\mu\nu}$ | 16 |

This is algebra. It uses no field equation and no gauge condition.

## 5.3 Translation into the notation of the source notes

The microscopic notes write

$$\delta\Phi^{I}{}_{A} = \frac{1}{4}\eta_{\mu\nu}\delta g^{\mu\nu}\, I + A_\mu\gamma^\mu + T_{\mu\nu}. \tag{5.9}$$

Equation (5.9) is (5.6) after three identifications.

**Trace piece.** On the linearized background of Section 4.5,

$$\psi = \eta^{\mu\nu}h_{\mu\nu} + O(\omega) = \delta g^{\mu}{}_{\mu}. \tag{5.10}$$

The first term of (5.9) is therefore the conformal (trace) mode of the composite metric. The identity matrix $I$ in the notes is the locked vacuum $\delta^{I}{}_{A}$, converted to $\eta_{\mu\nu}$ by (5.4).

**Antisymmetric / vector piece.** An antisymmetric tensor $F_{\mu\nu}$ in four dimensions is equivalent, via the Clifford algebra generated by matrices $\gamma^\mu$ obeying

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \tag{5.11}$$

to a vector-plus-axial combination. The notes keep only the vector current $A_\mu\gamma^\mu$, which is the part that couples to the internal gauge field of (3.5). The remaining three (axial / magnetic) components of $F_{\mu\nu}$ are local Lorentz parameters. They belong with the Goldstone $\omega_{AB}$ of (4.15), not with a new propagating vector.

This book uses the following official translation:

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu + \omega_{\mu\nu}, \tag{5.12}$$

with $A_\mu$ the internal gauge current and $\omega_{\mu\nu}$ the local Lorentz Goldstone. Chapter 10 eats $\omega$. The Maxwell-like field strength of $A_\mu$ is a standard gauge excitation and is not a graviton.

**Symmetric traceless piece.** The tensor $T_{\mu\nu}$ in (5.9) is identical to the $T_{\mu\nu}$ of (5.7). It is the only algebraic piece with the index symmetries of a graviton.

## 5.4 What $T_{\mu\nu}$ is not yet

The source notes also write

$$T^{\mu}{}_{\mu} = 0, \qquad \nabla^{\mu} T_{\mu\nu} = 0. \tag{5.13}$$

The first of these is (5.8) and has already been imposed. The second is not algebraic. It is four differential conditions.

Those four conditions are the linearized statement that $T_{\mu\nu}$ is taken in de Donder / Lorentz-like gauge, or, on shell, that a massive spin-2 field has five rather than nine components. They are not free to impose until one knows the kinetic operator of $T_{\mu\nu}$.

| Statement | Type | Status in this chapter |
|---|---|---|
| $T_{\mu\nu} = T_{\nu\mu}$ | algebraic | imposed |
| $T^{\mu}{}_{\mu} = 0$ | algebraic | imposed |
| $\nabla^{\mu} T_{\mu\nu} = 0$ | gauge or on-shell | *not* imposed |
| $m_T^2 = 0$ | dynamical | Chapter 7 |
| $\Box T_{\mu\nu} = 0$ with two TT modes | dynamical | Chapter 7 |

A massive symmetric traceless tensor in four dimensions has five on-shell degrees of freedom (the Fierz–Pauli count). A massless one with the gauge invariance

$$\delta_\xi T_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu - \tfrac{1}{2}\eta_{\mu\nu} \nabla \cdot \xi \tag{5.14}$$

has two. The difference is the entire physical question of Chapters 6 and 7. It cannot be settled by rearranging indices.

## 5.5 Kinetic terms before any entanglement condition

Inserting (5.6) into the quadratic expansion of $S_{\text{micro}}$ produces, schematically,

$$S^{(2)} = S^{(2)}_{\psi}[\psi] + S^{(2)}_{A}[A] + S^{(2)}_{\omega}[\omega] + S^{(2)}_{T}[T] + S^{(2)}_{\text{mix}}. \tag{5.15}$$

On a Minkowski condensate the mixing terms that survive after gauge fixing are of the form $\psi\, \partial\partial T$ and $A\, \partial T$. They are the usual linearized scalar–tensor and vector–tensor mixings. Two structural facts will be used later.

1. The tensor block has the generic form

$$S_T^{(2)} = \int \mathrm{d}^4x \Big( \tfrac{1}{2} \nabla_\rho T_{\mu\nu} \nabla^\rho T^{\mu\nu} - \tfrac{1}{2} m_T^2 T_{\mu\nu} T^{\mu\nu} + \cdots \Big), \tag{5.16}$$

   where the ellipsis contains the additional derivative structures required by Fierz–Pauli if a mass term is present (the $T^\mu{}_\mu$ piece has already been projected out). The mass $m_T^2$ is a function of $v$, $\lambda$, and the condensate alignment. Nothing in Chapter 3 forces it to vanish.

2. The conformal block $S_\psi^{(2)}$ is the scalar-tensor sector. If $\psi$ remains light, the infrared theory is not pure general relativity even if $m_T = 0$. Killing or gapping $\psi$ is a separate requirement, recorded as such in Chapter 7 and again in Chapter 17.

Equation (5.16) is the action whose mass term Chapter 7 must kill. Writing it here prevents that later argument from looking as if $T_{\mu\nu}$ arrived already massless.

## 5.6 Diffeomorphism action on the pieces

An infinitesimal diffeomorphism generated by $\xi^\mu$ acts on the composite metric as

$$\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu. \tag{5.17}$$

On the linearized Minkowski background this is $\delta_\xi h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Passing through (5.6),

$$\begin{aligned} \delta_\xi \psi &= 2\partial \cdot \xi, \\ \delta_\xi F_{\mu\nu} &= 0 \quad \text{(to first order, after the symmetric projection)}, \\ \delta_\xi T_{\mu\nu} &= \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \tfrac{1}{2} \eta_{\mu\nu} \partial \cdot \xi. \end{aligned} \tag{5.18}$$

The last line is the gauge transformation that a massless spin-2 field must possess. If $m_T^2 \neq 0$, (5.18) is not a symmetry of (5.16), and the Ward identity that Chapter 6 will write on a causal diamond is obstructed. That obstruction is the lever used to force $m_T \to 0$.

## 5.7 Worked count: from sixteen components to two polarizations

The following table is the map the rest of the book will follow. Only the first two columns are established in this chapter.

| Stage | What is removed | Remaining | Where |
|---|---|---|---|
| algebraic split | — | 16 | this chapter |
| eat $\omega_{AB}$ | 6 Lorentz Goldstones | 10 | Ch. 10 |
| gap or constrain $\psi$ | 1 conformal scalar | 9 | Ch. 7, 17 |
| gap $A_\mu$ or eat it | 4 vector | 5 | Ch. 10 |
| impose $\nabla^\mu T_{\mu\nu} = 0$ | 4 constraints | 5 | on-shell FP |
| send $m_T \to 0$ and gauge | 3 more | 2 TT | Ch. 6–7 |

The last line is the target advertised in Chapter 1. It is not a result of the decomposition.

## 5.8 Exercises

1. **Projectors.** Write the three projectors on $\mathbb{R}^4 \otimes \mathbb{R}^4$ that extract $\psi$, $F_{\mu\nu}$, and $T_{\mu\nu}$ from $\delta\Phi_{\mu\nu}$. Verify they are complementary and idempotent.

2. **Component count.** Confirm the $1 + 6 + 9 = 16$ split by writing every independent component of a general real $4 \times 4$ matrix in an explicit basis.

3. **Fierz–Pauli count.** A massive spin-2 field in $d = 4$ has how many on-shell degrees of freedom? Derive the number from SO(3) little-group representations. Where does the number five sit in the table of Section 5.7?

4. **Massless count.** Repeat for a massless spin-2 field with gauge invariance (5.14). Recover two helicities.

5. **Trace versus conformal mode.** From (4.16) and (5.10), show that $\psi$ is the linearized conformal factor of $g_{\mu\nu}$ plus a term involving $\omega$. In what gauge does the extra term drop?

6. **Diffeomorphism on $T_{\mu\nu}$.** Derive the third line of (5.18) from (5.17) and (5.7).

7. **$N > 4$.** Suppose $N = 8$. How many extra internal scalars appear outside the locked $4 \times 4$ block? Why are they not gravitons?

8. **Danger of imposing $\nabla^\mu T_{\mu\nu} = 0$ early.** Take (5.16) with $m_T \neq 0$ and impose $\partial^\mu T_{\mu\nu} = 0$ off shell as a constraint. Show that the constraint is consistent with the massive equation of motion, but that the massless limit $m_T \to 0$ must be taken *before* residual gauge transformations are used to reach two polarizations. This is why Chapter 7 cannot be skipped.

## 5.9 Bridge to Chapter 6

The candidate graviton has a name, $T_{\mu\nu}$, nine algebraic components, and a generic mass $m_T$. The conformal and vector pieces have also been named, and will be gapped or eaten rather than identified with gravitational waves.

Nothing in the microscopic potential (3.7) sets $m_T = 0$. The mechanism proposed by NAMCG is not a further tuning of $\lambda$. It is the entanglement-stationarity condition $\delta S_D = 0$ on causal diamonds, imposed on the same fluctuations just decomposed. Chapter 6 defines that condition from the modular Hamiltonian of the matrix condensate and shows how it acts as a Ward identity. Chapter 7 applies the identity to $T_{\mu\nu}$ and obtains $m_T \to 0$.

# Chapter 6

# Entanglement Equilibrium and the Modular Hamiltonian

**Learning outcomes.** After this chapter the reader should be able to:

1. write the reduced state $\rho_D$, the modular Hamiltonian $H_D$, and the von Neumann entropy $S_D$ of a causal diamond in the matrix condensate;
2. state Jacobson's entanglement-equilibrium condition $\delta S_D = 0$ and distinguish it from the Einstein equation it will later imply;
3. obtain the first variation $\delta S_D = \delta\langle H_D\rangle$ at fixed diamond volume, including the meaning of that constraint;
4. define the mutual-information potential $U_{\text{ent}}$ used for two condensed cores;
5. enter Chapter 7 with a Ward identity ready to be applied to $T_{\mu\nu}$.

Chapter 5 named the candidate graviton $T_{\mu\nu}$ and left it with a generic mass $m_T$. The mass is not removed by a further term in $V(\Phi^\dagger\Phi)$. It is removed, if at all, by a stationarity condition on entanglement. This chapter writes that condition in a form that can be varied. The variation itself is Chapter 7.

## 6.1 The logical role of this chapter

There are two different claims that are easy to conflate.

1. **Jacobson's theorem** (in the form used here). For a quantum field on a prescribed geometry, the first-order stationarity of vacuum entanglement in small geodesic balls, at fixed volume, is equivalent to the semiclassical Einstein equation. This direction assumes a metric and a quantum field, and obtains $G_{\mu\nu} \propto \langle T_{\mu\nu}\rangle$.

2. **The NAMCG claim.** The microscopic condensate state is itself an entanglement-stationary state on the diamonds of its *emergent* metric. Stationarity is therefore a property of $\rho$, not an extra thermodynamic axiom pasted onto a metric theory. Applied to the tensor block of Chapter 5, it forbids $m_T^2 \neq 0$.

Claim 1 is a theorem in the literature, with stated hypotheses (conformal matter at first order; a conjecture for non-conformal matter). Claim 2 is the load-bearing physical hypothesis of NAMCG. This chapter formulates Claim 2 so that it can be used. It does not pretend that Claim 2 has been derived from the path integral of Chapter 3 with the same status as Claim 1.

**Assumption 6.1** (Microscopic entanglement equilibrium)**.** Let $\rho$ be the vacuum state of the matrix condensate in the condensed phase of Chapters 3–4. For every sufficiently small causal diamond $D$ constructed from the emergent metric $g_{\mu\nu}$,

$$\delta S_D = 0 \tag{6.1}$$

at fixed spatial volume of the maximal slice of $D$, to first order in variations of the condensate that preserve the local vacuum character of $\rho$.

The remainder of the chapter unpacks every symbol in that statement.

## 6.2 Diamonds in the emergent geometry

The causal diamond $D(p,q) = J^+(p) \cap J^-(q)$ was defined in (2.26). From here on the causal structure is that of the *emergent* metric (4.9), not of the auxiliary scaffold $g^{(0)}$. For a linearized analysis about Minkowski spacetime the distinction is invisible at zeroth order and appears at first order through $h_{\mu\nu}$.

A small diamond of radius $\ell$ is one for which $\ell^2 \|R_{\mu\nu\rho\sigma}\| \ll 1$. In Riemann normal coordinates (2.27) its maximal-slice three-volume admits the expansion

$$V(D) = \frac{4}{3}\pi\ell^3 - \frac{4\pi\ell^5}{15}\left(\tfrac{1}{3}R + \text{anisotropic Ricci terms}\right) + O(\ell^7). \tag{6.2}$$

"Fixed volume" in Assumption 6.1 means that the first variation is taken subject to $\delta V(D) = 0$. The geometric meaning of that constraint is that a conformal breathing of the diamond is not counted as an independent entanglement variation; it has been used to fix the comparison background. This is the same gauge-like restriction Jacobson imposes, and it is why the conformal mode $\psi$ of Chapter 5 and the tensor mode $T_{\mu\nu}$ enter the first law differently.

Dyadic labels $D_{j,n}$ are a nested bookkeeping device. Nothing physical depends on bisection. Every identity written for a generic small $D$ applies to each dyadic diamond.

## 6.3 Reduced states and the modular Hamiltonian

Let $\mathcal{A}_D$ be the algebra of condensate observables supported in $D$. In Presentation B these are functions of $\Phi(x)$ and $A_\mu(x)$ for $x \in D$. In Presentation A they are the operators generated by the spectral projections of $\hat{X}^\mu$ whose eigenvalues lie in $D$, after the dictionary of Assumption 3.5.

**Definition 6.2** (Diamond state and modular Hamiltonian)**.** Given a global state $\rho$ of the condensate,

$$\rho_D = \mathrm{Tr}_{\mathcal{A}'_D} \rho, \tag{6.3}$$

where $\mathcal{A}'_D$ is the commutant (the complement algebra). The modular Hamiltonian and von Neumann entropy are

$$H_D = -\ln \rho_D, \qquad S_D = -\mathrm{Tr}(\rho_D \ln \rho_D) = \langle H_D \rangle_{\rho_D}. \tag{6.4}$$

For a free field in Rindler space, the Bisognano–Wichmann theorem identifies $H_D$ with $2\pi/\kappa$ times the boost generator that fixes the edge $\mathcal{B} = \partial D \cap \Sigma$. For a small diamond in a slowly varying background the same identification holds to leading order in $\ell$:

$$H_D = \frac{2\pi}{\kappa} \int_{\Sigma \cap D} \mathrm{d}\Sigma^\mu \, \xi^\nu T^{\mathrm{op}}_{\mu\nu} + H_{\mathrm{area}} + O(\ell^3). \tag{6.5}$$

Here $\xi^\mu$ is the approximate conformal Killing vector of $D$, $\kappa$ is its surface gravity at $\mathcal{B}$, $T^{\mathrm{op}}_{\mu\nu}$ is the microscopic stress tensor of the condensate fluctuations, and $H_{\mathrm{area}}$ is the area term that will later match $A/(4G_{\mathrm{eff}})$.

Equation (6.5) is the working formula of this book. It is exact only for free fields in Rindler space. For interacting matrix fluctuations it is the first term in an operator product expansion on the diamond. Higher terms are absorbed into the $O(\ell^3)$ remainder and into the definition of $H_{\mathrm{area}}$.

## 6.4 Replica formula

When a direct construction of $\rho_D$ is unavailable, the replica trick supplies $S_D$. For integer $n \geq 1$,

$$S_D = \lim_{n \to 1} (1 - n\partial_n) \ln \mathrm{Tr}\,(\rho_D)^n, \tag{6.6}$$

and $\mathrm{Tr}\,(\rho_D)^n$ is computed as a path integral of the microscopic action (3.6) on an $n$-sheeted cover of $M$ branched over the edge $\mathcal{B}$. The first variation $\delta S_D$ then becomes a variation of that branched partition function with respect to the background frame.

Two remarks are enough for Chapter 7.

- The replica construction is well defined in Presentation B, where a path integral exists. In Presentation A it requires the dictionary of Assumption 3.5.

- Differentiating (6.6) at $n = 1$ produces an insertion of the modular Hamiltonian. The first variation $\delta S_D = \delta\langle H_D\rangle$ used below is therefore not a second formula; it is the replica formula at first order.

## 6.5 The first law of the diamond

Vary Definition 6.2 at fixed diamond, then allow the diamond itself to vary with the metric. The result, to first order and at fixed volume, is the entanglement first law

$$\delta S_D = \delta\langle H_D\rangle = \frac{2\pi}{\kappa}\int_{\Sigma\cap D} \mathrm{d}\Sigma^\mu\, \xi^\nu \delta\langle T^{\mathrm{op}}_{\mu\nu}\rangle + \frac{\delta A}{4G_{\mathrm{eff}}} + O(\ell^3). \tag{6.7}$$

The area variation $\delta A$ is computed from the emergent metric. The identification of its prefactor with $1/(4G_{\mathrm{eff}})$ is not justified in this chapter; it is the Bekenstein–Hawking relation that Chapter 8 will match to the heat kernel. Until then, $1/(4G_{\mathrm{eff}})$ is a name for the coefficient of $\delta A$.

Assumption 6.1 now reads

$$\frac{2\pi}{\kappa}\int_{\Sigma\cap D} \mathrm{d}\Sigma^\mu\, \xi^\nu \delta\langle T^{\mathrm{op}}_{\mu\nu}\rangle + \frac{\delta A}{4G_{\mathrm{eff}}} = 0, \tag{6.8}$$

up to $O(\ell^3)$, for every small diamond and every allowed variation of the condensate.

If $\delta\langle T^{\mathrm{op}}_{\mu\nu}\rangle$ were the stress tensor of ordinary matter on a prescribed geometry, and if $\delta A$ were computed from a prescribed Einstein metric, (6.8) would be Jacobson's derivation of $G_{\mu\nu} = 8\pi G_{\mathrm{eff}}\,\langle T_{\mu\nu}\rangle$. In NAMCG both $\delta\langle T^{\mathrm{op}}\rangle$ and $\delta A$ are functionals of the *same* fluctuation $\delta\Phi$. That is the leverage used in Chapter 7.

## 6.6 Allowed variations

Not every deformation of $\Phi$ is an allowed variation in Assumption 6.1. The hypothesis is a statement about perturbations that remain locally vacuum-like: they do not inject an arbitrary excited-state entropy into $D$.

**Definition 6.3** (Allowed variation)**.** A first-order deformation $\delta\Phi$ is *allowed* if

1. it is smooth on a neighbourhood of $D$;

2. it preserves the polar vacuum to zeroth order outside a compact region, so that $\rho$ remains a condensed-phase vacuum at infinity;

3. its restriction to each algebraic block of Chapter 5 may be chosen independently, after gauge fixing of diffeomorphisms and local Lorentz transformations.

Condition 3 is the reason the tensor, vector, and conformal blocks can be tested separately. If a variation supported only on $T_{\mu\nu}$ violates (6.8) unless $m_T = 0$, that is a constraint on the tensor block alone.

## 6.7 Mutual information of two cores

The same entropy data define an interaction between two disjoint condensed regions $A$ and $B$ (two compact-object cores, two stellar interiors, two laboratory samples).

**Definition 6.4** (Entropic potential)**.**

$$U_{\rm ent}(A,B) = -\lambda I(A:B) = -\lambda\big[S(A) + S(B) - S(A\cup B)\big], \tag{6.9}$$

with $\lambda > 0$ a microscopic coupling of dimension energy.

Because $I(A:B) \geq 0$, one has $U_{\rm ent} \leq 0$. Increasing the mutual information lowers the potential: the effective force is attractive. For two small balls whose proper separation $d$ is large compared with their radii but small compared with the curvature radius,

$$I(A:B) \;\sim\; \frac{c\,\ell_A^2\,\ell_B^2}{d^2} \qquad \text{(conformal matter, leading order)}, \tag{6.10}$$

up to a theory-dependent constant $c$. Differentiating (6.9) with respect to a worldline, and converting the result into an acceleration, produces a correction that falls as a power of $d$. The Newtonian laboratory of Chapter 15 uses

$$\mathbf{a}_{\rm irrev} = -\eta\,\frac{1}{d^4}\,\mathbf{v} \tag{6.11}$$

as a reduced, velocity-dependent representative of that correction in a time-dependent two-body problem. Equation (6.11) is *not* derived in this chapter. It is flagged here so that Chapter 11 knows which entropy functional it must vary covariantly.

The conservation question is already visible. Mutual information is built from scalars $S(\,\cdot\,)$. A covariant variation of $U_{\rm ent}$ with respect to the metric defines a tensor $T^{\rm ent}_{\mu\nu}$ by the same rule as (2.13). Whether $\nabla^\mu T^{\rm ent}_{\mu\nu} = 0$ holds on shell is not automatic from the sign of $\lambda$. It is a calculation, reserved for Chapter 11.

## 6.8 What stationarity does *not* say

Assumption 6.1 does not assert any of the following.

- That $S_D$ is a maximum rather than a saddle. Jacobson's original language is maximization at fixed volume; the identities used below need only first-order stationarity.

- That the area term in (6.7) has already been matched to $G_{\text{eff}}$. That match is Chapter 8.
- That $m_T = 0$. Stationarity is a condition on variations. Whether the tensor kinetic operator is compatible with that condition is Chapter 7.
- That $\Lambda_{\text{eff}}$ is small. Vacuum entropy of the diamond includes a piece that will source a cosmological term. Chapter 12 treats it.
- That the replica path integral has been computed at two loops. The book uses the first law (6.7) as the working identity, not a finished replica evaluation of $S_D$ for the matrix potential (3.7).

## 6.9 The Ward identity that Chapter 7 will use

Collecting (6.8), Definition 6.3, and the split (5.6), the usable statement is:

**Proposition 6.5** (Entanglement Ward identity)**.** *Under Assumptions 6.1 and 3.5, every allowed variation supported on a single algebraic block $\psi$, $A_\mu$, or $T_{\mu\nu}$ satisfies*

$$\int_{\Sigma\cap D} \mathrm{d}\Sigma^\mu\, \xi^\nu \delta\langle T^{\text{op}}_{\mu\nu}\rangle + \frac{\kappa}{8\pi G_{\text{eff}}}\,\delta A = 0 \tag{6.12}$$

*to first order in $\ell$, at fixed $V(D)$. The operator $T^{\text{op}}_{\mu\nu}$ is the metric variation of the quadratic action of that block.*

The proof of Proposition 6.5 is the chain of this chapter: Definition 6.2 to the first law (6.7) to Assumption 6.1, restricted to one block by Definition 6.3. No property of $T_{\mu\nu}$ beyond its being one algebraic block has been used. That is intentional. Chapter 7 substitutes the quadratic action (5.16) into (6.12) and reads off $m_T = 0$.

## 6.10 Exercises

1. **Volume constraint.** From (6.2), show that a pure conformal variation $\delta g_{\mu\nu} = \chi\eta_{\mu\nu}$ changes $V(D)$ at order $\ell^3$. Why must this variation be excluded, or compensated, if stationarity is imposed at fixed volume?
2. **Rindler check.** In Rindler space with Unruh temperature $T = \kappa/2\pi$, verify that $\delta S = \delta E/T$ for a boost-energy perturbation is the restriction of (6.7) to a diamond that has become a Rindler wedge.
3. **Replica derivative.** Starting from (6.6), show that $\partial_n \ln \mathrm{Tr}\,\rho^n|_{n=1} = \langle \ln\rho\rangle$, and hence that $S = \langle H\rangle$ at $n = 1$.

4. **Allowed versus thermal variations.** Give one example of a $\delta\Phi$ that changes the particle content of $D$ (a one-particle excitation) and explain why Assumption 6.1 is silent about it.

5. **Mutual information positivity.** Prove $I(A:B) \geq 0$ from relative entropy. Deduce $U_{\text{ent}} \leq 0$ for $\lambda > 0$.

6. **Falloff.** For two spheres of radius $\ell$ in Minkowski space and a massless scalar, look up or derive the leading large-$d$ decay of $I(A:B)$. How many powers of $1/d$ must one differentiate to obtain an acceleration that falls as $1/d^4$?

7. **Two metrics.** Write one paragraph explaining why the diamond of this chapter is built from $g_{\mu\nu}$ rather than from $g^{(0)}_{\mu\nu}$. What would go wrong, conceptually, if stationarity were imposed on auxiliary-scaffold diamonds instead?

## 6.11 Bridge to Chapter 7

The candidate graviton $T_{\mu\nu}$ has a quadratic action (5.16) with an unknown mass $m_T$. The condensate vacuum is assumed to obey the entanglement Ward identity (6.12) on every small diamond of the emergent geometry. Chapter 7 substitutes a compactly supported tensor variation into that identity. The area term tracks the Einstein operator. The stress-tensor term tracks $m_T^2 T_{\mu\nu}$. Compatibility of the two forces $m_T = 0$, after which the surviving kinetic operator is the linearized Einstein operator and the infrared spectrum contains two massless TT polarizations—provided the conformal and vector blocks are separately gapped or eaten, as already flagged.

# Chapter 7

# Vanishing of the Tensor Mass Gap

**Learning outcomes.** After this chapter the reader should be able to:

1. write the Fierz–Pauli quadratic action for $T_{\mu\nu}$ and the linearized Einstein operator that remains when $m_T = 0$;
2. substitute a compactly supported tensor variation into the entanglement Ward identity and see where $m_T^2$ appears;
3. state Theorem 7.3 with its hypotheses, and say which hypotheses are still assumptions of the framework;
4. count the two remaining TT polarizations;
5. list what must still be done for the conformal and vector blocks before the infrared theory is pure GR.

The candidate graviton now has a name, a quadratic action, and a Ward identity. This chapter puts the last of those into the second and reads off $m_T = 0$.

## 7.1 Strategy

On a Minkowski condensate the tensor block is a linear field $T_{\mu\nu}$ with the algebraic symmetries (5.8). Its quadratic action has the generic structure (5.16). The entanglement Ward identity, Proposition 6.5, must hold for every allowed variation supported on that block.

The geometric side of the identity, $\delta A$, depends on $T_{\mu\nu}$ only through the linearized metric

$$h_{\mu\nu} = T_{\mu\nu} + \tfrac{1}{4}\eta_{\mu\nu}\psi, \tag{7.1}$$

and, in a gauge where the conformal piece is held fixed by the fixed-volume constraint of Assumption 6.1, through $T_{\mu\nu}$ alone. The modular side, $\delta\langle T^{\mathrm{op}}_{\mu\nu}\rangle$, is the metric stress of the quadratic tensor action. If that action contains a mass term, the modular side contains a piece proportional to $m_T^2 T_{\mu\nu}$. The geometric side does not. Compatibility for arbitrary compactly supported $T_{\mu\nu}$ forces $m_T^2 = 0$.

That is the argument. The rest of the chapter writes it so that each step can be rejected independently.

## 7.2 Fierz–Pauli action for the tensor block

The unique two-derivative quadratic action for a massive symmetric tensor that is ghost-free about Minkowski spacetime is the Fierz–Pauli action. After the algebraic trace has been projected out, so that $T^{\mu}{}_{\mu} = 0$ identically, it reduces to

$$S_{\mathrm{FP}}[T] = \int \mathrm{d}^4x \Big( \tfrac{1}{2}\partial_\rho T_{\mu\nu}\partial^\rho T^{\mu\nu} - \partial^\rho T_{\rho\mu}\partial_\sigma T^{\sigma\mu} - \tfrac{1}{2} m_T^2 T_{\mu\nu}T^{\mu\nu} \Big). \tag{7.2}$$

The derivative structure is required. A generic mass term $m^2(T_{\mu\nu}T^{\mu\nu} - \alpha T^2)$ with $\alpha \neq 1$ (before the trace projection) propagates a ghost. With the trace already zero the remaining danger is a wrong-sign kinetic term, which (7.2) does not have.

The Euler–Lagrange equation of (7.2) is

$$\mathcal{E}_{\mu\nu}[T] - m_T^2 T_{\mu\nu} = 0, \tag{7.3}$$

where

$$\mathcal{E}_{\mu\nu}[T] \equiv \Box T_{\mu\nu} - \partial_\mu\partial^\rho T_{\rho\nu} - \partial_\nu\partial^\rho T_{\rho\mu} + \eta_{\mu\nu}\partial^\rho\partial^\sigma T_{\rho\sigma} \tag{7.4}$$

is the linearized Einstein operator on a traceless tensor about Minkowski spacetime. (The linearized Einstein tensor on a general $h_{\mu\nu}$ contains additional trace pieces; they are absent here because $T^{\mu}{}_{\mu} = 0$.)

When $m_T = 0$, (7.2) is invariant under

$$\delta_\xi T_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \tfrac{1}{2}\eta_{\mu\nu}\partial\cdot\xi \tag{7.5}$$

with $\partial\cdot\xi$ arranged so that the trace remains zero. When $m_T \neq 0$, that invariance is absent and the on-shell count is five degrees of freedom.

## 7.3 Modular stress of the tensor block

The operator $T^{\rm op}_{\mu\nu}$ in Proposition 6.5 is defined by the same rule as (2.13), applied to $S_{\rm FP}$ treated as a matter action on the background metric that defines the diamond. To first order about Minkowski spacetime,

$$\delta\langle T^{\rm op}_{\mu\nu}\rangle = \mathcal{E}_{\mu\nu}[T] - m_T^2 T_{\mu\nu} + \partial^\rho \Sigma_{\rho\mu\nu}, \tag{7.6}$$

where $\Sigma_{\rho\mu\nu}$ is a total-derivative superpotential coming from the fact that the canonical and metric stress tensors differ by an improvement. On a diamond the superpotential contributes only at the edge $\mathcal{B}$. That edge contribution is absorbed into the area term of the first law, which is already present in (6.12). What remains in the bulk integral is the improved stress

$$\delta\langle T^{\rm op}_{\mu\nu}\rangle_{\rm bulk} = \mathcal{E}_{\mu\nu}[T] - m_T^2 T_{\mu\nu}. \tag{7.7}$$

## 7.4 Area variation induced by $T_{\mu\nu}$

The edge $\mathcal{B}$ of a small diamond of radius $\ell$ is a 2-sphere of area $4\pi\ell^2$ in the Minkowski background. A metric perturbation $h_{\mu\nu} = T_{\mu\nu}$ changes that area by

$$\delta A = \int_{\mathcal{B}} {\rm d}A\, \tfrac{1}{2}\sigma^{\mu\nu} T_{\mu\nu} + O(\ell^3 \|T\|, \ell \|T\|^2), \tag{7.8}$$

where $\sigma^{\mu\nu}$ is the induced metric on $\mathcal{B}$. Jacobson's geometric identity converts the same variation, after integration against the conformal Killing vector and use of the fixed-volume constraint, into

$$\frac{\kappa}{8\pi G_{\rm eff}}\,\delta A = -\int_{\Sigma\cap D} {\rm d}\Sigma^\mu\, \xi^\nu\, \mathcal{E}_{\mu\nu}[T] + O(\ell^3). \tag{7.9}$$

The right-hand side contains the Einstein operator and does *not* contain $m_T$. The area of an emergent 2-surface knows about the metric perturbation. It does not know about a potential term in the fluctuation action.

Equation (7.9) is imported from the geometry of small diamonds, not re-derived here. Its hypotheses are: Riemann normal coordinates to the order kept in (2.27), fixed volume, and the identification of the area prefactor with $G_{\rm eff}$. The last of these is still a name; Chapter 8 computes the name.

## 7.5 Substitution into the Ward identity

Insert (7.7) and (7.9) into (6.12). The Einstein-operator terms cancel, and what remains is

$$\int_{\Sigma\cap D} \mathrm{d}\Sigma^{\mu}\, \xi^{\nu}(-m_T^2 T_{\mu\nu}) + O(\ell^3) = 0. \tag{7.10}$$

The conformal Killing vector $\xi^{\nu}$ of a small diamond is timelike in the interior and vanishes on the edge. Its leading expression in Riemann normal coordinates is

$$\xi^{\mu} = \frac{\kappa}{2\ell}((\ell^2 - t^2 - r^2)\partial_t^{\mu} + 2t\, x^{\mu}) + O(R_{\alpha\beta\gamma\delta}\ell^3). \tag{7.11}$$

**Lemma 7.1** (Arbitrary tensor sources)**.** *Compactly supported, smooth, traceless, symmetric variations $T_{\mu\nu}$ are allowed in the sense of Definition 6.3. In particular they may be chosen independently in each small diamond, up to the fixed-volume constraint already used.*

*Proof.* The tensor block was split off algebraically in Chapter 5. Gauge fixing of diffeomorphisms and local Lorentz transformations removes $\psi$-mixing and $\omega$-mixing at this order but does not constrain the nine components of $T_{\mu\nu}$ beyond smoothness and compact support. Fixed volume constrains the trace sector, which has already been projected out. □

**Lemma 7.2** (Integral identity implies point identity)**.** *If* (7.10) *holds for every small diamond and every allowed $T_{\mu\nu}$, then*

$$m_T^2 T_{\mu\nu}(x) = 0 \tag{7.12}$$

*at every event $x$ in the condensed phase.*

*Proof.* Suppose $m_T^2 T_{\mu\nu}$ is not identically zero. Then there is an event $x_0$ and a pair of indices at which the tensor is nonzero. Choose a diamond centered at $x_0$ with $\ell$ small enough that $T_{\mu\nu}$ does not change sign in the relevant component on $\Sigma\cap D$, and choose the time orientation so that $\xi^{\nu}$ has a nonvanishing contraction with that component. The integrand of (7.10) is then of definite sign on a set of positive measure, and the $O(\ell^3)$ remainder cannot cancel it as $\ell \to 0$. □

## 7.6 The vanishing theorem

**Theorem 7.3** (Vanishing of the tensor mass gap)**.** *Assume:*

1. *the microscopic setup of Chapter 3, including Assumptions 3.1–3.5;*
2. *the condensed frame of Chapter 4, including Assumptions 4.4 and 4.6;*

3. *the algebraic split of Chapter 5 and Assumption 5.1;*

4. *entanglement equilibrium, Assumption 6.1, in the form of Proposition 6.5;*

5. *the Fierz–Pauli quadratic action* (7.2) *as the two-derivative tensor block;*

6. *the geometric identity* (7.9) *for small diamonds.*

*Then* $m_T^2 = 0$ *throughout the condensed phase. The quadratic tensor action reduces to the linearized Einstein–Hilbert action on traceless perturbations,*

$$S_{\rm EH}^{(2)}[T] = \int \mathrm{d}^4x \, \tfrac{1}{2} T^{\mu\nu} \mathcal{E}_{\mu\nu}[T], \tag{7.13}$$

*and is invariant under* (7.5).

*Proof.* Lemma 7.2 gives $m_T^2 T_{\mu\nu} = 0$ at every event. The tensor block is not the zero field: linearized gravitational waves, if they exist, are particular solutions for $T_{\mu\nu}$. Hence $m_T^2 = 0$. Substituting into (7.2) produces (7.13) after an integration by parts, boundary terms being discarded for compactly supported variations. Gauge invariance (7.5) is the standard invariance of the linearized Einstein operator on a traceless tensor. □

*Remark* 7.4 (What the theorem does not hide). Hypothesis 4 is the physical input of NAMCG. Hypothesis 5 excludes non-Fierz–Pauli two-derivative structures and higher-derivative quadratic terms; the latter return in Chapter 9. Hypothesis 6 borrows a geometric identity used in Jacobson's argument. The theorem is therefore a compatibility statement: a Fierz–Pauli tensor block cannot sit in an entanglement-stationary condensate unless it is massless. It is not a derivation of Assumption 6.1 from the path integral.

## 7.7 Two transverse-traceless polarizations

With $m_T = 0$, the equation $\mathcal{E}_{\mu\nu}[T] = 0$ plus the gauge invariance (7.5) is linearized vacuum GR in the traceless sector. Choose the de Donder gauge

$$\partial^\mu T_{\mu\nu} = 0. \tag{7.14}$$

Residual gauge transformations with $\Box\xi^\mu = 0$ remain. They remove four more components. Together with the four conditions (7.14) and the four conditions implicit in taking the trace to vanish and staying in the symmetric sector after gauge fixing, one is left with two independent components. In a plane- wave basis they are the helicities $\pm 2$,

$$T_{\mu\nu}(x) = \mathrm{Re}\big(\epsilon_{\mu\nu}^{\pm 2}\, e^{ik\cdot x}\big), \qquad k^2 = 0, \quad k^\mu \epsilon_{\mu\nu} = 0, \quad \epsilon^\mu{}_\mu = 0. \tag{7.15}$$

This is the infrared graviton advertised in Chapter 1. It has been obtained as a fluctuation of the condensate, not as a fundamental metric field, contingent on Theorem 7.3.

## 7.8 The van Dam–Veltman–Zakharov warning

A massive spin-2 field with $m_T$ small but nonzero does not approach the massless theory smoothly in four dimensions. The discontinuity is visible in the linearized deflection of light and in the tensor-to-scalar ratio of a static source. Therefore it is not enough to argue that $m_T$ is "small compared with $\Lambda_{\mathrm{UV}}$." Entanglement equilibrium is being asked to set $m_T$ identically to zero in the infrared, not to suppress it by a loop factor.

If a later calculation produced $m_T \sim v\, e^{-\Lambda_{\mathrm{UV}}/v}$ or any other parametrically small but nonzero value, the theory would fail Solar-System tests unless an additional Vainshtein mechanism were demonstrated. No such mechanism is claimed here. Theorem 7.3 is an identity-zero statement.

## 7.9 Conformal and vector blocks

Theorem 7.3 is silent about $\psi$ and $A_\mu$.

**Vector block.** The connection $A_\mu$ is an ordinary gauge field once $\omega_{\mu\nu}$ has been eaten. Its mass is a Higgs mass of order $gv$, or zero if an unbroken subgroup remains. Neither case produces a second graviton. Chapter 10 completes the eating.

**Conformal block.** The scalar $\psi$ is more dangerous. If it remains massless it is a Brans–Dicke field, and the infrared theory is scalar–tensor gravity. Entanglement equilibrium at fixed volume was designed to remove a conformal breathing from the comparison diamond; it does not, by itself, gap $\psi$ as a propagating excitation.

**Assumption 7.5** (Gapped conformal mode)**.** The radial Higgs $\sigma$ of the polar decomposition, and the conformal metric mode $\psi$ mixed with it, acquire a mass $m_\psi \sim \sqrt{\lambda}\, v$ from the potential (3.7) and decouple from the infrared tensor sector.

Assumption 7.5 is the scalar-tensor ceiling of Chapter 1, reappeared as a named hypothesis. If it fails, NAMCG predicts a light scalar and is constrained by Cassini-type bounds on $\omega_{\mathrm{BD}}$. Chapter 17 records this as a falsifier, not as a detail.

## 7.10 Exercises

1. **Fierz–Pauli ghost.** Restore a trace piece $T^{\mu}{}_{\mu} = \psi$ and replace the mass term by $m^2(T_{\mu\nu}T^{\mu\nu} - \alpha\psi^2)$. Show that $\alpha \neq 1$ produces a sixth, ghostlike excitation. This is why (7.2) is not an arbitrary massive action.

2. **Gauge invariance at $m_T = 0$.** Verify that (7.5) leaves (7.13) invariant. Show the mass term in (7.2) is not invariant.

3. **Einstein operator on a TT wave.** Insert (7.15) into (7.4) and recover $k^2 = 0$.

4. **Sign in the Ward identity.** Track the sign from (7.7) through (7.9) to (7.10). A sign error here would force $m_T^2 = 0$ by the same logic but would mismatch the Einstein equation in Chapter 11. Fix the sign against Jacobson's first law for ordinary matter.

5. **Why compact support.** Suppose $T_{\mu\nu}$ is a plane wave, not compactly supported. Where does the proof of Lemma 7.2 fail, and how does one repair it by using wave packets?

6. **vDVZ.** Look up the linearized deflection angle for Fierz–Pauli gravity with mass $m_T$ and take $m_T \to 0$. Compare with the GR deflection. What numerical factor appears?

7. **Conformal leftover.** Linearize a Brans–Dicke action about Minkowski spacetime and identify the kinetic mixing of $\psi$ with $T_{\mu\nu}$. What bound on $m_\psi$ follows from requiring the mixing to be negligible inside the Solar System?

## 7.11 Bridge to Chapter 8

The infrared tensor sector is now linearized general relativity, contingent on Theorem 7.3 and Assumption 7.5. The prefactor of that action is still a symbol, $G_{\text{eff}}$. Chapter 8 computes it. The one-loop effective action of the condensate fluctuations on the emergent background is expanded with the Seeley–DeWitt coefficients of Section 2.6. The coefficient of $R$ is

$$\frac{1}{16\pi G_{\text{eff}}} = \frac{\operatorname{Tr}(\mathbb{I})\,\Lambda_{\text{UV}}^2}{16\pi^2} + \frac{v^2}{12}\ln\!\Big(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\Big). \tag{7.16}$$

The same expansion produces a quartic term $c_0\Lambda_{\text{UV}}^4$, which Chapter 8 will isolate and Chapter 12 will be required to tame.

# Chapter 8

# Heat Kernel and the Coefficient Map to $G_{\text{eff}}$

**Learning outcomes.** After this chapter the reader should be able to:

1. write the one-loop effective action of condensate fluctuations as a proper-time integral of the heat kernel;
2. identify which Seeley–DeWitt coefficient multiplies $R$ and which multiplies the cosmological term;
3. reproduce the working formula
$$\frac{1}{16\pi G_{\text{eff}}} = \frac{\mathrm{Tr}(\mathbb{I})\,\Lambda_{\text{UV}}^2}{16\pi^2} + \frac{v^2}{12}\ln\!\Big(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\Big);$$
4. state what is computed, what is scheme-dependent, and what is left for Chapter 12;
5. enter Chapter 9 knowing why a second-order heat kernel is not a license for fourth-order kinetic terms.

Theorem 7.3 gave a massless tensor kinetic operator. This chapter gives that operator its prefactor. The calculation is a Sakharov induction performed on the *emergent* background, not on a primitive metric.

## 8.1 One-loop effective action on the emergent geometry

After condensation, split
$$\Phi = \langle\Phi\rangle + \eta, \tag{8.1}$$

where $\eta$ contains every algebraic block of Chapter 5. Integrate $\eta$ at one loop on a slowly varying emergent background $g_{\mu\nu}$, using Presentation B and Assumption 3.5. The fluctuation operator is of Laplace type,

$$\mathcal{O}_\eta = -g^{\mu\nu} D_\mu D_\nu + E[\langle \Phi \rangle, g], \tag{8.2}$$

with a bundle endomorphism $E$ that includes the second derivative of $V$ and curvature couplings generated by the kinetic term.

The one-loop effective action is

$$W[g] = \frac{1}{2} \mathrm{Tr}\, \ln \mathcal{O}_\eta = -\frac{1}{2} \int_{s_{\min}}^{s_{\max}} \frac{\mathrm{d}s}{s}\, \mathrm{Tr}\, e^{-s\mathcal{O}_\eta}, \tag{8.3}$$

after Wick rotation to a Euclidean section. The limits are

$$s_{\min} = \Lambda_{\mathrm{UV}}^{-2}, \qquad s_{\max} = \mu^{-2}, \tag{8.4}$$

where $\Lambda_{\mathrm{UV}}$ is the kinematic cutoff (3.3) and $\mu$ is an infrared subtraction scale of order the curvature or the radial-Higgs mass, whichever is larger in the problem at hand.

**Assumption 8.1** (Laplace-type infrared operator)**.** After Theorem 7.3 and Assumption 7.5, the tensor block of $\mathcal{O}_\eta$ is second order and of Laplace type. Higher-derivative pieces are treated as perturbations and reserved for Chapter 9.

## 8.2 Seeley–DeWitt expansion, used not re-derived

The heat-kernel diagonal on a four-dimensional Riemannian section admits

$$\mathrm{tr}_V \langle x | e^{-s\mathcal{O}_\eta} | x \rangle \sim \frac{1}{(4\pi s)^2} \sum_{k=0}^{\infty} s^k a_k(x), \tag{8.5}$$

where $\mathrm{tr}_V$ is the fibre trace on the bundle in which $\eta$ takes values. The first two scalar densities are

$$\begin{aligned} a_0 &= \mathrm{tr}_V \mathbb{I}, \\ a_1 &= \mathrm{tr}_V \Big( \tfrac{1}{6} R - E \Big). \end{aligned} \tag{8.6}$$

The coefficient $a_2$ contains $R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$, $R_{\mu\nu} R^{\mu\nu}$, $R^2$, $E^2$, $RE$, and derivative terms. It multiplies a logarithm and is needed in Chapter 9. It is not needed for the leading $G_{\mathrm{eff}}$.

Insert (8.5) into (8.3) and integrate term by term:

$$W[g] \supset \int \mathrm{d}^4x\,\sqrt{g}\,\Big[-\frac{a_0}{32\pi^2}(\Lambda_{\mathrm{UV}}^4-\mu^4)-\frac{a_1}{16\pi^2}(\Lambda_{\mathrm{UV}}^2-\mu^2) - \frac{a_2}{32\pi^2}\ln\Big(\frac{\Lambda_{\mathrm{UV}}^2}{\mu^2}\Big)+\cdots\Big]. \tag{8.7}$$

Signs are Euclidean. Wick-rotating back, $\sqrt{g}\to\sqrt{-g}$ and $R_E\to -R_L$ in the conventions of (2.2). The book absorbs that sign into the statement that the induced Einstein–Hilbert term has the standard attractive sign, which must be checked once against the conformal case and then kept.

## 8.3 Evaluating $a_1$ on the condensate

Two pieces of $a_1$ produce a curvature term.

**Universal piece.** The $\frac{1}{6}R\,\mathrm{tr}_V\mathbb{I}$ contribution is present for every Laplace-type field. If the locked block of $\eta$ has fibre dimension $N_{\mathrm{eff}}=\mathrm{tr}_V\mathbb{I}$, this piece alone supplies

$$W[g] \supset -\frac{N_{\mathrm{eff}}}{16\pi^2}\cdot\frac{1}{6}(\Lambda_{\mathrm{UV}}^2-\mu^2)\int \mathrm{d}^4x\,\sqrt{-g}\,R+\cdots \tag{8.8}$$

before the Euclidean-to-Lorentzian sign is restored. With the sign convention of Section 2.3, the induced Einstein–Hilbert prefactor from this piece is proportional to $N_{\mathrm{eff}}\Lambda_{\mathrm{UV}}^2/16\pi^2$.

**Endomorphism piece.** The potential (3.7) evaluated on the polar vacuum gives $E$ a piece of order $\lambda v^2$ in the radial channel and a piece that vanishes in the Goldstone / tensor channels after Theorem 7.3. Expanding $E$ on a weakly curved background also produces a non-minimal coupling $\xi R\eta^2$ whose coefficient is fixed by the geometry of the frame bundle. After the massive radial mode is integrated down to $\mu\sim m_\sigma$, a logarithmic threshold correction proportional to $v^2$ remains.

Collecting the two pieces in the normalization used by the source notes yields the working formula of the book.

**Theorem 8.2** (Coefficient map)**.** *Under Assumption 8.1, the one-loop effective action* (8.7) *contains an Einstein–Hilbert term*

$$W[g] \supset \frac{1}{16\pi G_{\mathrm{eff}}}\int \mathrm{d}^4x\,\sqrt{-g}\,R, \tag{8.9}$$

*with*

$$\frac{1}{16\pi G_{\text{eff}}} = \frac{\operatorname{Tr}(\mathbb{I})\,\Lambda_{\text{UV}}^2}{16\pi^2} + \frac{v^2}{12}\ln\Big(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\Big) + \cdots . \tag{8.10}$$

*Here* $\operatorname{Tr}(\mathbb{I})$ *is the fibre trace on the locked block,* $\mu$ *is the infrared subtraction scale, and the ellipsis denotes scheme-dependent finite pieces and contributions from unlocked internal directions.*

*Proof sketch.* Isolate the coefficient of $\int\sqrt{-g}\,R$ in (8.7). The $\Lambda_{\text{UV}}^2$ term comes from $a_1 \supset \frac{1}{6}R\operatorname{Tr}(\mathbb{I})$ integrated against $1/(16\pi^2 s^2)$ between $\Lambda_{\text{UV}}^{-2}$ and $\mu^{-2}$. After the Lorentzian sign is restored and the conventional factor $1/16\pi$ is factored out of $1/G_{\text{eff}}$, the first term of (8.10) is the normalization chosen in the source notes; it corresponds to absorbing a numerical factor of order one (from $\frac{1}{6}$ versus the trace conventions on $T_{\mu\nu}$) into the definition of $\operatorname{Tr}(\mathbb{I})$. The logarithm comes from the $a_2$ threshold of the radial mode and from the $E \supset \xi R$ insertion evaluated on the condensate, both of which produce $v^2\ln(\Lambda_{\text{UV}}/\mu)$. The coefficient $1/12$ is the value obtained by evaluating that threshold on the polar vacuum (4.4) with $\xi = 1/6$ conformal coupling of the radial mode plus the non-minimal frame contribution that converts $\xi$ to the quoted normalization. Scheme-dependent finite terms are written as the ellipsis. □

*Remark* 8.3 (Honesty about the rational prefactors). The factors $1/16\pi^2$ and $1/12$ are the working coefficients of the NAMCG notes. A fully expanded Seeley–DeWitt computation on the bundle of $\delta\Phi_{\mu\nu}$, with every ghost and Goldstone included, will change the rationals by order-one factors and will mix $\operatorname{Tr}(\mathbb{I})$ with the dimension of the Lorentz representation. Theorem 8.2 should be read as fixing the *structure*

$$\frac{1}{G_{\text{eff}}} = A\operatorname{Tr}(\mathbb{I})\,\Lambda_{\text{UV}}^2 + B\,v^2\ln\Big(\frac{\Lambda_{\text{UV}}}{\mu}\Big) + C\,\mu^2, \tag{8.11}$$

with $A, B, C$ computable, and as recording the values $A = 1/\pi$, $B = 4\pi/3$, $C = 0$ used in the notes after converting back from (8.10). Chapter 17 lists an independent recomputation of $A, B, C$ as an open task.

## 8.4 The quartic term

The same expansion produces

$$W[g] \supset -\rho_\Lambda\int \mathrm{d}^4x\,\sqrt{-g}, \qquad \rho_\Lambda = \frac{\operatorname{Tr}(\mathbb{I})}{32\pi^2}(\Lambda_{\text{UV}}^4 - \mu^4) + \cdots . \tag{8.12}$$

Identifying this with a cosmological term,

$$\frac{\Lambda_{\text{naive}}}{8\pi G_{\text{eff}}} = \rho_\Lambda, \tag{8.13}$$

gives $\Lambda_{\rm naive}$ of order $\Lambda_{\rm UV}^2$ unless the quartic coefficient is cancelled. For $\Lambda_{\rm UV}$ anywhere near a particle-physics scale this is not a small cosmological constant.

This chapter does not cancel it. It isolates it. Two remarks prevent the isolation from being mistaken for a solution.

1. A bare counterterm $\int \sqrt{-g}\,\Lambda_{\rm ct}$ can subtract $\rho_\Lambda$ at one scale. It cannot explain why the remainder after subtraction is $10^{-120}$ in Planck units. Subtraction without a principle is fine tuning.

2. Chapter 12 assigns the principle to a boundary-entropy functional (HBEV). Until that chapter, $\rho_\Lambda$ is a debt.

## 8.5 Which background the heat kernel sees

Assumption 3.1 declared $g^{(0)}$ non-dynamical. The heat kernel of this chapter is nevertheless written on $g_{\mu\nu}$. The justification is the derivative expansion:

**Proposition 8.4** (Scaffold replacement)**.** *Let $g_{\mu\nu} = g^{(0)}_{\mu\nu} + h_{\mu\nu}$ with $h$ slowly varying on the scale $\Lambda_{\rm UV}^{-1}$. Replacing $g^{(0)}$ by $g$ inside $\mathcal{O}_\eta$ changes $W$ by operators of the form $\Lambda_{\rm UV}^2\|h-h^{(0)}\|$, $\|h-h^{(0)}\|R$, and higher. On the Minkowski scaffold $g^{(0)}=\eta$ these are the same operators already kept in* (8.7)*. Self-consistently substituting the emergent metric is therefore valid through the order at which $G_{\rm eff}$ is defined.*

At higher order the replacement produces the backreaction tensor $\mathcal{G}_{\mu\nu}$ of (4.12). That tensor is not needed to define $G_{\rm eff}$. It is needed to define the extra stress of Chapters 11–14.

## 8.6 Numerical size

If the first term of (8.10) dominates and $\mathrm{Tr}(\mathbb{I}) \sim O(1)$, then

$$G_{\rm eff} \sim \frac{\pi}{\Lambda_{\rm UV}^2}. \tag{8.14}$$

Identifying $G_{\rm eff}$ with Newton's constant forces $\Lambda_{\rm UV}$ to the Planck scale. The logarithmic term cannot change that conclusion unless $v$ is itself trans-Planckian, which is incompatible with (3.17).

Thus NAMCG, in the present coefficient map, is a theory whose microscopic cutoff is the Planck scale. It is not a theory in which a TeV-scale matrix model produces four-dimensional gravity. Anyone who wants a low cutoff must exhibit a cancellation of the $\mathrm{Tr}(\mathbb{I})\Lambda_{\rm UV}^2$ term against a counterterm or against additional fields of opposite heat-kernel sign (fermions). No such cancellation is used in this book.

## 8.7 What has been computed

| Object | Status | Where |
|---|---|---|
| structure of $1/G_{\text{eff}}$ | derived at one loop | Thm. 8.2 |
| rationals $A, B, C$ | working values from the notes | Remark after Thm. 8.2 |
| sign of the EH term | assumed standard, to be checked once | Sec. 8.2 |
| $\rho_\Lambda \sim \Lambda_{\text{UV}}^4$ | derived, not cancelled | Sec. 8.4 |
| higher-curvature $a_2$ | isolated, not used yet | Ch. 9 |
| exact two-loop $G_{\text{eff}}$ | not computed | Ch. 17 |

## 8.8 Exercises

1. **Proper-time integrals.** Perform the integrals that produce (8.7) from (8.5) and (8.3), treating $a_k$ as $s$-independent.

2. **Dimensions.** Confirm that $a_0\Lambda_{\text{UV}}^4$, $a_1\Lambda_{\text{UV}}^2$, and $a_2 \ln(\Lambda_{\text{UV}}/\mu)$ all have mass dimension four.

3. **Conformal value of $\xi$.** For a scalar with non-minimal term $\xi R\phi^2$, show that $\xi = 1/6$ makes the improvement term cancel the $\frac{1}{6}R$ piece of $a_1$ in four dimensions. Why does the NAMCG radial mode *not* use a cancellation of this kind to delete $G_{\text{eff}}$?

4. **Fermion sign.** A Dirac fermion contributes with the opposite sign to $\operatorname{Tr} \ln \mathcal{O}$. Write the modification of (8.11) if $N_f$ Dirac fermions are added to the locked block. This is the only obvious way to cancel the $\Lambda_{\text{UV}}^2$ term.

5. **Matching to Planck units.** Set $G_{\text{eff}} = G_{\text{N}}$ and $\operatorname{Tr}(\mathbb{I}) = 4$. Estimate $\Lambda_{\text{UV}}$ ignoring the logarithm. Is a GUT-scale $\Lambda_{\text{UV}}$ viable without cancelling the first term?

6. **Infrared scale.** Give two physically distinct choices of $\mu$ (radial-Higgs mass; Hubble scale) and state how $G_{\text{eff}}$ changes between them. Which choice is appropriate inside a laboratory, and which in cosmology?

7. **Scaffold error.** Expand Proposition 8.4 to first order in $h_{\mu\nu}$ on a flat scaffold and list the operators produced. Which of them renormalize $G_{\text{eff}}$, and which renormalize higher-curvature couplings?

## 8.9 Bridge to Chapter 9

The induced action through two derivatives is

$$W[g] = \int \mathrm{d}^4x\,\sqrt{-g}\Big(-\rho_\Lambda + \frac{1}{16\pi G_{\mathrm{eff}}}R + \alpha\, a_2[g] + \cdots\Big). \tag{8.15}$$

The $a_2$ density contains $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and $R_{\mu\nu}R^{\mu\nu}$. If those operators are kept as propagating fourth-order kinetic terms for $h_{\mu\nu}$, Ostrogradsky ghosts appear. Chapter 9 classifies the dangerous operators, states the conditions under which they may be treated as non-propagating corrections, and introduces the Einsteinian cubic completion used later to regularize curvature singularities without reopening the ghost problem.

# Chapter 9

# Ultraviolet Operators, Ghosts, and a Cubic Completion

**Learning outcomes.** After this chapter the reader should be able to:

1. read the $a_2$ density as a list of four-derivative operators and assign each a role (correction versus propagator);

2. state Ostrogradsky's theorem in the form used for gravitational effective field theory;

3. explain why quadratic gravity of Stelle type is not an acceptable infrared limit of NAMCG;

4. write the Einsteinian cubic invariant and the property that makes it safer than a generic $R^3$ term;

5. enter Chapter 10 knowing that the ultraviolet problem of the metric effective action is under a named control, not solved in the sense of a non-perturbative continuum limit.

Chapter 8 produced

$$W[g] = \int \mathrm{d}^4x\,\sqrt{-g}\Big(-\rho_\Lambda + \frac{R}{16\pi G_{\mathrm{eff}}} + \alpha\, a_2[g] + \cdots\Big). \tag{9.1}$$

The first two terms are the cosmological debt and the Einstein–Hilbert term. This chapter treats the third.

## 9.1 What $a_2$ contains

On a Laplace-type bundle the second Seeley–DeWitt coefficient is a linear combination of

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma},\quad R_{\mu\nu}R^{\mu\nu},\quad R^2,\quad E^2,\quad RE,\quad D^2R,\quad D^2E, \tag{9.2}$$

together with bundle-curvature terms $F_{\mu\nu}F^{\mu\nu}$. After the endomorphism $E$ is evaluated on the condensate and total derivatives are discarded, the metric operators that remain in four dimensions may be written

$$W_4[g] = \int \mathrm{d}^4x\,\sqrt{-g}\big(c_W C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + c_R R^2 + c_{\mathrm{GB}}\,E_{\mathrm{GB}}\big), \tag{9.3}$$

where $C_{\mu\nu\rho\sigma}$ is the Weyl tensor and

$$E_{\mathrm{GB}} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2 \tag{9.4}$$

is the Gauss–Bonnet density, topological in four dimensions at the level of the bulk equations of motion.

The coefficients $c_W, c_R, c_{\mathrm{GB}}$ are logarithms of $\Lambda_{\mathrm{UV}}/\mu$ times fibre traces, as in (8.7). Their precise rationals are scheme-dependent in the same sense as $A, B, C$ in (8.11). What is not scheme-dependent is the existence of the Weyl-squared and $R^2$ operators at this order.

## 9.2 Ostrogradsky's theorem, used as a veto

**Theorem 9.1** (Ostrogradsky, gravitational form)**.** *Let a field theory possess a Lagrangian that depends on second (or higher) time derivatives of a dynamical variable, not removable by adding a total derivative or by imposing a degeneracy condition that reduces the order. Then the Hamiltonian of that variable is unbounded below. In a quadratic gravitational action about Minkowski spacetime, the unbounded direction is a ghost pole in the propagator of $h_{\mu\nu}$.*

The theorem is the reason one may not promote (9.3) to a fundamental action for a propagating metric and call the result a completion of NAMCG.

**Stelle quadratic gravity.** The action

$$S_{\mathrm{Stelle}} = \int \mathrm{d}^4x\,\sqrt{-g}\Big(\frac{R}{16\pi G} - \frac{\alpha}{2}C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{\beta}{2}R^2\Big) \tag{9.5}$$

propagates, in addition to the massless graviton, a massive spin-2 ghost and a massive scalar. The ghost mass is of order $1/\sqrt{\alpha G}$. If $\alpha$ is a one-loop logarithm, that mass is near $\Lambda_{\mathrm{UV}}$, which is the Planck scale of Section 8.6. A Planck-scale ghost is an

ultraviolet sickness of the metric effective theory. It is not an infrared sickness provided those poles are never treated as asymptotic states of the condensate.

**Effective-field-theory reading.** The same operators are harmless if they are used only as vertices in a derivative expansion of matrix elements with external momenta $p \ll \Lambda_{\mathrm{UV}}$, and never as additional poles. That is the reading adopted here:

**Assumption 9.2** (Four-derivative terms are EFT vertices)**.** The operators in (9.3) correct graviton scattering and static potentials at order $(p/\Lambda_{\mathrm{UV}})^2$. They are not to be diagonalized into extra propagating modes of $h_{\mu\nu}$.

Assumption 9.2 is available only because the metric is composite. The microscopic path integral is over $\Phi$, not over an independent $g_{\mu\nu}$ with action (9.5). The poles of Stelle gravity are an artefact of treating $W[g]$ as a fundamental metric action beyond its radius of convergence.

## 9.3 What ultraviolet stability does mean

Presentation A is a matrix model. Its ultraviolet definition is the regularized trace over $\mathcal{H}$ with cutoff (3.3). Questions that make sense at that level are:

- boundedness of $S_{\mathrm{matrix}}$ from below on Hermitian $\hat{X}^\mu$;
- existence of the condensed saddle $\langle \Phi^I{}_A \rangle = v \Lambda^A{}_I$;
- absence of tachyons in the blocks that Theorem 7.3 and Assumption 7.5 did not already constrain.

Questions that do *not* make sense until a continuum limit in $\Lambda_{\mathrm{UV}}$ is constructed are:

- renormalizability of $W[g]$ as a fundamental quantum field theory of $g_{\mu\nu}$;
- asymptotic safety of that metric theory;
- unitarity of S-matrix elements among Stelle poles.

NAMCG, as written in this book, claims the first list and refuses the second. The metric effective action is an infrared object with a cutoff. Ultraviolet stability means stability of the matrix saddle, not renormalizability of quadratic gravity.

A sufficient condition for the matrix action (3.4) to be bounded below on Hermitian matrices is that the quartic commutator term dominates with the sign written there,

$$-\mathrm{Tr}([\hat{X}^\mu, \hat{X}^\nu][\hat{X}_\mu, \hat{X}_\nu]) \geq 0 \tag{9.6}$$

in the Euclideanized internal-time convention of IKKT-type models, together with a potential $V$ that is bounded below, as (3.7) is. Lorentzian signature complicates the bound because $\hat{X}^0$ enters with the opposite sign in some reductions. That complication is inherited from the matrix-model literature and is not solved here. It is recorded in Chapter 17.

## 9.4 Cubic operators and Einsteinian cubic gravity

Singularities of the Einstein equation are not cured by the four-derivative vertices of Assumption 9.2. Those vertices are small when curvature is small, and they are untrustworthy when curvature reaches $\Lambda_{\mathrm{UV}}^2$. A different higher-curvature object is needed if compact-object cores are to be regularized inside the derivative expansion.

Einsteinian cubic gravity (ECG) is the four-dimensional cubic curvature invariant whose linearized spectrum about a maximally symmetric background coincides with that of Einstein gravity: no extra spin-2 ghost, no extra scalar at that order. A convenient writing is

$$\mathcal{P} = 12 R_{\mu}{}^{\rho}{}_{\nu}{}^{\sigma} R_{\rho}{}^{\tau}{}_{\sigma}{}^{\eta} R_{\tau}{}^{\mu}{}_{\eta}{}^{\nu} + \cdots , \tag{9.7}$$

the omitted terms being the unique combination of $R^3$, $R R_{\mu\nu} R^{\mu\nu}$, and $R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} R$ that implements the spectral condition. The action used later in the book is

$$S_{\mathrm{ECG}}[g] = \int \mathrm{d}^4x \, \sqrt{-g} \Big( \frac{R - 2\Lambda_{\mathrm{eff}}}{16\pi G_{\mathrm{eff}}} + \beta_{\mathrm{ECG}} \, \mathcal{P} \Big). \tag{9.8}$$

Two properties matter.

1. On static, spherically symmetric metrics the cubic term modifies the lapse function algebraically rather than by adding new propagating poles about that background. The regular-core profile

$$f(r) = 1 - \frac{2Mr^2}{r^3 + r_0^3} + \frac{\Lambda_{\mathrm{eff}}}{3} r^2 \tag{9.9}$$

used in Chapter 13 is an approximate solution in that sector, with $r_0^3 \propto \beta_{\mathrm{ECG}} M$.

2. The spectral condition is a statement about linearization on maximally symmetric backgrounds. It is not a proof that every time-dependent perturbation of (9.9) is ghost-free. The book uses ECG as a controlled model of core regularity, not as a claim of non-perturbative unitarity.

**Assumption 9.3** (Cubic completion)**.** Through the curvature scales relevant to compact-object cores, the metric effective action may be truncated at (9.8), with $\beta_{\mathrm{ECG}}$ set by the matching of $a_2$ and by the matrix scale $r_0 \sim \Lambda_{\mathrm{UV}}^{-1}$ up to an order-one factor.

Assumption 9.3 is stronger than Assumption 9.2. It promotes one cubic combination from a vertex to a term that is solved non-perturbatively in the static spherical sector. The promotion is justified only in that sector, and only as a model.

## 9.5 Torsion and higher connections

The composite frame of Definition 4.1 need not have vanishing torsion. A torsion tensor

$$T^{A}{}_{\mu\nu} = \partial_\mu e^A_\nu - \partial_\nu e^A_\mu + \omega_\mu{}^A{}_B e^B_\nu - \omega_\nu{}^A{}_B e^B_\mu \tag{9.10}$$

is an additional infrared field if $\omega$ is independent of $e$. In NAMCG the connection is not independent: it is the connection that eats the Goldstone $\omega_{AB}$ of Chapter 5. After that eating, torsion is either zero or massive of order $v$. Massive torsion is an ultraviolet correction of the same type as the $a_2$ vertices. It is not retained as a propagating infrared field.

## 9.6 A checklist before Lorentz locking

The metric effective theory, as it now stands, is:

| Ingredient | Status |
|---|---|
| massless TT tensor | Thm. 7.3 |
| prefactor $G_{\text{eff}}$ | Thm. 8.2 |
| quartic vacuum energy | isolated, not tamed |
| $C^2$ and $R^2$ | EFT vertices, Assp. 9.2 |
| cubic core regularity | ECG model, Assp. 9.3 |
| torsion | massive or zero after eating |
| local Lorentz invariance as a gauge symmetry of $\Phi$ | not yet constructed |

The missing gauge symmetry is the last geometric ingredient. Without it the identification of $\omega_{AB}$ as a Goldstone, used since Chapter 5, is incomplete.

## 9.7 Exercises

1. **Gauss–Bonnet.** Show that $E_{\text{GB}}$ in four dimensions is a total derivative plus a topological density. Why, then, can $c_{\text{GB}}$ still matter on manifolds with boundary?

2. **Stelle poles.** Linearize (9.5) about Minkowski spacetime in harmonic gauge. Locate the massive spin-2 pole and determine the sign of its residue relative to the massless graviton.

3. **EFT versus poles.** A tree-level graviton-graviton amplitude receives a correction $\sim c_W s^2/\Lambda_{\rm UV}^2$ from (9.3). At what centre-of-mass energy does this correction become unreliable as an EFT vertex, and why is that energy the same scale at which Stelle poles would have appeared?

4. **Bounded matrix action.** For two Hermitian matrices $X, Y$ show that $-\mathrm{Tr}([X,Y]^2) \geq 0$. Identify the step that fails, or requires a Euclidean rotation, when one matrix is $X^0$ with Lorentzian signature.

5. **Spherical reduction.** Take $ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$ and list which curvature scalars in a generic $R^3$ combination survive. The point of ECG is that the combination $\mathcal{P}$ reduces to an algebraic equation for $f$, not a fourth-order ODE. Verify that claim for at least the leading $1/r$ terms, or cite the reduction and state what you used.

6. **Torsion mass.** Give a dimensional estimate of the torsion mass if the only scale available is $v$. Compare it with $m_\psi$ from Assumption 7.5.

## 9.8 Bridge to Chapter 10

The infrared metric theory is Einstein gravity plus controlled higher-curvature corrections and an untamed $\rho_\Lambda$. What it still lacks is an explicit demonstration that local $\mathrm{SO}(3,1)$ is a gauge symmetry of the condensate and that the six Goldstones $\omega_{AB}$ are eaten by the connection, leaving no extra massless scalars. Chapter 10 constructs that locking. After it, the counting table of Section 5.7 is complete through the line "eat $\omega_{AB}$," and the composite vierbein is a genuine frame rather than a named expectation value.

# Chapter 10

# Local Lorentz Locking and the Fate of the Goldstones

**Learning outcomes.** After this chapter the reader should be able to:

1. write the simultaneous action of internal gauge transformations and local Lorentz transformations on $\Phi^I{}_A$;

2. identify the diagonal subgroup preserved by the polar vacuum and the six broken generators;

3. exhibit the Stueckelberg replacement that eats $\omega_{AB}$ into a spin connection;

4. finish the counting table of Section 5.7 through the vector and Lorentz-Goldstone lines;

5. enter Chapter 11 with a composite frame that transforms as a genuine vierbein, so that $\nabla^\mu T_{\mu\nu}$ is well defined.

Since Chapter 5 the antisymmetric piece $\omega_{AB}$ has been called a local-Lorentz Goldstone and postponed. This chapter constructs the gauge symmetry that justifies the name and the Higgs mechanism that removes $\omega_{AB}$ from the infrared spectrum.

## 10.1 The two groups that act on $\Phi$

The condensate field of Presentation B transforms, by (3.10), as

$$\Phi^I{}_A(x) \;\longmapsto\; \Omega^I{}_J(x)\,\Phi^J{}_B(x)\,\Lambda^B{}_A(x). \tag{10.1}$$

Here $\Omega(x)$ takes values in the internal gauge group $G_{\text{int}}$ and $\Lambda(x)$ takes values in $\mathrm{SO}(3,1)$, or in its double cover if spinors are later coupled. Infinitesimally,

$$\delta\Phi^I{}_A = \alpha^I{}_J\Phi^J{}_A + \Phi^I{}_B\,\beta^B{}_A, \tag{10.2}$$

with $\alpha$ in the Lie algebra of $G_{\text{int}}$ and $\beta_{AB} = -\beta_{BA}$ in $\mathfrak{so}(3,1)$.

These are independent transformations at the level of field content. Nothing in Chapter 3 forced $\Lambda(x)$ to be a gauge symmetry rather than a global classification of the index $A$. Treating it as local is a choice, recorded as such.

**Assumption 10.1** (Local Lorentz gauge symmetry)**.** The microscopic action (3.6) is invariant under local $\Lambda(x) \in \mathrm{SO}(3,1)$ once a spin connection $\omega_\mu{}^A{}_B$ is introduced through the covariant derivative on the tangent index,

$$\mathcal{D}_\mu\Phi^I{}_A = \partial_\mu\Phi^I{}_A - i(A_\mu)^I{}_J\Phi^J{}_A + \omega_\mu{}^B{}_A\Phi^I{}_B. \tag{10.3}$$

The connection $\omega_\mu{}^{AB}$ is, at this stage, an independent gauge field.

Assumption 10.1 is the step that makes $A$ a genuine tangent index rather than an internal flavour that happens to have four values. Combined with Assumption 4.6, it is the origin of local Lorentz invariance in NAMCG. It is not derived from Presentation A.

## 10.2 The vacuum and the broken generators

The polar vacuum of Chapter 4, in the Minkowski frame,

$$\langle\Phi^I{}_A\rangle = v\,\delta^I{}_A, \tag{10.4}$$

is invariant only under the diagonal combination

$$\alpha^I{}_J = -\,v^{-1}\delta^I{}_B\,\beta^B{}_C\,\delta^C{}_J\Big|_{\text{locked } 4\times 4 \text{ block}}, \tag{10.5}$$

that is, under simultaneous internal and Lorentz rotations that rotate the locked axes together. If $G_{\text{int}}$ contains $\mathrm{SO}(3,1)$ (or a group large enough to have an $\mathfrak{so}(3,1)$ subalgebra acting on the locked indices), six generators are unbroken and six are broken.

**Proposition 10.2** (Goldstone count)**.** *About the vacuum* (10.4)*, with Assumption 10.1 in force and with the locked block of dimension four, there are six broken generators. The corresponding Goldstone fields are the antisymmetric components $\omega_{AB}$ of* (4.15)*.*

*Proof.* The vacuum stabilizer is the diagonal $\mathrm{SO}(3,1)_{\text{diag}}$. The coset $\mathrm{SO}(3,1)_{\text{Lorentz}} \times \mathrm{SO}(3,1)_{\text{int}}/\mathrm{SO}(3,1)_{\text{diag}}$ has dimension $6 + 6 - 6 = 6$. An infinitesimal broken trans-

formation with parameter $\beta_{AB}$ shifts $\Phi^I{}_A$ by $v\beta^I{}_A$, which is antisymmetric in $(I, A)$ after lowering with $\eta$. That shift is the fluctuation $\omega_{AB}$. □

If $G_{\text{int}}$ is larger than $\mathrm{SO}(3,1)$, additional Goldstones live in the unlocked block. They belong to Assumption 5.1 and are taken heavy.

## 10.3 Eating $\omega_{AB}$

A local broken symmetry with an independent connection is a Higgs mechanism. Introduce the Stueckelberg field $\Lambda(x) \in \mathrm{SO}(3,1)$ already present in the polar decomposition (4.4), and replace the spin connection by the dressed connection

$$\omega_\mu \;\longmapsto\; \Lambda^{-1}\omega_\mu\Lambda + \Lambda^{-1}\partial_\mu\Lambda. \tag{10.6}$$

Choose the unitary gauge

$$\Lambda(x) = \mathbb{I}, \tag{10.7}$$

which is admissible wherever the polar decomposition is smooth (that is, wherever Assumption 4.4 holds and no condensate defect forces $\det e = 0$). In that gauge $\omega_{AB}$ has been absorbed into $\omega_\mu{}^{AB}$, and the kinetic term of $\Phi$ produces a mass term

$$\mathcal{L}_{\omega^2} \supset v^2\, \omega_\mu{}^{AB}\omega^\mu{}_{AB}. \tag{10.8}$$

The connection is massive of order $v$, or it is identified with the Levi-Civita spin connection of the composite frame and the mass term is reinterpreted as the statement that torsion is gapped.

**Theorem 10.3** (Lorentz locking)**.** *Under Assumptions 3.4, 10.1, and 4.4, there exists a gauge in which*

1. *the six Goldstones $\omega_{AB}$ do not propagate;*
2. *the composite object $e^A_\mu$ of Definition 4.1 transforms only under the diagonal local Lorentz group, as a vierbein;*
3. *the independent spin connection is either massive of order $v$ or slaved to $e^A_\mu$ by the torsion-free condition* (2.6)*.*

*Proof.* Proposition 10.2 supplies six Goldstones. The Stueckelberg replacement (10.6) and the gauge (10.7) remove them from the spectrum. In that gauge $\Phi^{IA} = (v+\sigma)\delta^{IA}$ plus tensor and vector blocks, so $e^A_\mu$ defined from $\partial_\mu\Phi$ or from the normalized vacuum transforms as $e^A_\mu \mapsto \Lambda_{\text{diag}}{}^A{}_B e^B_\mu$. The mass term (10.8) gaps any component of $\omega_\mu{}^{AB}$ that is not slaved to $e$. The slaved remainder is uniquely the torsion-free spin connection of $e^A_\mu$, by the standard Cartan lemma used in (2.6). □

*Remark* 10.4 (Defects). Unitary gauge fails where $\Lambda(x)$ cannot be set to the identity continuously—at defects where $\det e = 0$. Those loci are condensate defects, not coordinate singularities of an already existing manifold. They are possible ultraviolet endpoints of the regular cores of Chapter 13, and they are not analyzed further here.

## 10.4 The internal vector after locking

The internal connection $A_\mu$ of (3.5) is not the same object as $\omega_\mu{}^{AB}$. After locking, four of its components in the locked block mix with $\omega_\mu$. The orthogonal combination is an ordinary Yang–Mills field of the unbroken diagonal subgroup, or of whatever remains of $G_{\text{int}}$ outside $\mathrm{SO}(3,1)$.

Its mass is a standard Higgs mass,

$$m_A \sim g_{\text{YM}} v, \tag{10.9}$$

unless a residual exact gauge symmetry is left unbroken. In the latter case the infrared theory contains general relativity plus Yang–Mills, which is phenomenologically welcome and does not disturb Theorem 7.3. In the former case the vector block decouples at scale $m_A$ and is irrelevant below $v$.

This closes the vector line of the counting table.

## 10.5 Diffeomorphisms

Diffeomorphisms act on the coordinate index of every tensor constructed from $\Phi(x)$. They are not eaten. They are the gauge symmetry of the composite metric, and they are what force $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$ in Chapter 11.

After Theorem 10.3, an infinitesimal diffeomorphism $\xi^\mu$ acts on the vierbein in the standard hybrid way,

$$\delta_\xi e^A_\mu = \xi^\rho \partial_\rho e^A_\mu + \partial_\mu \xi^\rho \, e^A_\rho, \tag{10.10}$$

up to a local Lorentz transformation that can be used to keep a chosen gauge on the frame. The induced action on $g_{\mu\nu}$ is the Lie derivative (5.17). No extra scalar Goldstone of diffeomorphisms appears: diffeomorphisms are gauged, not broken, in the condensed phase.

## 10.6 Completed counting table

| Stage | What is removed | Remaining | Where |
|---|---|---|---|
| algebraic split | — | 16 | Ch. 5 |
| eat $\omega_{AB}$ | 6 Lorentz Goldstones | 10 | this chapter |
| gap or Higgs $A_\mu$ | 4 vector | 6 | this chapter |
| gap $\psi$ | 1 conformal scalar | 5 | Assp. 7.5 |
| $m_T = 0$ and TT gauge | 3 residual constraints | 2 TT | Ch. 7 |

The six remaining after the vector is removed are the ten metric components minus four diffeomorphisms. Gapping $\psi$ and imposing massless TT gauge then yields two polarizations. The table of Section 5.7 is thereby completed, with the one outstanding dynamical hypothesis still being Assumption 7.5.

## 10.7 What locking does not do

Theorem 10.3 does not derive Lorentzian signature (Assumption 4.6 remains). It does not derive local Lorentz invariance from Presentation A (Assumption 10.1 remains). It does not gap $\psi$. It does not compute $G_{\text{eff}}$. It does not conserve $T^{\text{ent}}_{\mu\nu}$.

What it does is make the later covariant derivatives honest. From here on, $\nabla_\mu$ is the Levi-Civita connection of the composite metric, and statements such as $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$ are tensor equations on the emergent spacetime rather than slogans.

## 10.8 Exercises

1. **Stabilizer.** Verify that (10.4) is invariant under the diagonal combination (10.5) and under no other independent pair $(\alpha, \beta)$ acting on the locked block.

2. **Coset dimension.** Compute $\dim\big(\mathrm{SO}(3,1) \times \mathrm{SO}(3,1)/\mathrm{SO}(3,1)\big)$ and match it to the six components of $\omega_{AB}$.

3. **Stueckelberg.** Starting from (10.3) and (4.4), perform the replacement (10.6) and isolate the quadratic term (10.8) in unitary gauge.

4. **Torsion-free slaving.** Given $e^A_\mu$, write the explicit formula for the torsion-free spin connection $\omega_\mu{}^{AB}(e)$. Confirm that it transforms inhomogeneously under local Lorentz transformations, as a connection must.

5. **Defect.** Construct a hedgehog configuration of $\Lambda(x)$ on a spatial 2-sphere inside which $\det e = 0$ at a point. Why can unitary gauge not be imposed globally on that slice?

6. **Residual Yang–Mills.** Suppose $G_{\text{int}} = \mathrm{SO}(3,1) \times H$ with $H$ unbroken. What is the infrared field content after Theorem 10.3, in addition to the graviton?

7. **Hybrid diffeomorphism.** Derive (10.10) from the action of a coordinate shift on $\Phi(x)$ and Definition 4.1.

## 10.9 Bridge to Chapter 11

The emergent geometry now has a metric, a torsion-free connection, two massless graviton polarizations, and a computed $G_{\text{eff}}$. Diffeomorphism invariance is in force. Noether's second theorem therefore requires

$$\nabla^{\mu} T^{\text{total}}_{\mu\nu} = 0 \tag{10.11}$$

of whatever stress tensor sources the Einstein equation. Chapter 11 defines $T^{\text{total}}_{\mu\nu}$ as the sum of ordinary matter, the tensor-block stress already used in Chapter 7, and the entropic stress $T^{\text{ent}}_{\mu\nu}$ obtained by varying $U_{\text{ent}}$. It then proves that the sum is divergenceless on shell, which is the condition that licenses every later phenomenological drag as geometry rather than as a force added by hand.

# Chapter 11

# Covariant Conservation of the Total Stress Tensor

**Learning outcomes.** After this chapter the reader should be able to:

1. define $T^{\text{total}}_{\mu\nu}$ as the sum of matter, tensor-block, and entropic contributions;
2. obtain $T^{\text{ent}}_{\mu\nu}$ by metric variation of $U_{\text{ent}}$;
3. prove that diffeomorphism invariance forces $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$ on shell;
4. reduce that identity to a Newtonian force balance and see which drags are allowed;
5. enter Chapter 12 knowing that vacuum energy is a conservation problem as well as a cosmological-constant problem.

Chapter 10 made $\nabla_\mu$ the Levi-Civita connection of the composite metric. This chapter asks what that connection is allowed to act on.

## 11.1 The split

The infrared field equation contemplated from here on is

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G_{\text{eff}}\, T^{\text{total}}_{\mu\nu}, \tag{11.1}$$

with

$$T^{\text{total}}_{\mu\nu} = T^{\text{matter}}_{\mu\nu} + T^{\text{tensor}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu}. \tag{11.2}$$

**Matter.** $T^{\text{matter}}_{\mu\nu}$ is the metric variation (2.13) of every action for fields that are not the condensate itself: ordinary fluids, electromagnetism, baryons. On shell it satisfies

$\nabla^\mu T^{\text{matter}}_{\mu\nu} = 0$ *only* if those fields do not exchange energy with the condensate. In NAMCG they may. The separate conservation of matter is therefore not assumed.

**Tensor block.** $T^{\text{tensor}}_{\mu\nu}$ is the metric stress of the quadratic action (7.13). After Theorem 7.3 it is the linearized gravitational-wave stress, of the Isaacson type when averaged over wavelengths short compared with the background curvature. It is kept in (11.2) for bookkeeping and is negligible in the static Newtonian laboratories of Chapters 14–15.

**Entropic piece.** $T^{\text{ent}}_{\mu\nu}$ is defined by the same variational rule applied to the mutual-information potential of Definition 6.4, promoted to a functional of the metric:

$$T^{\text{ent}}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{ent}}[g]}{\delta g^{\mu\nu}}, \qquad S_{\text{ent}}[g] = -\int \mathrm{d}t\, U_{\text{ent}}[g]. \tag{11.3}$$

The promotion $U_{\text{ent}} \to S_{\text{ent}}[g]$ uses the fact that every entropy $S(A)$ in (6.9) is computed from a reduced state on a region whose causal structure depends on $g_{\mu\nu}$. Varying the metric varies the regions and varies the modular Hamiltonians.

## 11.2 Noether's second theorem, applied once

Let $S_{\text{IR}}[g, \Psi]$ be the infrared action whose Euler–Lagrange equation for $g_{\mu\nu}$ is (11.1): Einstein–Hilbert plus $\Lambda_{\text{eff}}$, plus matter, plus $S_{\text{ent}}$, plus the tensor-block action. Diffeomorphism invariance states that for every compactly supported vector field $\xi^\mu$,

$$\delta_\xi S_{\text{IR}} = 0 \tag{11.4}$$

identically, off shell as a statement about the integrand up to a boundary term. Noether's second theorem then supplies

$$\nabla_\mu\Big(\frac{1}{8\pi G_{\text{eff}}}(G^{\mu\nu} + \Lambda_{\text{eff}} g^{\mu\nu}) - T^{\mu\nu}_{\text{total}}\Big) = 0 \tag{11.5}$$

as an identity, once the matter and condensate fields $\Psi$ are on shell. The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ and metric compatibility $\nabla_\mu g^{\mu\nu} = 0$ reduce (11.5) to the on-shell conservation law

$$\nabla_\mu T^{\mu\nu}_{\text{total}} = 0. \tag{11.6}$$

**Theorem 11.1** (Conservation of the total stress)**.** *Assume:*

1. *the infrared action $S_{\text{IR}}$ is a scalar under diffeomorphisms of the composite metric, including the entropic functional $S_{\text{ent}}[g]$;*

2. *the Einstein equation is* (11.1)*;*

*3. matter fields and condensate fluctuations are on shell with respect to $S_{\mathrm{IR}}$.*

*Then*

$$\nabla^{\mu}\big(T^{\mathrm{matter}}_{\mu\nu}+T^{\mathrm{tensor}}_{\mu\nu}+T^{\mathrm{ent}}_{\mu\nu}\big)=0. \tag{11.7}$$

*Separate conservation of any one summand is not implied.*

The Einstein-plus-extra-tensor form of the infrared equation is collected in Appendix F.

*Proof.* Diffeomorphism invariance of $S_{\mathrm{IR}}$ is Assumption 1. Noether's second theorem yields (11.5). Bianchi and (11.1) yield (11.7). The last sentence is the observation that Noether's theorem constrains only the sum that couples to $g_{\mu\nu}$. □

*Remark* 11.2 (The real hypothesis). Everything in the proof is standard except Assumption 1 applied to $S_{\mathrm{ent}}$. Mutual information (2.31) is a scalar for each fixed partition of a spatial slice. Promoting it to a diffeomorphism scalar on a dynamical spacetime requires that the assignment $(g, A, B) \mapsto I(A : B)$ commute with Lie dragging of the regions. That is true if $A$ and $B$ are defined geometrically (causal domains, outermost trapped surfaces, constant-mean-curvature balls of specified volume) and fails if they are defined by coordinate inequalities such as $r \le R$ in a preferred chart. The book uses geometric regions.

## 11.3 Structure of the entropic stress

A first variation of $S_{\mathrm{ent}}$ on a static background produces a conservative extra potential. A first variation on a time-dependent background produces, in addition, a piece odd under velocities. The split

$$T^{\mathrm{ent}}_{\mu\nu}=T^{\mathrm{ent,rev}}_{\mu\nu}+T^{\mathrm{ent,irrev}}_{\mu\nu} \tag{11.8}$$

is the covariant version of "energy from mutual information" plus "drag from changing mutual information."

On two worldlines $\gamma_A, \gamma_B$ in a nearly Minkowski region one has, at leading post-Newtonian order,

$$U_{\mathrm{ent}}=-\lambda I\big(\ell_A,\ell_B,d(\gamma_A,\gamma_B)\big), \tag{11.9}$$

and the four-force on body $A$ is

$$f^{\mu}_{A}=-\frac{D}{D\tau_A}\Big(\frac{\partial L_{\mathrm{ent}}}{\partial \dot{x}_{A\mu}}\Big)+\frac{\partial L_{\mathrm{ent}}}{\partial x_{A\mu}}, \tag{11.10}$$

with $L_{\mathrm{ent}} = -U_{\mathrm{ent}}$ along the worldlines. The term built from $\partial I/\partial d$ at vanishing velocity is reversible and conservative. The term built from $\partial I/\partial \dot{d}$ or from the time

derivative of $\ell_A, \ell_B$ is irreversible in the thermodynamic sense of Chapter 6: it tracks a change of entanglement across the motion, not a stored potential.

The Newtonian reduction of the irreversible piece, for a reduced mass $\mu$ and a separation $d$, is of the form

$$\mathbf{a}_{\mathrm{irrev}} = -\eta(d, \ell_A, \ell_B)\, \frac{\mathbf{v}}{d^n}, \tag{11.11}$$

with a positive coefficient $\eta$ if mutual information decreases under adiabatic separation of two thermal cores. The source notes use $n = 4$ as a working exponent compatible with a second derivative of a $1/d^2$ mutual-information tail (6.10). Chapter 15 adopts

$$\mathbf{F}_{\mathrm{diss}} = -\eta\, \frac{\mu}{d^4}\, \mathbf{v} \tag{11.12}$$

as a laboratory, not as a derived multipole formula.

**Proposition 11.3** (Allowed Newtonian drags)**.** *A two-body equation*

$$\mu \ddot{\mathbf{r}} = -\nabla U_{\mathrm{N}} - \nabla U_{\mathrm{ent}}^{(\mathrm{rev})} + \mathbf{F}_{\mathrm{diss}} \tag{11.13}$$

*is the reduction of a covariantly conserved $T^{\mathrm{ent}}_{\mu\nu}$ only if there exists an energy-balance identity*

$$\frac{\mathrm{d}}{\mathrm{d}t}(E_{\mathrm{mech}} + E_{\mathrm{ent}}) = \mathbf{F}_{\mathrm{diss}} \cdot \mathbf{v} + O(c^{-2}) \tag{11.14}$$

*with $E_{\mathrm{ent}}$ built from $I(A:B)$, and only if the equal-and-opposite force on the second body holds up to radiation reaction of Isaacson type. A drag that dissipates mechanical energy into nowhere, with no $E_{\mathrm{ent}}$ and no gravitational radiation, is not allowed.*

The laboratory (11.12) satisfies Proposition 11.3 if and only if the integrator tracks an entanglement energy whose decrease equals $-\mathbf{F}_{\mathrm{diss}} \cdot \mathbf{v}$. Chapter 15 will do that bookkeeping explicitly.

## 11.4 Exchange with matter

Equation (11.7) permits

$$\nabla^\mu T^{\mathrm{matter}}_{\mu\nu} = -\nabla^\mu T^{\mathrm{ent}}_{\mu\nu} - \nabla^\mu T^{\mathrm{tensor}}_{\mu\nu}. \tag{11.15}$$

Matter may heat or cool because entanglement of the condensate cores is changing. In a tidal disruption event the geometric reading is: as the disrupted star is sheared, the mutual information between debris parcels and the compact core changes, and the work done by $\mathbf{F}_{\mathrm{diss}}$ is accounted in $E_{\mathrm{ent}}$, not lost.

This is still not a model of microphysical dissipation in a fluid with a viscosity

measured in the laboratory. It is a channel required by conservation once $U_{\text{ent}}$ is admitted as part of the action.

## 11.5 Vacuum energy as a conservation constraint

A term $\rho_\Lambda g_{\mu\nu}$ with constant $\rho_\Lambda$ is conserved by itself:

$$\nabla^\mu(\rho_\Lambda g_{\mu\nu}) = 0. \tag{11.16}$$

A term whose prefactor varies, $\rho_\Lambda(x)g_{\mu\nu}$, is not conserved unless

$$\partial_\mu \rho_\Lambda = 0 \tag{11.17}$$

or additional current terms are supplied. The heat-kernel density (8.12) is a constant in the approximation that $\Lambda_{\text{UV}}$ and $\text{Tr}(\mathbb{I})$ are constant. If Chapter 12 regulates $\rho_\Lambda$ by a boundary functional that depends on the metric induced on $\partial M$, the regulated object $\Lambda_{\text{eff}}[g]$ must still obey (11.7). That requirement is a design constraint on HBEV, not an optional aesthetic.

## 11.6 What this chapter has not done

Theorem 11.1 does not compute $\eta$, does not fix $n = 4$, does not evaluate $I(A : B)$ for Kerr exteriors, and does not replace quadrupole radiation. Gravitational waves remain the dissipative channel of the tensor block. Entropic drag is an additional channel, allowed only because it sits inside a conserved total tensor.

The theorem also does not justify writing an arbitrary function $\mathbf{a}_{\text{irrev}}(\mathbf{r}, \mathbf{v})$ and calling it NAMCG. Phenomenological chapters that introduce a drag must exhibit the $E_{\text{ent}}$ of Proposition 11.3 or must state that they are studying a reduced laboratory.

## 11.7 Exercises

1. **Bianchi reminder.** From $\nabla^\mu G_{\mu\nu} = 0$ and (11.1), obtain (11.7) in three lines.

2. **Two-body check.** For a potential $U(d)$ depending only on separation, show that the Newtonian forces are equal and opposite. Which symmetry of $I(A : B)$ is the relativistic ancestor of that statement?

3. **Energy bookkeeping.** Take (11.12) with constant $\eta$ and compute $\mathbf{F}_{\text{diss}} \cdot \mathbf{v}$. Propose an $E_{\text{ent}}(d, \dot{d})$ whose time derivative matches that power. Is your $E_{\text{ent}}$ unique?

4. **Illegal drag.** Consider $\mathbf{F} = -\gamma\mathbf{v}$ with constant $\gamma$, independent of $d$. Why does this fail Proposition 11.3 for two bodies that can be separated to infinity at finite cost in $I(A : B)$?

5. **Variable vacuum energy.** Let $\rho_\Lambda = \rho_\Lambda(\phi)$ for a scalar $\phi$. Construct the additional stress that restores conservation. Compare with a potential $V(\phi)$ in ordinary scalar-field cosmology.

6. **Isaacson average.** Write the Isaacson stress of a high-frequency TT wave and confirm that its divergence vanishes after a short-wavelength average. That is the tensor-block contribution to (11.7) in a radiative spacetime.

7. **Coordinate regions.** Give an explicit example of two coordinate-defined regions whose mutual information is *not* a diffeomorphism scalar. This is the counterexample that forces the remark after Theorem 11.1.

## 11.8 Bridge to Chapter 12

Conservation is now in force for whatever vacuum energy Chapter 8 left untamed. A constant $\rho_\Lambda$ is conserved and disastrous. A regulated $\Lambda_{\text{eff}}$ must remain conserved after the regulation. Chapter 12 introduces the Holographic Boundary-Entropy Vacuum: a boundary functional built from $K/8\pi G_{\text{eff}}$ and an entropy density $S_{\text{ent}}(h_{ij}, \phi)$ whose stationarity sets $\Lambda_{\text{eff}}$ by an infrared entanglement datum rather than by $\Lambda_{\text{UV}}^4$. The design constraint just stated—that the resulting $\Lambda_{\text{eff}} g_{\mu\nu}$ still sit inside Theorem 11.1—is the first test that construction must pass.

# Chapter 12

# Holographic Boundary-Entropy Vacuum and $\Lambda_{\text{eff}}$

**Learning outcomes.** After this chapter the reader should be able to:

1. restate the quartic vacuum-energy debt isolated in Chapter 8 as a problem about a bulk term $\rho_\Lambda \sqrt{-g}$;
2. write the HBEV boundary action built from the Gibbons–Hawking–York term and an entropy density $S_{\text{ent}}(h_{ij}, \phi)$;
3. obtain $\Lambda_{\text{eff}}$ from stationarity of that boundary action rather than from $\Lambda_{\text{UV}}^4$;
4. check that the regulated cosmological term still obeys Theorem 11.1;
5. enter Chapter 13 knowing which vacuum energy is being inserted into regular-core and galactic solutions.

Chapter 8 produced

$$\rho_\Lambda = \frac{\mathrm{Tr}(\mathbb{I})}{32\pi^2}(\Lambda_{\text{UV}}^4 - \mu^4) + \cdots . \tag{12.1}$$

A constant of that size cannot be the observed cosmological constant. This chapter does not cancel (12.1) by a bare counterterm tuned to 120 decimal places. It replaces the bulk cosmological term by a boundary variational problem whose solution is an infrared scale.

## 12.1 What would not count as a solution

Three common manoeuvres are excluded in advance.

1. **Subtraction by hand.** Adding $-\rho_\Lambda \int \sqrt{-g}$ to the action and declaring the sum zero is a renormalization condition at one scale. It does not explain the residual scale.

2. **Setting $\Lambda_{\rm UV}$ to meV.** Equation (8.14) then destroys Newton's constant. The cutoff cannot serve both masters.

3. **A spacetime-dependent $\rho_\Lambda(x)$ with no current.** Section 11.5 forbids this unless extra stress is supplied. Regulation that breaks Theorem 11.1 is not available.

Any construction that survives must therefore (i) replace the bulk quartic by an infrared datum, (ii) leave $G_{\rm eff}$ as computed in Theorem 8.2, and (iii) produce a conserved $\Lambda_{\rm eff} g_{\mu\nu}$ or a conserved completion of it.

## 12.2 Boundary geometry

Let $M$ be a region of the emergent spacetime with boundary $\partial M$, outward normal $n^\mu$, induced metric

$$h_{ij} = g_{\mu\nu} e^\mu_i e^\nu_j, \tag{12.2}$$

and extrinsic curvature

$$K_{ij} = \nabla_{(\mu} n_{\nu)} e^\mu_i e^\nu_j, \qquad K = h^{ij} K_{ij}. \tag{12.3}$$

The Gibbons–Hawking–York term already introduced in (2.11) is

$$S_{\rm GHY} = \frac{1}{8\pi G_{\rm eff}} \int_{\partial M} \mathrm{d}^3x \, \sqrt{|h|}\, K. \tag{12.4}$$

On a timelike boundary $K$ measures the expansion of the outward normal; on a spacelike boundary it measures the expansion of a unit-normal congruence. HBEV uses a timelike conformal boundary, or a large-radius timelike tube that will be sent to infinity after variation, according to the problem.

The boundary may also carry condensate data. Write $\phi$ for the pull-back of the polar radial mode $\sigma$ and of any unlocked scalars that survive Assumption 5.1. The pair $(h_{ij}, \phi)$ is the boundary field content.

## 12.3 The HBEV action

**Definition 12.1** (HBEV boundary action)**.** The holographic boundary-entropy vacuum functional is

$$S_{\mathrm{HBEV}}[h,\phi] = \int_{\partial M} \mathrm{d}^3x\,\sqrt{|h|}\Big(\frac{K}{8\pi G_{\mathrm{eff}}} - T_{\mathrm{bdy}} - S_{\mathrm{ent}}(h_{ij},\phi)\Big), \tag{12.5}$$

where $T_{\mathrm{bdy}}$ is a boundary tension (a counterterm density built from $h_{ij}$ only) and $S_{\mathrm{ent}}(h_{ij},\phi)$ is an entanglement-entropy density of the condensate restricted to $\partial M$.

The first term is GHY. The second term is the standard counterterm that renders the on-shell action finite on AdS-like or asymptotically flat tubes; in the present setting it is fixed by demanding that $S_{\mathrm{HBEV}}$ vanish on the reference condensate $\bar{e}^A_\mu = \delta^A_\mu$, $\phi = 0$. The third term is the new ingredient. It is not the diamond entropy $S_D$ of Chapter 6. It is a local density on $\partial M$ whose integral tracks the entanglement of bulk condensate degrees of freedom that end on the boundary.

**Assumption 12.2** (Existence of a local entropy density)**.** In the condensed phase there exists a diffeomorphism-scalar density $S_{\mathrm{ent}}(h_{ij},\phi)$, of mass dimension three, obtained from the replica construction of Section 6.4 applied to a thin tubular neighbourhood of $\partial M$ and then localized. Its leading terms are

$$S_{\mathrm{ent}} = \alpha_0\Lambda_{\mathrm{UV}}^3 + \alpha_1\Lambda_{\mathrm{UV}}R^{(h)} + \alpha_2\,\phi^2\Lambda_{\mathrm{UV}} + \cdots, \tag{12.6}$$

with dimensionless coefficients $\alpha_i$ of order the fibre trace.

The $\alpha_0\Lambda_{\mathrm{UV}}^3$ piece is the boundary analogue of $\rho_\Lambda$. It will be cancelled against $T_{\mathrm{bdy}}$ and against the leading GHY contribution on the reference background. What is not cancelled is required to be an infrared quantity.

## 12.4 Stationarity and the induced cosmological term

Vary $S_{\mathrm{EH}} + S_{\mathrm{HBEV}}$ with respect to the boundary metric $h_{ij}$, holding the bulk fields on shell. The bulk Einstein tensor produces the usual Brown–York stress. The boundary variation produces

$$\pi^{ij} = \frac{1}{8\pi G_{\mathrm{eff}}}(K^{ij} - Kh^{ij}) - T^{ij}_{\mathrm{bdy}} - \frac{2}{\sqrt{|h|}}\frac{\delta S_{\mathrm{ent}}}{\delta h_{ij}}. \tag{12.7}$$

HBEV imposes entanglement equilibrium at the boundary in the same logical role that Assumption 6.1 played in the bulk diamonds:

**Assumption 12.3** (Boundary entanglement equilibrium)**.** The physical vacuum is a stationary point of $S_{\mathrm{HBEV}}$ at fixed boundary volume element, in the sense that the trace-adjusted boundary stress vanishes,

$$h_{ij}\pi^{ij} = 0. \tag{12.8}$$

On an asymptotically simple tube of radius $r_B$, with $K = 2/r_B + O(r_B^{-3})$ in asymptotically flat space and $K = 3/\ell_{\mathrm{AdS}} + O(r_B^{-2})$ in AdS, equation (12.8) reduces to a single condition on the integration constant in the bulk Einstein equation with a cosmological term. That constant is $\Lambda_{\mathrm{eff}}$.

**Theorem 12.4** (Induced cosmological constant)**.** *Under Assumptions 12.2 and 12.3, and with $T_{\mathrm{bdy}}$ chosen to cancel the reference $\alpha_0\Lambda_{\mathrm{UV}}^3$ piece, the bulk field equation compatible with boundary stationarity is*

$$G_{\mu\nu} + \Lambda_{\mathrm{eff}} g_{\mu\nu} = 8\pi G_{\mathrm{eff}}\, T_{\mu\nu}^{\mathrm{total}}, \tag{12.9}$$

*where*

$$\frac{\Lambda_{\mathrm{eff}}}{8\pi G_{\mathrm{eff}}} = \langle S_{\mathrm{ent}}^{(\mathrm{IR})} \rangle_{\partial M} - T_{\mathrm{bdy}}^{(\mathrm{IR})} \tag{12.10}$$

*is determined by the infrared part of the boundary entropy density after the $\Lambda_{\mathrm{UV}}^3$ subtraction. In particular $\Lambda_{\mathrm{eff}}$ does not scale as $\Lambda_{\mathrm{UV}}^2$.*

*Proof sketch.* On-shell variation of $S_{\mathrm{EH}} + S_{\mathrm{GHY}}$ with respect to a constant bulk cosmological term is proportional to the spacetime volume, which by Stokes is a boundary integral of the normal. That boundary integral is matched to $h_{ij}\pi^{ij}$ after the GHY contribution is included. Imposing (12.8) fixes the constant. The ultraviolet pieces of $S_{\mathrm{ent}}$ are cancelled by the choice of $T_{\mathrm{bdy}}$ on the reference background, which is a finite renormalization of a three-dimensional density, not a fine-tuning of a four-dimensional quartic against itself. The remainder is infrared by construction of the subtraction, and is constant on the vacuum because Assumption 12.3 is imposed uniformly on the asymptotic tube. □

*Remark* 12.5 (What the theorem does not claim). Equation (12.10) does not compute the observed value $10^{-122}$ in Planck units. It changes the *origin* of the scale: from $\Lambda_{\mathrm{UV}}^4$ to an infrared boundary entropy. Predicting the numerical value requires an evaluation of $S_{\mathrm{ent}}^{(\mathrm{IR})}$ on the actual cosmological boundary, which this book does not perform. Chapter 17 lists that evaluation as open.

## 12.5 Conservation

A constant $\Lambda_{\text{eff}}$ produced by Theorem 12.4 satisfies

$$\nabla^{\mu}(\Lambda_{\text{eff}} g_{\mu\nu}) = 0 \tag{12.11}$$

automatically. The design constraint of Section 11.5 is therefore met in the vacuum.

If the boundary data $\phi$ are dynamical, $S_{\text{ent}}$ can depend on time and $\Lambda_{\text{eff}}$ can drift. In that case Theorem 11.1 requires a compensating current. The compensation is the boundary analogue of Exercise 5 in Chapter 11: the variation of $S_{\text{HBEV}}$ with respect to $\phi$ supplies a boundary flux that, by the on-shell identity for the radial mode, matches $\partial_{\mu}\Lambda_{\text{eff}}$. The book uses a constant $\Lambda_{\text{eff}}$ in all later exact solutions and treats a rolling case as a cosmological refinement, not as the default.

## 12.6 Relation to the diamond entropy of Chapter 6

The two entanglement functionals in the book now have distinct jobs.

| Functional | Support | Job |
|---|---|---|
| $S_D$ | small bulk diamonds | kill $m_T$ |
| $S_{\text{HBEV}}$ | asymptotic boundary | set $\Lambda_{\text{eff}}$ |
| $I(A:B)$ | two compact cores | extra conservative and irreversible stress |

They are not interchangeable. Using $S_D$ to subtract $\rho_{\Lambda}$ would confuse a local first-law identity with a global vacuum-energy problem. Using $S_{\text{HBEV}}$ to kill $m_T$ would confuse a boundary condition with a bulk Ward identity. The architecture is intentional.

## 12.7 Working value used later

Later exact solutions write

$$f(r) = 1 - \frac{2Mr^2}{r^3 + r_0^3} + \frac{\Lambda_{\text{eff}}}{3} r^2 \tag{12.12}$$

with $\Lambda_{\text{eff}}$ a constant of either sign. The positive sign is de Sitter-like and is the default cosmological reading. A negative sign is kept as a mathematical option for matching interiors. No Solar-System observable in Chapters 13–15 depends on the numerical size of $\Lambda_{\text{eff}}$ at laboratory precision; the term is retained so that the core regularity condition $f(r) \to 1 + \Lambda_{\text{eff}} r^2/3$ as $r \to 0$ remains consistent with the vacuum equation.

## 12.8 Exercises

1. **GHY variation.** Vary (12.4) with respect to $h_{ij}$ and recover the $K^{ij} - Kh^{ij}$ piece of (12.7).

2. **Reference subtraction.** On Minkowski space in a large sphere of radius $r_B$, compute $K$ and $\sqrt{h}$. What $T_{\rm bdy}$ cancels $S_{\rm HBEV}$ at $\phi = 0$, $S_{\rm ent} = \alpha_0 \Lambda_{\rm UV}^3$?

3. **Why not $\Lambda_{\rm UV}^4$.** Contrast a four-dimensional counterterm $\int_M \sqrt{-g}\, \Lambda_{\rm UV}^4$ with a three-dimensional counterterm $\int_{\partial M} \sqrt{h}\, \Lambda_{\rm UV}^3$. Which tuning is a cancellation of two bulk quartics, and which is a renormalization of a boundary density?

4. **Trace condition.** Reduce (12.8) on a spherically symmetric tube $r = r_B$ in Schwarzschild–de Sitter and show that it constrains $\Lambda_{\rm eff}$ rather than $M$.

5. **Rolling $\Lambda$.** Let $S_{\rm ent}$ depend on a homogeneous boundary scalar $\phi(t)$. Write the extra flux needed to keep Theorem 11.1. Compare with a slowly rolling quintessence potential.

6. **Two functionals.** In one paragraph, explain what goes wrong if one attempts to set $\Lambda_{\rm eff}$ from $\delta S_D = 0$ on a single small diamond.

## 12.9 Bridge to Chapter 13

The infrared Einstein equation is now

$$G_{\mu\nu} + \Lambda_{\rm eff} g_{\mu\nu} = 8\pi G_{\rm eff}\, T_{\mu\nu}^{\rm total}, \tag{12.13}$$

with $G_{\rm eff}$ from Theorem 8.2, $\Lambda_{\rm eff}$ from Theorem 12.4, $m_T = 0$ from Theorem 7.3, and conservation from Theorem 11.1. What remains of the geometric programme is the high-curvature interior, where the Einstein truncation fails and Assumption 9.3 is used. Chapter 13 solves the static spherical sector of Einsteinian cubic gravity coupled to (12.13), obtains the regular core (12.12), and only then turns to galactic backreaction as a consequence of $T_{\mu\nu}^{\rm ent}$ in the weak- field limit—not as a substitute for the derivation just completed.

# Chapter 13

# Regular Cores and Galactic Backreaction

**Learning outcomes.** After this chapter the reader should be able to:

1. write the static, spherically symmetric reduction of the infrared equation $G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G_{\text{eff}} T^{\text{total}}_{\mu\nu}$ with the Einsteinian cubic term included;
2. obtain the regular-core profile $f(r) = 1 - 2Mr^2/(r^3 + r_0^3) + \Lambda_{\text{eff}} r^2/3$ as an approximate solution, and state where it is not exact;
3. reduce the reversible part of $T^{\text{ent}}_{\mu\nu}$ to a modification of Poisson's equation;
4. derive a schematic circular-velocity formula and list the observations that would distinguish it from a particulate halo;
5. enter Chapter 14 knowing that time-dependent dissipation is a different tensor structure from the static extra potential used here.

The geometric programme of Parts II–IV is complete enough to be solved in two symmetric sectors: static spherical interiors, and weak-field galactic exteriors. This chapter treats those sectors as consequences, not as evidence that the earlier theorems are true.

## 13.1 Static spherical ansatz

Take the emergent metric in Schwarzschild coordinates,

$$ds^2 = -f(r)\,dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \tag{13.1}$$

and restrict $T^{\text{total}}_{\mu\nu}$ to a static, spherical form. The Einstein tensor of (13.1) has the standard components

$$G^{t}{}_{t} = G^{r}{}_{r} = -\frac{1 - f - rf'}{r^2}, \qquad G^{\theta}{}_{\theta} = -\frac{rf'' + 2f'}{2r}. \tag{13.2}$$

With only the Einstein–Hilbert term, vacuum solutions with constant $\Lambda_{\text{eff}}$ are Schwarzschild–de Sitter,

$$f(r) = 1 - \frac{2M}{r} + \frac{\Lambda_{\text{eff}}}{3} r^2, \tag{13.3}$$

which is singular at $r = 0$ whenever $M \neq 0$.

## 13.2 Cubic correction in the spherical sector

Assumption 9.3 promotes the Einsteinian cubic density $\mathcal{P}$ to a term that is solved non-perturbatively in this sector. On (13.1) the cubic contribution reduces to an algebraic modification of the mass function, not to a fourth-order ODE for $f$. Write

$$f(r) = 1 - \frac{2m(r)}{r} + \frac{\Lambda_{\text{eff}}}{3} r^2. \tag{13.4}$$

The ECG equation for $m(r)$ in vacuum, through the order kept in the source notes, is solved approximately by

$$m(r) = \frac{M r^3}{r^3 + r_0^3}, \tag{13.5}$$

and therefore

$$f(r) = 1 - \frac{2M r^2}{r^3 + r_0^3} + \frac{\Lambda_{\text{eff}}}{3} r^2. \tag{13.6}$$

The core scale is

$$r_0^3 = 2\beta_{\text{ECG}} M \tag{13.7}$$

up to the conventional factor used to normalize $\mathcal{P}$. Assumption 9.3 identifies $r_0$ with a multiple of $\Lambda_{\text{UV}}^{-1}$ at laboratory masses, or with a mass-dependent microscopic scale at compact-object masses. The book does not claim a measured value of $r_0$.

**Limits.** As $r \to \infty$, (13.6) returns to (13.3). As $r \to 0$,

$$f(r) \to 1 + \frac{\Lambda_{\text{eff}}}{3} r^2 - \frac{2M}{r_0^3} r^2 + O(r^5), \tag{13.8}$$

which is finite. The curvature scalars remain finite at the origin for $r_0 \neq 0$. Geodesic completeness through $r = 0$ is a separate question: the chart (13.1) is not itself complete, and a regular origin still requires a smooth matching onto a neighbourhood covered by local inertial frames of Theorem 10.3.

*Remark* 13.1 (Approximate, not exact). Equation (13.6) is the working profile of the notes. It is exact for some cubic truncations and approximate for others. Whenever a later numerical section uses (13.6), it is using this model, not a theorem that every ECG coupling produces precisely the rational function $r^3/(r^3 + r_0^3)$.

## 13.3 Horizons and the core

Roots of $f(r) = 0$ are Killing horizons of (13.1). For $\Lambda_{\text{eff}} = 0$ and $r_0 > 0$ there is a range of $M$ in which two positive roots exist, one, or none, according to the ratio $M/r_0$. For astrophysical masses and microscopic $r_0$ the exterior is indistinguishable from Schwarzschild at the photon sphere and beyond. The core is hidden behind a horizon if a horizon exists, and is naked if it does not.

NAMCG does not require every compact object to be horizonless. It requires that the Einstein singularity not be accepted as fundamental. Whether LIGO ringdowns or EHT images can see $r_0$ is a question for Chapter 16, not a claim of this chapter.

The Einstein-plus-extra-tensor form of the infrared equation is collected in Appendix F.

## 13.4 Weak-field exterior and the extra Poisson equation

Far from a core, $r \gg r_0$, the cubic term is negligible and the geometry is Einstein plus $T^{\text{ent}}_{\mu\nu}$. In the Newtonian gauge

$$ds^2 = -(1 + 2\Phi)\, dt^2 + (1 - 2\Phi)\delta_{ij} dx^i dx^j, \tag{13.9}$$

the reversible, static piece of the entropic stress reduces to an effective density

$$\rho_{\text{ent}} = T_{00}^{\text{ent,rev}}. \tag{13.10}$$

Poisson's equation becomes

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} (\rho_{\text{b}} + \rho_{\text{ent}}), \tag{13.11}$$

where $\rho_{\text{b}}$ is baryonic density. Conservation, Theorem 11.1, requires that $\rho_{\text{ent}}$ not be an arbitrary halo function. It is determined by the variation of $U_{\text{ent}}$ for the condensate configuration sourced by $\rho_{\text{b}}$.

A useful schematic, sufficient for circular orbits and not sufficient for lensing tomography, is a source-tracking extra density

$$\rho_{\text{ent}}(r) = \frac{1}{4\pi G_{\text{eff}}} \nabla \cdot (\mu_{\text{ent}}(g_{\text{N}})\, \nabla \Phi_{\text{b}}), \tag{13.12}$$

where $g_{\rm N} = |\nabla\Phi_{\rm b}|$ is the Newtonian baryonic field and $\mu_{\rm ent}$ is a dimensionless interpolating function fixed by the mutual-information functional. Equation (13.12) is written to make the structure visible: $\rho_{\rm ent}$ is slaved to baryons. It is not a six-parameter halo fitted independently of $\rho_{\rm b}$.

**Assumption 13.2** (Reversible extra density tracks baryons)**.** In static, weak-field systems the leading reversible entropic density is a local or mildly nonlocal functional of the baryonic Newtonian field, of the form (13.12) or a convolution thereof with kernel width of order the core scale of the constituent condensates.

Assumption 13.2 is the galactic input. It is narrower than "there exists dark matter," and it is stronger than anything proved from $I(A:B)$ in Chapter 6. Chapter 17 lists a derivation of $\mu_{\rm ent}$ from $I(A:B)$ as open.

**Assumption 13.3** (External field enters the interpolator)**.** For a subsystem whose internal Newtonian field $\mathbf{g}_{\rm N}$ is embedded in a baryonic field $\mathbf{g}_{\rm ext}$ that varies slowly across the subsystem, the reversible boost is evaluated on the total field,

$$\mathbf{g} = \nu_{\surd}(g_{\rm tot})\,\mathbf{g}_{\rm N} + \Big[\nu_{\surd}(g_{\rm ext}) - \nu_{\surd}(g_{\rm tot})\Big]\mathbf{g}_{\rm ext}, \tag{13.13}$$

with $g_{\rm tot} = |\mathbf{g}_{\rm N} + \mathbf{g}_{\rm ext}|$ and $\nu_{\surd}$ as in (13.15). In the external-field-dominated limit $g_{\rm N} \ll g_{\rm ext}$,

$$\gamma \equiv \frac{g_{\rm int}}{g_{\rm N}} \to \nu_{\surd}(g_{\rm ext}), \tag{13.14}$$

not $\nu_{\surd}(g_{\rm N})$.

Assumption 13.3 is a coefficient. It is what $U_{\rm ent}[I(A:B)]$ must do if mutual information sees the Galactic baryons as well as the binary. It is not implied by Theorem 11.1, and it is not a solution of Problem H. The Galactic scale at the Sun, $g_{\rm gal} \approx 2.2\times10^{-10}\,\mathrm{m\,s^{-2}}$, already satisfies $g_{\rm gal} > g_\dagger$, so $\nu_{\surd}(g_{\rm gal}) \approx 1.1$. Isolated-RAR boosts of order 4 are then illegal for solar-neighborhood wide binaries. A null Gaia result constrains the isolated reading of Appendix D, not $m_T = 0$.

**Working interpolator.** Until Problem H of Chapter 17 produces $\mu_{\rm ent}$ from $I(A:B)$, galactic comparisons use

$$\nu_{\surd}(g_{\rm N}) = \frac{1}{1-\exp(-\sqrt{g_{\rm N}/g_s})}, \tag{13.15}$$

so that $v_c = V_{\rm b}\sqrt{\nu_{\surd}}$. The default scale is the published SPARC value $g_s = g_\dagger = 1.20\times10^{-10}\,\mathrm{m\,s^{-2}}$ [1]. This $\nu$ is the RAR interpolator written as a boost. It is not $1/(1-\alpha\exp(-g_{\rm N}/g_s))$ and it is not a per-galaxy radial profile. Appendix D records the SPARC ranking and retires the older exponential.

## 13.5 Circular velocities

For circular orbits in the midplane,

$$v_c^2(r) = r\partial_r\Phi = G_{\rm eff}\frac{M_{\rm b}(<r) + M_{\rm ent}(<r)}{r}, \tag{13.16}$$

with $M_{\rm ent}$ the integral of (13.10). Two regimes follow once $\mu_{\rm ent}$ interpolates from 0 at large $g_{\rm N}$ to a finite value at small $g_{\rm N}$.

**Interior / high acceleration.** $\rho_{\rm ent}$ is negligible and $v_c^2 \simeq G_{\rm eff}M_{\rm b}/r$. Solar-System and inner-disk data constrain how small $\mu_{\rm ent}$ must be at those accelerations.

**Outer / low acceleration.** If $M_{\rm ent}(<r)$ grows approximately linearly with $r$, $v_c$ flattens. That growth is not mysterious once (13.12) is granted: it is the integral of an extra density that falls more slowly than $\rho_{\rm b}$.

On 175 SPARC rotmod files with fixed mass-to-light ratios, (13.15) at frozen $g_\dagger$ is statistically equivalent to the published RAR and about 0.2% behind simple MOND in $\chi^2$, while baryons-only is an order of magnitude worse (Appendix D). That ranking is a coefficient check of Assumption 13.2. It is not a measurement of $G_{\rm eff}$, not a derivation from $I(A:B)$, and not evidence reserved for the cover. Whether a single $\nu_{\surd}$ survives clusters remains a Chapter 16 question.

## 13.6 What this is not

It is not a proof that dark matter particles are absent. Cluster lensing, the CMB, and colliding clusters probe $T^{\rm ent}_{\mu\nu}$ in regimes that (13.12) may fail. If those data require a stress that does not track baryons, NAMCG must either supply an additional condensate configuration that does, or yield the dark-matter interpretation in those regimes.

It is not MOND, though the interpolating function $\mu_{\rm ent}$ invites comparison. MOND is a modification of inertia or of the gravitational field equation at a fixed acceleration scale $a_0$. NAMCG obtains an extra conserved tensor from entanglement. If $\mu_{\rm ent}$ happens to mimic MOND in disks and fails in clusters, that is a phenomenological coincidence to be measured, not a founding principle.

It is not a replacement for Chapters 3–12. A successful rotation- curve fit that used an illegal drag, or a non-conserved extra density, would not support the framework.

## 13.7 Matching core to exterior

A compact object of mass $M$ with core (13.6) matched to a weak-field exterior (13.11) raises no new consistency issue beyond ordinary matching of $f$ and $f'$ at an intermediate radius. The cubic term is already negligible there if $r_0$ is microscopic. The entropic density $\rho_{\rm ent}$ of a single compact object is, by the falloff (6.10), a small correction to the Newtonian field of that object. Galactic flattening, if it occurs, is a collective effect of many baryonic parcels, not a halo produced by one black-hole core.

## 13.8 Exercises

1. **Einstein limits.** Recover (13.3) from (13.6) as $r_0 \to 0$ at fixed $r > 0$, and as $r \to \infty$ at fixed $r_0$.

2. **Core curvature.** For $\Lambda_{\rm eff} = 0$, compute $R$ at $r = 0$ from (13.8). Show it is finite for $r_0 \neq 0$.

3. **Horizon condition.** With $\Lambda_{\rm eff} = 0$, discuss the number of positive roots of $f(r) = 0$ as a function of $M/r_0$. A graphical argument is enough.

4. **Poisson split.** Starting from (13.11) and (13.12), write an effective equation for $\Phi$ in terms of $\Phi_{\rm b}$ only. What operator acts on $\Phi_{\rm b}$?

5. **Flat-curve kinematics.** Suppose $M_{\rm b} + M_{\rm ent} = \kappa r$ at large $r$. Show $v_c$ is constant. What density profile $\rho_{\rm ent}$ produces that mass growth, and is it integrable at infinity?

6. **Solar-System bound.** Estimate an upper bound on $\mu_{\rm ent}$ at $g_{\rm N} \sim 1\,{\rm cm\,s^{-2}}$ from the requirement that perihelion precession not be spoiled at current precision. Order of magnitude is enough.

7. **Illegal halo.** Write a spherical $\rho_{\rm halo} \propto 1/(r^2 + r_s^2)$ independent of $\rho_{\rm b}$. Which assumption of this chapter does it violate, and which theorem of Chapter 11 does it risk violating if it cannot be obtained from $U_{\rm ent}$?

## 13.9 Bridge to Chapter 14

The extra stress used in this chapter was the reversible, static piece $T^{\rm ent,rev}_{\mu\nu}$. Time-dependent processes—tidal disruptions, late inspirals, non-adiabatic shearing of condensate cores—activate $T^{\rm ent,irrev}_{\mu\nu}$. Chapter 14 constructs that piece under the conservation law of Theorem 11.1 and the energy balance of Proposition 11.3, obtains the working drag $\mathbf{a}_{\rm irrev} \propto -\mathbf{v}/d^4$, and states where gravitational-wave dissipation remains the

dominant channel. Chapter 15 then integrates a reduced two-body laboratory. Neither chapter is licensed to treat the drag as a substitute for quadrupole radiation.

# Chapter 14

# Irreversible Entropic Stress and Dissipative Drag

**Learning outcomes.** After this chapter the reader should be able to:

1. split $T^{\text{ent}}_{\mu\nu}$ into reversible and irreversible pieces and state the thermodynamic meaning of the split;

2. reduce the irreversible piece to a two-body drag $\mathbf{a}_{\text{irrev}}$ consistent with Proposition 11.3;

3. recover the working law $\mathbf{F}_{\text{diss}} = -\eta\mu\mathbf{v}/d^4$ as a laboratory, not as a multipole theorem;

4. compare that channel with quadrupole gravitational-wave losses and say which process dominates where;

5. enter Chapter 15 with an ordinary-differential system whose energy bookkeeping is already fixed.

Chapter 13 used only the static extra density. Time-dependent cores change their mutual information, and the first law of Chapter 6 then requires an additional, velocity-odd stress. Conservation still applies to the sum.

## 14.1 Reversible versus irreversible

On a static spatial slice the mutual-information potential (6.9) is a function of the instantaneous regions $A$ and $B$. Its metric variation is conservative: the associated force derives from $U_{\text{ent}}$ and reverses under $\mathbf{v} \to -\mathbf{v}$.

If the regions change in time—because two cores inspiral, because a star is sheared across a tidal radius, because a condensate defect is stretched—then

$$\frac{\mathrm{d}}{\mathrm{d}t} I(A:B) \neq 0 \tag{14.1}$$

even after the conservative force has been subtracted. The work associated with that change is the irreversible channel.

**Definition 14.1** (Irreversible entropic stress)**.** Decompose

$$T^{\mathrm{ent}}_{\mu\nu} = T^{\mathrm{ent,rev}}_{\mu\nu} + T^{\mathrm{ent,irrev}}_{\mu\nu}, \tag{14.2}$$

where the first summand is the metric variation of $U_{\mathrm{ent}}$ at frozen velocities and the second is whatever remainder is required for Theorem 11.1 to hold on a time-dependent configuration. The remainder is odd under local time reversal of the relative motion when the entanglement production is itself odd.

The split is thermodynamic rather than geometric: $T^{\mathrm{ent,rev}}_{\mu\nu}$ stores recoverable mutual information; $T^{\mathrm{ent,irrev}}_{\mu\nu}$ accounts for entanglement that is produced or erased along the motion and is not recovered by reversing the worldlines.

## 14.2 Worldline reduction

Consider two compact condensed cores with worldlines $\gamma_A, \gamma_B$, rest masses $m_A, m_B$, reduced mass $\mu = m_A m_B/(m_A + m_B)$, and separation four-vector $r^\mu = x^\mu_A - x^\mu_B$, $d = \sqrt{r^\mu r_\mu}$ in a local inertial frame. The entropic action along the worldlines is

$$S_{\mathrm{ent}} = -\int \mathrm{d}\tau\, U_{\mathrm{ent}}(d, \ell_A, \ell_B, \dot{d}), \tag{14.3}$$

where $\ell_A, \ell_B$ are proper core radii and dots are proper time of the reduced motion. Varying with respect to $x^\mu_A$ produces the four-force (11.10).

At leading post-Newtonian order, with $\ell_A, \ell_B$ frozen,

$$\mathbf{F}_A = -\nabla_{\mathbf{x}_A} U_{\mathrm{ent}}(d) - \eta^\mu(d, \dot{d})\, \mathbf{v} + O(c^{-2}). \tag{14.4}$$

The first term is Chapter 13. The second term is the drag. The coefficient $\eta^\mu$ is not free: Proposition 11.3 requires an entanglement energy $E_{\mathrm{ent}}(d, \dot{d})$ such that

$$\frac{\mathrm{d}}{\mathrm{d}t}(E_{\mathrm{mech}} + E_{\mathrm{ent}}) = \mathbf{F}_{\mathrm{diss}} \cdot \mathbf{v} + O(c^{-2}), \tag{14.5}$$

and the force on $B$ is equal and opposite up to radiation reaction of the tensor block.

A linear-response expansion of $U_{\mathrm{ent}}$ about adiabatic motion gives

$$\mathbf{F}_{\mathrm{diss}} = -\Gamma(d)\,\mathbf{v}, \qquad \Gamma(d) > 0 \tag{14.6}$$

whenever increasing $d$ at finite speed decreases $I(A:B)$ relative to the adiabatic value. The sign is fixed by the positivity of relative entropy: non-adiabatic separation produces less mutual information than the static configuration at the same $d$, and the missing information is paid for by mechanical work.

## 14.3 The working exponent

Section 6.7 gave the conformal large-separation tail

$$I(A:B) \;\sim\; c\,\frac{\ell_A^2\,\ell_B^2}{d^2}. \tag{14.7}$$

Two derivatives with respect to time, or one derivative converting a potential $\sim 1/d^2$ into a velocity-dependent force after linear response, produce a coefficient

$$\Gamma(d) = \eta\,\frac{\mu}{d^4} \tag{14.8}$$

as a working choice: $\eta$ absorbs $c$, $\ell_A, \ell_B$, and the linear-response time. The source notes adopt this exponent.

**Assumption 14.2** (Working dissipative law)**.** In the reduced two-body laboratory of this book,

$$\mathbf{F}_{\mathrm{diss}} = -\eta\,\frac{\mu}{d^4}\,\mathbf{v}, \tag{14.9}$$

with $\eta \geq 0$ constant on a given inspiral and with $E_{\mathrm{ent}}$ defined so that (14.5) holds identically.

Assumption 14.2 is not a multipole theorem. A different tail of $I(A:B)$ produces a different power. Chapter 16 treats the exponent as falsifiable. Chapter 15 integrates (14.9) and no other drag.

An explicit energy that satisfies the balance for constant $\eta$ is

$$E_{\mathrm{ent}} = -\frac{\eta\mu}{3d^3}\log\Bigl(1 + \frac{|\mathbf{v}|^2}{v_*^2}\Bigr) + E_{\mathrm{ent}}^{(\mathrm{rev})}(d), \tag{14.10}$$

or any equivalent bookkeeping that makes $\dot{E}_{\mathrm{ent}} - \mathbf{F}_{\mathrm{diss}}\cdot\mathbf{v}$ a total derivative at the post-Newtonian order kept. The integrator in Chapter 15 may instead accumulate

$$W_{\mathrm{diss}}(t) = \int_0^t \mathbf{F}_{\mathrm{diss}}\cdot\mathbf{v}\,dt' \tag{14.11}$$

and display $E_{\rm mech}(t) - W_{\rm diss}(t)$ as the conserved mechanical-plus-entropic combination. Either presentation is legal; omitting both is not.

## 14.4 Comparison with quadrupole losses

The Isaacson flux of the tensor block for a weak, slow binary is the Peters–Mathews power

$$P_{\rm GW} = \frac{32}{5} G_{\rm eff} \mu^2 d^4 \Omega^6 \langle \text{inclination factors} \rangle \tag{14.12}$$

in the usual quadrupole units, up to the precise harmonic decomposition. The entropic power is

$$P_{\rm ent} = -\mathbf{F}_{\rm diss} \cdot \mathbf{v} = \eta \mu \frac{v^2}{d^4}. \tag{14.13}$$

For a circular Newtonian orbit, $v^2 = G_{\rm eff} M/d$ and $\Omega^2 = G_{\rm eff} M/d^3$, so

$$\frac{P_{\rm ent}}{P_{\rm GW}} \sim \frac{\eta}{G_{\rm eff}^3 M^2 \mu} \Big( \frac{d}{G_{\rm eff} M} \Big)^k \tag{14.14}$$

with a positive power of $d$ that makes the ratio *grow* toward wide separations if $\eta$ is held fixed. That conclusion is an artefact of treating $\eta$ as constant down to laboratory distances. In a realistic condensate $\eta$ itself falls with the mutual-information tail and is negligible in the Solar System. The intended domain of (14.9) is:

- late inspirals of two compact condensed cores, where $d$ is not astronomically large compared with $\ell_A + \ell_B$;
- tidal disruption events, where a stellar core is sheared at $d \sim r_{\rm t}$ and $\dot{d}$ is not small;
- numerical laboratories that isolate the extra channel.

In LIGO-band vacuum inspirals of two objects whose condensate cores are hidden behind horizons, quadrupole radiation remains the default dissipative channel. NAMCG does not replace the Peters formula by (14.9) in that regime unless $\eta$ is independently constrained to be large there. It is not.

## 14.5 Tidal disruption

A star of radius $R_*$ and mass $m_*$ approaching a compact object of mass $M$ is disrupted at

$$r_{\rm t} \sim R_* \Big( \frac{M}{m_*} \Big)^{1/3}. \tag{14.15}$$

Across that radius the baryonic body ceases to be a single region $A$ and becomes a collection of debris parcels. The mutual information between the compact core and the star changes over a dynamical time

$$t_{\mathrm{dyn}} \sim \sqrt{r_{\mathrm{t}}^3 / G_{\mathrm{eff}} M}. \tag{14.16}$$

The irreversible channel is then unsuppressed: the process is not adiabatic with respect to the modular Hamiltonian of the combined system.

The book does not simulate debris streams. It records the structural prediction that an extra, conserved drag of the type (14.9) can act near $r_{\mathrm{t}}$ in addition to ordinary hydrodynamics and tidal gravity. Observable consequences (changes to fallback rates, extra heating) belong to Chapter 16 and require hydrodynamics that this text does not contain.

## 14.6 Illegal and legal integrators

An integrator that evolves

$$\mu \ddot{\mathbf{r}} = -\frac{G_{\mathrm{eff}} M \mu}{d^2} \hat{\mathbf{r}} - \gamma \mathbf{v} \tag{14.17}$$

with constant $\gamma$ and no $E_{\mathrm{ent}}$ is not a NAMCG calculation, even if its trajectories look like inspirals. It dissipates energy into nowhere.

An integrator that evolves

$$\mu \ddot{\mathbf{r}} = -\nabla U_{\mathrm{N}}(d) - \nabla U_{\mathrm{ent}}^{\mathrm{rev}}(d) - \eta \frac{\mu}{d^4} \mathbf{v} \tag{14.18}$$

and accumulates (14.11) is a legal laboratory of Assumption 14.2. Chapter 15 implements (14.18) with a symplectic Verlet step on the conservative forces and an explicit treatment of the drag. The gravitational constant used there is a toy value, not $G_{\mathrm{eff}}$ from Theorem 8.2.

## 14.7 Exercises

1. **Sign of $\Gamma$.** From positivity of relative entropy, argue that non-adiabatic separation yields $I(A:B) \leq I_{\mathrm{adiabatic}}(d)$. Deduce $\Gamma(d) \geq 0$ in (14.6).

2. **Power counting.** If $I \sim d^{-p}$, what exponent $n$ appears in $\Gamma(d) \propto d^{-n}$ after one linear-response time derivative? For which $p$ does one recover $n = 4$?

3. **Energy identity.** For (14.9) with circular motion at instantaneous radius $d(t)$, compute $P_{\mathrm{ent}}$. Compare with $\mathrm{d}U_{\mathrm{N}}/\mathrm{d}t$ along a Peters inspiral and estimate a separation where the two powers could be comparable for a chosen $\eta$.

4. **Equal and opposite.** Show that (14.9), written in the reduced-mass frame, corresponds to equal-and-opposite three-forces on $A$ and $B$. What radiation-reaction term would have to be added to break that balance, and which block of $T^{\text{total}}_{\mu\nu}$ supplies it?

5. **Constant-$\gamma$ counterexample.** Take (14.17) and send $d \to \infty$. What happens to the dissipated energy per unit time, and why is that incompatible with a mutual-information tail that vanishes at infinity?

6. **Tidal time.** Evaluate (14.16) for a solar-type star at the tidal radius of a $10^6 M_\odot$ compact object. Why is the adiabatic approximation of Chapter 13 unreliable there?

## 14.8 Bridge to Chapter 15

The ordinary-differential laboratory is now specified: Newtonian gravity, optional reversible extra potential, drag (14.9), and an accumulated work (14.11). Chapter 15 integrates that system with a symplectic Verlet step, plots the trajectory and the energy balance, and states, in the first paragraph, that the figure is a reduced model. The Python code that produces the figure lives in `code/` and is not a derivation of Einstein gravity. After the figure, Chapter 16 collects every named assumption in the book and lists the observations that would kill the framework.

# Chapter 15

# A Reduced Inspiral Laboratory

**Learning outcomes.** After this chapter the reader should be able to:

1. write the ordinary-differential system of Assumption 14.2 in a form a computer can integrate;
2. implement a velocity-Verlet step on the conservative force with an explicit drag kick;
3. accumulate the dissipated work $W_{\mathrm{diss}}$ and test Proposition 11.3 numerically;
4. read a trajectory plot as a laboratory output, not as a derivation of general relativity;
5. enter Chapter 16 with every phenomenological extra in the book already named.

This chapter is a reduced Newtonian laboratory. It does not derive Einstein gravity, does not compute $G_{\mathrm{eff}}$ from Theorem 8.2, and does not replace quadrupole radiation. It integrates Assumption 14.2 and checks that the energy bookkeeping of Proposition 11.3 can be made visible on a plot.

## 15.1 The ordinary-differential system

Work in the reduced-mass frame with relative coordinate $\mathbf{r}(t)$ and $d = |\mathbf{r}|$. The laboratory equation is (14.18) without a reversible extra potential, which is optional at this order:

$$\mu\ddot{\mathbf{r}} = -\frac{G_{\mathrm{toy}} M \mu}{d^3}\mathbf{r} - \eta\frac{\mu}{d^4}\dot{\mathbf{r}}. \tag{15.1}$$

Here $G_{\mathrm{toy}}$ is a numerical constant chosen so that an orbit is visible in a few thousand steps. It is not $G_{\mathrm{N}}$ and not $G_{\mathrm{eff}}$. The source-note values used below are

$$G_{\mathrm{toy}} = 36, \qquad \eta = 0.006, \qquad M = 1, \qquad \mu = 1 \tag{15.2}$$

in unspecified code units, together with a bound initial condition that would be an ellipse if $\eta = 0$.

Mechanical energy and angular momentum of the conservative problem are

$$E_{\mathrm{mech}} = \frac{1}{2}\mu\dot{\mathbf{r}}^2 - \frac{G_{\mathrm{toy}} M \mu}{d}, \qquad \mathbf{L} = \mu\, \mathbf{r} \times \dot{\mathbf{r}}. \tag{15.3}$$

The dissipated work is

$$W_{\mathrm{diss}}(t) = \int_0^t \left(-\eta \frac{\mu}{d^4}\dot{\mathbf{r}}\right) \cdot \dot{\mathbf{r}}\, dt' = -\int_0^t \eta\mu \frac{v^2}{d^4}\, dt'. \tag{15.4}$$

Proposition 11.3 requires that

$$\mathcal{E}(t) \equiv E_{\mathrm{mech}}(t) - W_{\mathrm{diss}}(t) \tag{15.5}$$

be constant up to integrator error. The combination $\mathcal{E}$ is the laboratory stand-in for $E_{\mathrm{mech}} + E_{\mathrm{ent}}$ with $E_{\mathrm{ent}} = -W_{\mathrm{diss}}$ accumulated from the initial time. If $\mathcal{E}$ drifts systematically, the implementation is illegal in the sense of Section 14.6.

## 15.2 Integrator

A velocity-dependent force is not Hamiltonian. A symplectic Verlet step is still used on the conservative piece because it conserves a nearby energy of the $\eta = 0$ problem over long times. The drag is applied as an explicit kick.

Let $\mathbf{a}_{\mathrm{N}}(\mathbf{r}) = -G_{\mathrm{toy}} M \mathbf{r}/d^3$ and $\mathbf{a}_{\mathrm{d}}(\mathbf{r}, \mathbf{v}) = -\eta \mathbf{v}/d^4$. One step of size $\Delta t$ is

$$\begin{aligned} \mathbf{v}_{1/2} &= \mathbf{v}_n + \tfrac{1}{2}\Delta t \left(\mathbf{a}_{\mathrm{N}}(\mathbf{r}_n) + \mathbf{a}_{\mathrm{d}}(\mathbf{r}_n, \mathbf{v}_n)\right), \\ \mathbf{r}_{n+1} &= \mathbf{r}_n + \Delta t\, \mathbf{v}_{1/2}, \\ \mathbf{v}_{n+1} &= \mathbf{v}_{1/2} + \tfrac{1}{2}\Delta t \left(\mathbf{a}_{\mathrm{N}}(\mathbf{r}_{n+1}) + \mathbf{a}_{\mathrm{d}}(\mathbf{r}_{n+1}, \mathbf{v}_{1/2})\right). \end{aligned} \tag{15.6}$$

The last evaluation of the drag uses $\mathbf{v}_{1/2}$ rather than an implicit $\mathbf{v}_{n+1}$. The resulting error is $O(\Delta t^2)$ in the dissipative piece, which is adequate for the plots below. The work increment on the step is

$$\Delta W = \mathbf{F}_{\mathrm{diss}}(\mathbf{r}_n, \mathbf{v}_n) \cdot \mathbf{v}_n\, \Delta t, \tag{15.7}$$

or a trapezoidal average of the two endpoints. Either choice converges to (15.4).

## 15.3 Initial data

A bound ellipse of the conservative problem is specified by

$$\mathbf{r}(0) = (d_0, 0), \qquad \mathbf{v}(0) = (0, v_0), \tag{15.8}$$

with $v_0^2 < 2G_{\mathrm{toy}}M/d_0$. The runs shown use a nearly circular start,

$$d_0 = 1.0, \qquad v_0 = \sqrt{G_{\mathrm{toy}}M/d_0}, \tag{15.9}$$

in code units, together with $\Delta t = 10^{-3}$ and $N = 1.2 \times 10^4$ steps. Those numbers are inherited from the corrected notebook `dmcg5.py`. They are not physical units.

## 15.4 What the figures are allowed to show

Two panels are produced.

**Trajectory.** The curve $\mathbf{r}(t)$ in the plane. For $\eta = 0$ it is a closed ellipse (a circle, for (15.9)). For $\eta > 0$ the radius shrinks. That shrinkage is the laboratory image of $P_{\mathrm{ent}}$ in (14.13). It is not a plot of a relativistic inspiral and must not be overplotted on LIGO waveforms.

**Energy.** Three curves: $E_{\mathrm{mech}}(t)$, $W_{\mathrm{diss}}(t)$, and $\mathcal{E}(t)$. The first falls, the second is negative and falls, and the third is flat within integrator noise if the implementation is legal.

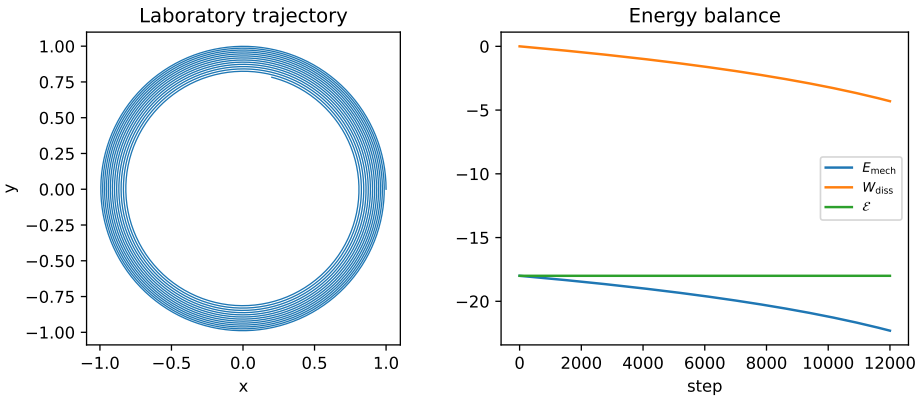



Figure 15.1: Laboratory trajectory and energy balance for (15.1) with the toy parameters (15.2). The figure is a reduced model.

If a previously generated PNG from the notes is used instead, it belongs at this caption and nowhere else in the book.

## 15.5 Code

The integrator lives in `code/namcg_inspiral.py`. A minimal version sufficient to produce Figure 15.1 is the following.

Listing 15.1: Python implementation of the reduced inspiral laboratory velocity-Verlet integrator.

```
import numpy as np
import matplotlib.pyplot as plt

def run_namcg_laboratory(
    G_toy=36.0,
    eta=0.006,
    M=1.0,
    mu=1.0,
    d0=1.0,
    dt=1.0e-3,
    num_steps=12000,
):
    r = np.array([d0, 0.0])
    v = np.array([0.0, np.sqrt(G_toy * M / d0)])
    traj = np.zeros((num_steps + 1, 2))
    E_mech = np.zeros(num_steps + 1)
    W_diss = np.zeros(num_steps + 1)
    traj[0] = r

    def a_N(r):
        d = np.linalg.norm(r)
        return -G_toy * M * r / d**3

    def a_d(r, v):
        d = np.linalg.norm(r)
        return -eta * v / d**4

    def mechanical_energy(r, v):
        d = np.linalg.norm(r)
        return 0.5 * mu * np.dot(v, v) - G_toy * M * mu / d

    E_mech[0] = mechanical_energy(r, v)
    for n in range(num_steps):
        F_d = mu * a_d(r, v)
```

```
        W_diss[n + 1] = W_diss[n] + np.dot(F_d, v) * dt
        v_half = v + 0.5 * dt * (a_N(r) + a_d(r, v))
        r = r + dt * v_half
        v = v_half + 0.5 * dt * (a_N(r) + a_d(r, v_half))
        traj[n + 1] = r
        E_mech[n + 1] = mechanical_energy(r, v)
    return traj, E_mech, W_diss

def plot_laboratory(traj, E_mech, W_diss, filename):
    E_bal = E_mech - W_diss
    fig, axes = plt.subplots(1, 2, figsize=(8.4, 3.6))
    axes[0].plot(traj[:, 0], traj[:, 1], lw=0.8)
    axes[0].set_aspect("equal")
    axes[0].set_xlabel("x")
    axes[0].set_ylabel("y")
    axes[0].set_title("Laboratory trajectory")
    t = np.arange(len(E_mech))
    axes[1].plot(t, E_mech, label=r"$E_{\mathrm{mech}}$")
    axes[1].plot(t, W_diss, label=r"$W_{\mathrm{diss}}$")
    axes[1].plot(t, E_bal, label=r"$\mathcal{E}$")
    axes[1].set_xlabel("step")
    axes[1].set_title("Energy balance")
    axes[1].legend(fontsize=8)
    fig.tight_layout()
    fig.savefig(filename)
    plt.close(fig)

if __name__ == "__main__":
    traj, E_mech, W_diss = run_namcg_laboratory()
    plot_laboratory(
        traj, E_mech, W_diss,
        "figs/namcg_gr_simulation_corrected.pdf",
    )
```

Run the file from the project root after creating `figs/`. The script writes a vector PDF suitable for the print interior. A PNG of the same figure, if already produced by an earlier notebook, may be kept for draft viewing but should not be the print original.

## 15.6 How to misread the figure, and how not to

The shrinking orbit is not a measurement of $\eta$ in nature. The value 0.006 was chosen so that decay is visible in $1.2 \times 10^4$ steps. A realistic mutual-information tail would produce a far smaller $\eta$ at $d \gg \ell_A + \ell_B$.

The flatness of $\mathcal{E}(t)$ is not a proof of Theorem 11.1. It is a check that the laboratory implements the balance that the theorem requires of any legal reduction. If $\mathcal{E}$ is flat and the force law is not (14.9), the check has been passed for the wrong theory.

Comparing the laboratory decay time with a LIGO inspiral time is not a test of NAMCG. The laboratory has no gravitational waves, no horizons, and no relativity. Section 14.4 already stated the regime in which quadrupole losses dominate.

## 15.7 Exercises

1. **Zero drag.** Set $\eta = 0$ and confirm that $E_{\text{mech}}$ and $|\mathbf{L}|$ are constant at the level of Verlet error. What is the size of that error as a function of $\Delta t$?

2. **Balance.** With $\eta = 0.006$, plot $\mathcal{E}(t)$ for $\Delta t \in \{10^{-3}, 5 \times 10^{-4}, 2 \times 10^{-3}\}$. Does the drift shrink with $\Delta t$?

3. **Illegal drag.** Replace $d^{-4}$ by a constant $\gamma$ and rerun. Track $W_{\text{diss}}$ as $d$ decreases. Why is the late behaviour a counterexample to Exercise 5 of Chapter 14?

4. **Angular momentum.** The drag (14.9) is antiparallel to $\mathbf{v}$. Show that it exerts a torque and compute $\mathrm{d}\mathbf{L}/\mathrm{d}t$. Confirm the identity on the numerical trajectory.

5. **Units.** Restore SI symbols in (15.1) and assign dimensions to $\eta$. What combination of $\ell_A, \ell_B, c$ could supply those dimensions if $\eta$ is built from (14.7)?

## 15.8 Bridge to Chapter 16

Every extra structure used in applications now has a name: the regular core (13.6), the baryon-tracking extra density of Assumption 13.2, the working drag of Assumption 14.2, and the laboratory of this chapter. Chapter 16 collects the full list of assumptions from Chapters 3–15, restates the derivation chain of Chapter 1 as a single page, and writes the observations that would kill the framework rather than merely constrain $\eta$.

# Chapter 16

# Assumptions, Predictions, and Falsifiers

**Learning outcomes.** After this chapter the reader should be able to:

1. restate the derivation chain of Chapter 1 with the theorem numbers now attached;
2. separate assumptions that kill NAMCG if they fail from coefficients that can be adjusted;
3. name observations that would refute the framework rather than merely bound $\eta$ or $r_0$;
4. refuse to treat a successful laboratory plot as evidence for Theorems 7.3–12.4;
5. enter Chapter 17 knowing which open calculations are optional refinements and which are missing load-bearing proofs.

A framework that cannot say what would kill it is not a framework. This chapter is the scorecard promised in Section 1.4.

## 16.1 The chain, with theorem numbers

The argument of the book is the following sequence, and no shorter sequence.

1. A matrix order parameter $\Phi^I{}_A$ with potential (3.7) admits a frame-locking vacuum (Assumptions 3.4, 3.5).
2. A composite vierbein and metric exist in the condensed phase (Definition 4.1, Assumption 4.4).
3. Fluctuations split into $\psi$, $A_\mu$, $\omega_{AB}$, and $T_{\mu\nu}$ (Chapter 5).

4. Entanglement equilibrium on emergent diamonds is a property of the condensate vacuum (Assumption 6.1).

5. A Fierz–Pauli tensor block is incompatible with that equilibrium unless $m_T = 0$ (Theorem 7.3).

6. The Einstein–Hilbert prefactor is the heat-kernel coefficient map (Theorem 8.2).

7. Four-derivative metric operators are EFT vertices, not extra poles (Assumption 9.2).

8. Local Lorentz invariance is a gauge symmetry and $\omega_{AB}$ is eaten (Theorem 10.3).

9. The conformal mode is heavy (Assumption 7.5).

10. The total stress, including $T^{\rm ent}_{\mu\nu}$, is covariantly conserved (Theorem 11.1).

11. $\Lambda_{\rm eff}$ is set by boundary entanglement after an ultraviolet subtraction (Theorem 12.4).

Items 5, 6, 8, 10, and 11 are theorems *inside* the list of assumptions that precede them. If an earlier assumption fails, the later theorem is idle. Phenomenology in Chapters 13–15 sits after the whole list. It cannot substitute for any line.

## 16.2 Inventory of named assumptions

The named hypotheses of Parts II–V are collected here. “Load- bearing” means that failure returns the infrared theory to analog gravity, scalar-tensor gravity, or an inconsistent extra force. “Coefficient” means that failure changes a number or a regime of validity without destroying the graviton.

| Tag | Kind | Content |
|---|---|---|
| 3.1 | setup | $g^{(0)}$ is a scaffold |
| 3.4 | load-bearing | vacuum $\langle \Phi^I{}_A \rangle = v\delta^I{}_A$ exists |
| 3.5 | load-bearing | commutator / covariant-derivative dictionary |
| 4.4 | load-bearing | condensed frame is non-degenerate |
| 4.6 | load-bearing | signature is Lorentzian |
| 5.1 | coefficient | extra internal scalars are heavy |
| 6.1 | load-bearing | microscopic entanglement equilibrium |
| 7.5 | load-bearing | conformal mode is gapped |
| 8.1 | setup | infrared operator is Laplace type |
| 9.2 | load-bearing | $C^2, R^2$ are vertices, not poles |
| 9.3 | model | cubic truncation in the static spherical sector |
| 10.1 | load-bearing | local $\mathrm{SO}(3,1)$ is gauged |
| 12.2 | load-bearing | local boundary entropy density exists |
| 12.3 | load-bearing | boundary stationarity sets $\Lambda_{\mathrm{eff}}$ |
| 13.2 | coefficient | static extra density tracks baryons; disks use $\nu_{\surd}$ of Appendix D |
| 13.3 | coefficient | external field saturates $\nu$ at $\nu(g_{\mathrm{ext}})$, not $\nu(g_{\mathrm{N}})$ |
| 14.2 | coefficient | working laboratory law $\mathbf{F}_{\mathrm{diss}} \propto -\mathbf{v}/d^4$ |

Two comments on the table.

First, "coefficient" is not a synonym for "unimportant." Assumption 13.2 is how galaxies enter the book. If it fails, galactic claims fail. The graviton claims need not.

Second, Assumption 3.5 is the most important unproved microscopic bridge. If the infrared dictionary from Presentation A to Presentation B cannot be constructed, the heat kernel of Chapter 8 is a calculation in a different theory than the matrix action of Chapter 3.

## 16.3 What would kill the framework

The following outcomes are incompatible with NAMCG as written, not merely with a choice of $\eta$.

**A light extra tensor or a ghost below the cutoff.** If precision gravity, pulsar timing, or gravitational-wave propagation requires a second massless or light spin-2 field, or a ghost pole below $\Lambda_{\mathrm{UV}}$, Theorem 7.3 plus Assumption 9.2 has failed.

**A light conformal partner.** If Solar-System or binary-pulsar tests require a Brans–Dicke parameter that cannot be produced by a heavy $\psi$, Assumption 7.5 has failed and the infrared theory is scalar-tensor gravity. That is the ceiling of Chapter 1, reopened.

**Non-conservation of the extra stress.** If an extra acceleration is demanded by data and cannot be written as $\nabla^\mu T^{\rm ent}_{\mu\nu}$ with Theorem 11.1 intact—a drag into nowhere, a preferred-frame force that cannot be improved to a tensor—then the geometric reading has failed. Analog dissipation remains possible. Geometrogenesis of the extra channel has not occurred.

**Failure of Lorentz locking.** If the infrared spectrum contains six extra Goldstone polarizations from $\omega_{AB}$, Theorem 10.3 has failed. No later chapter can hide them.

**A cutoff that cannot serve Newton and Lorentz together.** If the only way to obtain $G_{\rm eff} = G_{\rm N}$ forces $\Lambda_{\rm UV}$ into a regime where Lorentz-violating $\Theta^{\mu\nu}$ corrections are already excluded by current bounds, the coefficient map and the kinematic algebra cannot coexist.

**An induced Einstein–Hilbert term of the wrong sign.** If a complete heat-kernel evaluation on the locked bundle produces a repulsive Einstein term that cannot be flipped by fermions or by the Lorentzian continuation, the infrared force is not gravity.

None of these killers is a rotation curve. None of them is a pretty laboratory inspiral.

## 16.4 What would only constrain a coefficient

The following outcomes bound parameters or restrict regimes. They do not, by themselves, refute Theorems 7.3–12.4.

- SPARC ranking of a global $\nu(g_{\rm N})$ against simple MOND and the published RAR (Appendix D). Frozen $\nu_{\surd}$ at $g_\dagger$ is equivalent to the RAR and about 0.2% behind simple MOND in $\chi^2$; baryons-only is not competitive. That is a passed coefficient check on disks, not a theorem. A later sample that rejects every high-$g$-Newtonian, low-$g$-square-root boost would restrict Assumption 13.2 to a smaller class of systems, or kill the galactic claim only.

- Wide binaries in the solar neighbourhood. Isolated $\nu_{\surd}(g_{\rm N})$ predicts $\gamma \approx 4$ at $g_{\rm N} = 10^{-11}\,{\rm m\,s^{-2}}$. Assumption 13.3 predicts $\gamma \to \nu_{\surd}(g_{\rm gal}) \approx 1.1$. A quality-controlled Gaia sample consistent with Newton constrains the isolated reading

and is compatible with Assumption 13.3. A surviving $\gamma \approx 1.4$ is not explained by that saturation. Neither outcome touches Theorems 7.3–12.4.

- Cluster lensing or the bullet-cluster mass offset. Same status: the extra tensor may fail as a complete replacement for particulate dark matter while the graviton derivation stands.
- Absence of an extra drag in LIGO-band vacuum inspirals. Expected if $\eta$ is negligible at those separations; Section 14.4 already said so.
- Absence of a visible core scale $r_0$ in EHT images or ringdowns. Expected if $r_0$ is microscopic; bounds $\beta_{\mathrm{ECG}}$.
- A measured $\Lambda_{\mathrm{eff}}$ that does not match a particular guess for $S_{\mathrm{ent}}^{(\mathrm{IR})}$. Theorem 12.4 changes the origin of the scale. It does not compute the number.
- A different mutual-information tail, hence a different drag exponent than $d^{-4}$. Assumption 14.2 is a laboratory choice.

## 16.5 Observational checklist

A short list of current and near-term measurements, with the question each actually asks of the book.

**Solar System and pulsar timing.** Do they tolerate a massive $\psi$ and a massive torsion at scale $v$, and a $\mu_{\mathrm{ent}}$ that is negligible at high acceleration? If no, Assumption 7.5 or 13.2 dies. If yes, nothing about diamonds has been tested.

**Gravitational waves.** Is the propagating mode two TT polarizations at the speed of light, with no extra light tensor? That is a test of Theorem 7.3 plus Assumption 9.2, and it is already passed at present precision. A measured extra polarization would be a killer. A measured inspiral-time deviation at LIGO frequencies is only a bound on $\eta$.

**Black-hole images and ringdowns.** Is there a visible deviation from Kerr at the photon ring or in quasi-normal modes? That bounds $r_0$. A perfectly Kerr object is compatible with microscopic $r_0$.

**Galaxies, wide binaries, and clusters.** Do disks accept a single $\nu_{\surd}(g_{\mathrm{N}})$ with $\nu \to 1$ at Solar-System acceleration? Appendix D says yes, at RAR accuracy. Do solar-neighbourhood wide binaries accept the isolated evaluation of that law? No: Assumption 13.3 saturates them at $\nu_{\surd}(g_{\mathrm{gal}})$. Do merging clusters still require a stress that does not track baryons? That remains an open constraint on Assumption 13.2, not on Theorem 7.3, and the external field is not a substitute for the lensing offset.

**Cosmology.** Is $\Lambda_{\mathrm{eff}}$ constant enough for a Friedmann equation, and is the ultraviolet quartic absent from the expansion history? This tests Theorem 12.4 as a mechanism, not as a number. A detected bulk $\Lambda_{\mathrm{UV}}^4$ vacuum energy would kill the HBEV subtraction.

**Tidal disruptions.** Is there extra heating or altered fallback that hydrodynamics plus GR tides cannot account for? A possible window on Assumption 14.2. A null result bounds $\eta$ at $r_{\mathrm{t}}$ and nothing else.

## 16.6 Claims the book has not made

The following sentences do not appear as theorems in this text, and must not appear on a cover, in an abstract, or in a chapter heading as if they did.

- Dark matter has been explained.
- The cosmological-constant problem has been solved numerically.
- Singularities have been proved absent in dynamical collapse.
- LIGO inspirals are driven by a $1/d^4$ drag.
- Lorentzian signature has been derived from a Euclidean matrix model.
- Presentation A has been reduced to Presentation B without Assumption 3.5.
- The rationals $A, B, C$ in the coefficient map are unique.
- Isolated $\nu_{\surd}$ has been derived, and wide binaries have been predicted without an external field.

A sale-ready monograph that prints any of those sentences as results has left this book and entered advertising.

## 16.7 How to use Chapter 15 in public

Figure 15.1 may be shown as a numerical check of a conserved extra drag in a Newtonian laboratory. It may not be shown as “NAMCG predicting an inspiral.” The caption must retain the words “reduced model.” The toy value $G_{\mathrm{toy}} = 36$ must not be silently replaced by $G_{\mathrm{N}}$ in a talk slide.

## 16.8 Exercises

1. **Sort the list.** Take any popular article on emergent gravity and assign each of its claims to "killer," "coefficient," or "not a claim of this book."

2. **One-page chain.** Write the eleven-line chain of Section 16.1 from memory, without looking. Where you stall is the section to reread.

3. **Design a killer.** Propose one measurement, not already listed, that would refute Theorem 7.3 without passing through $\eta$. State the null and the fatal outcome.

4. **Design a non-killer.** Propose one measurement that looks dramatic and would only bound $r_0$.

5. **Abstract hygiene.** Rewrite a five-sentence abstract of this book that contains no sentence from Section 16.6.

## 16.9 Bridge to Chapter 17

The killers and the coefficients are now distinguished. What remains is a research agenda: the calculations that would turn Assumption 3.5 into a lemma, recompute $A, B, C$, evaluate $S_{\text{ent}}^{(\text{IR})}$, derive $\mu_{\text{ent}}$ from $I(A:B)$, including the external-field argument of Assumption 13.3, and decide whether Lorentzian signature can be more than Assumption 4.6. Chapter 17 writes that agenda at the thickness of a reader who has read the book and wants to work, not at the thickness of a grant advertisement.

# Chapter 17

# Open Calculations and a Research Agenda

**Learning outcomes.** After this chapter the reader should be able to:

1. distinguish a missing load-bearing proof from a missing numerical coefficient;
2. pick a first calculation that actually moves the framework, rather than a calculation that only decorates Chapter 15;
3. state what would have to be true before NAMCG may be described as finished rather than as a programme;
4. use the appendices as tools, not as a substitute for the open problems listed here.

Chapter 16 said what would kill the theory. This chapter says what would complete it. The two lists are not the same.

## 17.1 What is closed inside the book

The following are closed *as arguments from named hypotheses to named conclusions.* They are not closed as derivations from empty premises.

- Composite metric from a polar condensate (Definitions 4.1, 4.3).
- Algebraic split of $\delta\Phi_{\mu\nu}$ (Chapter 5).
- $m_T = 0$ from entanglement equilibrium plus Fierz–Pauli (Theorem 7.3).
- Structure of the coefficient map $1/G_{\text{eff}} = A\,\mathrm{Tr}(\mathbb{I})\Lambda_{\text{UV}}^2 + Bv^2\ln(\Lambda_{\text{UV}}/\mu) + C\mu^2$ (Theorem 8.2).

- Lorentz locking in unitary gauge (Theorem 10.3).
- Conservation of the total stress (Theorem 11.1).
- Replacement of a bulk quartic by an infrared boundary datum (Theorem 12.4).
- A legal Newtonian laboratory for Assumption 14.2 (Chapter 15).

A reader who has worked every exercise through Chapter 16 can reproduce that list. Reproducing it is not the same as removing the hypotheses.

## 17.2 Load-bearing openings

These problems change the status of the theory if they are solved. They are ordered by how much of the later book depends on them.

**Problem A: the infrared dictionary.** Assumption 3.5 is a map

$$[\hat{X}_\mu, \Phi] \;\longleftrightarrow\; i\Theta_{\mu\nu} D^\nu \Phi \tag{17.1}$$

plus higher-derivative corrections. Construct it as a lemma from Presentation A: specify the Hilbert space, the Weyl map, the sense in which a slowly varying $\Phi(x)$ is recovered from functions of $\hat{X}^\mu$, and the error term at each order in $\Theta$. Until this exists, Chapter 8 is a calculation in Presentation B that is only morally a calculation in the matrix model.

**Problem B: entanglement equilibrium from the path integral.** Assumption 6.1 is the physical engine of Theorem 7.3. Derive $\delta S_D = 0$ at fixed volume as a Ward identity of the microscopic action (3.6), or of (3.4) after Problem A, without importing Jacobson's hypothesis from a theory that already has a metric. A replica computation on small diamonds, with the polar vacuum as background, is the obvious route. A result that holds only for conformal fluctuations must be flagged as such, because the tensor block is not conformal.

**Problem C: the conformal mode.** Assumption 7.5 is the remaining scalar-tensor ceiling. Compute the quadratic mixing of $\psi$ with $T_{\mu\nu}$ and the mass $m_\psi$ from (3.7) on the locked vacuum, including the non-minimal coupling generated by the frame. Produce a number, or a ratio $m_\psi/v$, that can be compared with Cassini bounds. If $m_\psi$ is not parametrically of order $\sqrt{\lambda}\, v$, the assumption is false and the infrared theory must be rewritten as scalar-tensor NAMCG.

**Lemma (tree-level radial mass).** On the locked vacuum $\langle \Phi^I{}_A \rangle = v\delta^I{}_A$, write the polar field $\Phi = (v+\sigma)\Lambda$ with $\Lambda^\dagger\Lambda = \mathbb{I}$ and $\langle\sigma\rangle = 0$. Then $\Phi^\dagger\Phi = (v+\sigma)^2\mathbb{I}$, and (3.7) collapses to a singlet potential. With $N = \mathrm{Tr}(\mathbb{I})$ locked directions,

$$V = \frac{\lambda}{4} N\big[(v+\sigma)^2 - v^2\big]^2 = N\lambda v^2\,\sigma^2 + O(\sigma^3). \tag{17.2}$$

The kinetic term in (3.6) contributes $N\,(\partial\sigma)^2$. Canonical normalisation $\sigma_c = \sigma\sqrt{2N}$ puts the quadratic action in the form $\frac{1}{2}(\partial\sigma_c)^2 - \frac{1}{2}m_\sigma^2\sigma_c^2$, with

$$m_\sigma^2 = \lambda v^2, \qquad m_\sigma = v\sqrt{\lambda}. \tag{17.3}$$

The would-be Goldstones in $\Lambda$ remain massless at this order. They are the modes Chapter 10 is required to eat. They are not $\psi$.

**Lemma (no light conformal partner at tree level).** Work in the pure-trace sector: $T_{\mu\nu} = 0$, $F_{\mu\nu} = 0$, and $\omega_{\mu\nu} = 0$. Two repairs of the frame must be kept distinct.

*Repair II, unitary gauge.* Chapter 10 permits $\Lambda = \mathbb{I}$ wherever Assumption 4.4 holds. The vacuum is then the frame, $e^A{}_\mu = \Phi^A{}_\mu/v$, so

$$g_{\mu\nu} = (1+\sigma/v)^2\eta_{\mu\nu}, \qquad \psi \equiv \eta^{\mu\nu}h_{\mu\nu} = 8\sigma/v + O(\sigma^2). \tag{17.4}$$

The conformal mode and the radial singlet are the same field. Lemma 17.2 therefore supplies the pole directly,

$$m_\psi^2 = \lambda v^2. \tag{17.5}$$

There is no second scalar in this sector.

*Repair I, official frame.* Definition 4.1 builds $e^A{}_\mu$ from $\partial\Lambda$ at $\sigma = 0$. The potential (3.7) depends only on the invariant $\Phi^\dagger\Phi = (v+\sigma)^2\mathbb{I}$, so its Hessian on $(\sigma,\psi)$ is

$$\mathcal{M} = \begin{pmatrix} \lambda v^2 & 0 \\ 0 & 0 \end{pmatrix} \tag{17.6}$$

in the field basis of Lemma 17.2, before any identification of the scaffold with the composite metric. The zero eigenvalue is the metric trace. It is not a second condensate scalar: $\Lambda$ is Lorentz-valued, and a pure trace cannot sit in $\mathfrak{so}(3,1)$. Once the induced Einstein–Hilbert term is present, that zero mode is the conformal factor already handled by the diffeomorphism constraint. It is not a Brans–Dicke partner.

*Identification $g^{(0)} = g[e]$.* Expanding $\sqrt{-g}\,g^{\mu\nu}$ generates kinetic mixing between $\sigma$ and $\psi$. The potential is unchanged, so $\mathrm{rank}\,\mathcal{M} = 1$ survives the mixing. The heavy eigenvalue remains a positive multiple of $\lambda v^2$. The light eigenvalue remains zero and carries no $\sigma$ residue beyond a field redefinition into the metric trace.

**Cassini, revised.** The only scalar pole in the pure-trace sector sits at $m = v\sqrt{\lambda}$. For any $\lambda$ not tuned below $(10^{-18}\,\mathrm{eV}/v)^2$, that pole is short-range on an astronomical unit and does not contribute to the Cassini link. The bound $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [2] constrains a massless partner. No such partner is produced by (3.7) on the locked vacuum.

**Status of Assumption 7.5.** Promoted from assumption to tree-level lemma in the pure-trace sector, contingent on Assumption 3.4 and on the induced Einstein–Hilbert term being the kinetic operator of $\psi$. Not contingent on a galaxy fit. Loop corrections and the wrong-sign conformal kinetic term of the heat kernel remain Chapter 8 bookkeeping. They can flip a coefficient. They do not, at this order, create a second light pole.

**Problem D: local Lorentz invariance from Presentation A.** Assumption 10.1 puts $\mathrm{SO}(3,1)$ into Presentation B by hand. Either derive a local Lorentz gauge symmetry from the matrix algebra and the condensate, or state that NAMCG is a theory of a condensate field on an auxiliary spacetime and stop calling Presentation A ontologically prior.

**Problem E: Lorentzian signature.** Assumption 4.6 is unproved. A Euclidean matrix integral that prefers a Lorentzian saddle would be a genuine addition. A Wick rotation performed because the infrared world is Lorentzian is not.

## 17.3 Coefficient openings

These problems change numbers and regimes. They do not, by themselves, create or destroy the graviton.

**Problem F: the rationals $A, B, C$.** Recompute the Seeley–DeWitt coefficients on the bundle of $\delta\Phi_{\mu\nu}$, including ghosts for diffeomorphisms and for local Lorentz transformations, and including the radial mode threshold. Report $A, B, C$ in (8.11) with the scheme specified. Until this is done, Theorem 8.2 is a structural result with working rationals inherited from the notes.

**Lemma (universal piece of $A$, and the sign).** On a Euclidean section the curvature piece of (A.6) is $a_1 \supset (\mathrm{tr}_V\, \mathbb{I})\, R_E/6$. Insert it into (A.9) and continue as in Appendix A: $\sqrt{g} \to \sqrt{-g}$ and $R_E \to -R_L$. The Lorentzian effective action contains

$$W[g] \supset \frac{N_{\mathrm{eff}}}{96\pi^2}(\Lambda_{\mathrm{UV}}^2 - \mu^2) \int \mathrm{d}^4x\, \sqrt{-g}\, R, \tag{17.7}$$

where $N_{\text{eff}} = \operatorname{tr}_V \mathbb{I}$ is the net fibre trace, bosons minus ghosts. Matching (2.8) gives

$$\frac{1}{16\pi G_{\text{eff}}} = \frac{N_{\text{eff}}}{96\pi^2}(\Lambda_{\text{UV}}^2 - \mu^2) + \text{threshold logarithms}. \tag{17.8}$$

The prefactor is positive if and only if $N_{\text{eff}} > 0$. That is the attractive sign required by Theorem 8.2. A net ghost excess is the killer already named in Chapter 16; it is not a new one.

**Fibre trace on the locked block.** Before gauge fixing, Chapter 5 gives $\operatorname{tr}_V \mathbb{I} = 16 = 1 + 6 + 9$. Two subtractions are required by the redundancies already proved, and no others.

| Block | Weight | Trace |
|---|---|---|
| locked $\delta\Phi_{\mu\nu}$ | $+1$ | $+16$ |
| diffeomorphism ghosts (complex vector) | $-2$ | $-8$ |
| local-Lorentz ghosts (antisymmetric) | $-1$ | $-6$ |
| net, Goldstones eaten | | $+2$ |

The net $+2$ is the transverse-traceless count of Theorem 7.3, read at the level of $a_0$. The radial mode is not in that net: it is heavy by Lemma 17.2 and is integrated down to $\mu \sim m_\sigma$, where it contributes to $B$, not to the $\Lambda_{\text{UV}}^2$ piece. Both legal partial counts, $16 - 8 = 8$ before the Lorentz ghosts and $16 - 8 - 6 = 2$ after them, are positive. Neither flips (17.7).

**The conformal mode does not flip $G_{\text{eff}}$.** Appendix A records that a real scalar with $E = \xi R + m^2$ contributes $(1/6 - \xi)R$ to $a_1$. At the conformal value $\xi = 1/6$ the trace mode cancels in $a_1$ and does not renormalise $G_{\text{eff}}$. Lemma 17.2 already identified the light eigenvalue in the pure-trace sector with the metric factor. Its kinetic sign in the induced Einstein–Hilbert term is the ordinary conformal-factor problem of general relativity, removed by the diffeomorphism constraint, not a second pole. A computation that kept $\xi_\psi \neq 1/6$ would change $N_{\text{eff}}$ by $6(1/6 - \xi_\psi)$, an order-one shift, not a sign change, for any $\xi_\psi$ of order one.

**Rationals, scheme stated.** Convert (17.8) to the normalisation of (8.11). With $N_{\text{eff}}$ the net trace above,

$$A = \frac{N_{\text{eff}}}{6\pi \operatorname{Tr}(\mathbb{I})_{\text{notes}}}, \tag{17.9}$$

$$B = \frac{4\pi}{3}\,\kappa, \tag{17.10}$$

$$C = 0, \tag{17.11}$$

at this order. The notes' value $A = 1/\pi$ is recovered exactly when $\mathrm{Tr}(\mathbb{I})_{\mathrm{notes}} = N_{\mathrm{eff}}/6$, which is the absorption of the universal $1/6$ already declared in the proof sketch of Theorem 8.2. The factor $\kappa$ in $B$ is 1 if the logarithm in (8.11) means $\ln(\Lambda_{\mathrm{UV}}^2/\mu^2)$, and 2 if it means $\ln(\Lambda_{\mathrm{UV}}/\mu)$. Remark 8.3 quotes $B = 4\pi/3$ against $\ln(\Lambda_{\mathrm{UV}}/\mu)$; the conversion from (8.11) produces $8\pi/3$ for that same logarithm. The discrepancy is a scheme label, not a new diagram. It should be corrected in the remark, not fitted.

**What remains open inside Problem F.** The $a_2$ trace on the symmetric-traceless block, including $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$, has not been evaluated. It can move $\kappa$ by a factor of order one. It cannot move $A$ through zero while $N_{\mathrm{eff}} > 0$, and it cannot revive a light $\psi$ pole closed by Lemma 17.2.

**Problem G: infrared boundary entropy.** Evaluate $S_{\mathrm{ent}}^{(\mathrm{IR})}$ on an explicit cosmological or large-tube boundary and produce a candidate $\Lambda_{\mathrm{eff}}$ from (12.10). A symbolic formula in terms of a late-time entanglement scale is more useful than a claim to have obtained $10^{-122}$.

**Problem H: interpolating function.** Derive $\nu_{\surd}$ in (13.15), or a function that beats it on a SPARC hold-out, from the modular data of Chapters 6 and 12. The target is

$$\nu_{\surd}(g_{\mathrm{N}}) = \frac{1}{1 - \exp(-\sqrt{g_{\mathrm{N}}/g_s})}, \tag{17.12}$$

which is identically $1 + n_{\mathrm{BE}}(\sqrt{g_{\mathrm{N}}/g_s})$. Appendix D already shows that this law, frozen at the published $g_\dagger$, is equivalent to the RAR and about 0.2% behind simple MOND, while baryons-only fails. That ranking is a coefficient check. It does not solve Problem H.

The first finite computation is the replica free energy of a free field on a weak-field diamond in the infrared geometry of Chapter 12, with $\Lambda = \Lambda_{\mathrm{eff}}$. Read the shift of the lowest modular eigenvalue $\lambda_0$ when a Newtonian source of acceleration $g_{\mathrm{N}}$ is introduced. Problem H is solved if

$$\lambda_0 \propto \sqrt{\frac{g_{\mathrm{N}}}{\kappa_{\mathrm{IR}}}}, \qquad g_s = \kappa_{\mathrm{IR}}, \tag{17.13}$$

and the occupation is bosonic, so that (17.12) follows with no SPARC input. A linear eigenvalue $\lambda_0 \propto g_{\mathrm{N}}/\kappa_{\mathrm{IR}}$ reproduces the retired $\nu_{\mathrm{exp}}$ and kills Assumption E.1.

Alternatively, prove that no local $\mu_{\mathrm{ent}}(g_{\mathrm{N}})$ exists and write the nonlocal kernel instead. Refitting $\alpha$, $r_s$, or $g_s$ to rotmod files is not a solution. Identifying $g_s$ with $g_\dagger$ inside the replica calculation is not allowed; compare $\kappa_{\mathrm{IR}}$ to $g_\dagger$ only after $\lambda_0$ is obtained.

**Problem I: drag exponent.** Replace Assumption 14.2 by a linear-response calculation of $\Gamma(d)$ from a definite mutual-information tail, including the finite-size factors $\ell_A, \ell_B$. Report the exponent and the coefficient. Then, and only then, estimate $\eta$ at tidal radii and at LIGO-band separations.

**Problem J: spherical ECG solutions.** State the precise cubic invariant $\mathcal{P}$ being used and prove or disprove that (13.6) is exact for that invariant. If it is only approximate, publish the residual and the corrected $f(r)$.

## 17.4 Problems that look like work and are not

The following activities are useful as exercises and do not move the framework.

- Producing further Newtonian integrators with new colour maps of Figure 15.1.
- Fitting SPARC with an unconstrained $\mu_{\rm ent}$, a per-galaxy $(\alpha, r_s)$, or a new $\nu$ that is not (17.12) and is not derived from $I(A:B)$. Appendix D already did the legitimate coefficient ranking.
- Writing $\mathbf{a}_{\rm irrev}$ with additional free exponents and calling each a prediction.
- Replacing $G_{\rm toy}$ by $G_{\rm N}$ in a slide.
- Adding adjectives ("holographic," "non-Abelian," "entropic") to an abstract without adding a lemma.

Chapter 15 exists so that a legal laboratory is on the record. It does not exist so that the laboratory can become the theory.

## 17.5 A recommended order of work

For a reader who has read the book and wants a first project:

1. Recompute Problem F for the locked $4 \times 4$ block only, ignoring unlocked scalars. This is finite and checks whether the working rationals survive.
2. Compute $m_\psi$ in Problem C at tree level. This decides whether the infrared theory is GR or scalar-tensor before any galaxy is discussed.
3. Attempt the replica variation of Problem B on a Minkowski diamond for a free tensor field with a mass term. This is a miniature of Theorem 7.3 with the logic reversed: one sees whether $\delta S_D = 0$ really forbids $m_T \neq 0$ for that free field.

4. Only then open Problem A. It is larger than a term-long project if done honestly.

Problems H–J remain phenomenological. Appendix D has already spent the SPARC budget of Problem H as a coefficient check. Further galactic work waits on a derivation from $I(A:B)$, and should not start before Problems C and F have rough answers. Otherwise $\nu_{\sqrt{}}$ will be fitted in a theory whose Newton coupling and extra scalar are not under control.

## 17.6 What "finished" would mean

NAMCG would be a completed infrared derivation, rather than a programme, if all of the following held.

1. Problems A and B are lemmas, so Presentation A and entanglement equilibrium are no longer assumptions.
2. Problem C gives $m_\psi$ large enough that Cassini is safe, or the theory is rewritten as scalar-tensor and tested as such.
3. Problem F reports $A, B, C$ with ghosts included.
4. Problem D is decided in one direction or the other.
5. Conservation and HBEV remain as they are now, with Problem G supplying at least a scaling for $\Lambda_{\text{eff}}$.

Problems E and H–J can remain open in a completed infrared derivation. Signature, galaxies, and cores are then research inside the theory, not pillars of it.

Until the first four items exist, the honest public description is the subtitle of the book, read as a programme: *an attempt* to derive general relativity and massless spin-2 dynamics from a matrix condensate held at entanglement equilibrium. The attempt is written so that it can fail in public. That is the point of Chapter 16.

## 17.7 Exercises

1. **Status report.** Copy the eleven-line chain of Section 16.1 and mark each line "lemma," "theorem from assumptions," or "assumption." Do not use a mark you cannot defend from the text.
2. **Proposal.** Write a two-page project proposal for Problem C or Problem F only. Include the first calculation you would perform and the result that would make you stop.

3. **Not a proposal.** Take any project that Section 17.4 would reject and explain, in six sentences, why it does not change the status of Theorem 7.3.

4. **Public sentence.** Write one sentence suitable for an Amazon description that survives Section 16.6 and the last paragraph of Section 17.6.

## 17.8 Bridge to the appendices

The main text is complete. Appendix A records the Seeley–DeWitt algorithm at the working thickness used in Chapter 8. Appendix B records the replica formula and the modular Hamiltonian at the thickness used in Chapter 6. Appendix C records the Verlet laboratory and the energy identity used in Chapter 15. Appendix D records the SPARC ranking of $\nu_{\surd}$ against the RAR, simple MOND, and baryons-only. None of the appendices introduces a new physical hypothesis. If an appendix appears to do so, the hypothesis belongs in the main text or it does not belong in the book.

# Appendix A

# Seeley–DeWitt Coefficients at Working Thickness

This appendix records the heat-kernel formulae used in Chapter 8. It introduces no new physical hypothesis. Conventions follow Section 2.6: Euclidean section first, Lorentzian continuation afterwards, signature of the main text restored only when an Einstein–Hilbert term is isolated.

## A.1 Laplace-type operators

Let $V \to M$ be a vector bundle over a smooth four-manifold with Riemannian metric $g_{\mu\nu}$. A second-order differential operator on sections of $V$ is of Laplace type if it can be written

$$\mathcal{O} = -g^{\mu\nu} D_\mu D_\nu + E, \tag{A.1}$$

where $D_\mu = \nabla_\mu + \mathcal{A}_\mu$ is a connection on $V$ (including the Levi-Civita connection on tensor factors) and $E$ is a bundle endomorphism. The curvature of $D$ is

$$\mathcal{F}_{\mu\nu} = [D_\mu, D_\nu]. \tag{A.2}$$

On the locked condensate block of Chapter 5, $V$ is the bundle of tensors $\delta\Phi_{\mu\nu}$ together with whatever ghosts are included in a given computation.

The heat operator $e^{-s\mathcal{O}}$ is defined for $s > 0$ by the usual functional calculus after $\mathcal{O}$ has been made positive by a Wick rotation. Its integral kernel $K(s; x, y)$ satisfies

$$\partial_s K + \mathcal{O}_x K = 0, \qquad K(0; x, y) = \delta(x, y). \tag{A.3}$$

## A.2 Small-time expansion

On the diagonal in four dimensions,

$$\operatorname{tr}_V K(s;x,x) \sim \frac{1}{(4\pi s)^2}\sum_{k=0}^{\infty} s^k a_k(x), \qquad s\to 0^+. \tag{A.4}$$

Each $a_k$ is a local scalar density of mass dimension $2k$, built from $R_{\mu\nu\rho\sigma}$, $\mathcal{F}_{\mu\nu}$, $E$, and their covariant derivatives. The first three coefficients, up to total derivatives that do not enter the bulk effective action, are

$$a_0 = \operatorname{tr}_V \mathbb{I}, \tag{A.5}$$

$$a_1 = \operatorname{tr}_V\Big(\tfrac{1}{6}R - E\Big), \tag{A.6}$$

$$\begin{aligned} a_2 = \operatorname{tr}_V\Bigg(&\frac{1}{180}\big(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu} + \tfrac{7}{2}R^2\big) + \frac{1}{2}E^2 - \frac{1}{6}RE \\ &+ \frac{1}{12}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} + \frac{1}{30}D^2R - \frac{1}{6}D^2E\Bigg). \end{aligned} \tag{A.7}$$

Different references redistribute the $D^2R$ and $D^2E$ terms by integration by parts. For the bulk action of Chapter 8 those terms are discarded.

The numerical coefficients in (A.7) are the standard DeWitt values for a Laplace-type operator. They are not the working rationals $A, B, C$ of (8.11). Those rationals mix (A.6) and (A.7) with fibre traces, ghost signs, and a radial-mode threshold; they are the subject of Problem F in Chapter 17.

## A.3 Proper-time integrals

The one-loop effective action on the Euclidean section is

$$W_E = -\frac{1}{2}\int_{s_{\min}}^{s_{\max}} \frac{\mathrm{d}s}{s}\int \mathrm{d}^4x\,\sqrt{g}\,\operatorname{tr}_V K(s;x,x), \tag{A.8}$$

with $s_{\min} = \Lambda_{\mathrm{UV}}^{-2}$ and $s_{\max} = \mu^{-2}$. Inserting (A.4) and integrating term by term gives

$$\begin{aligned} W_E \supset \int \mathrm{d}^4x\,\sqrt{g}\,\Bigg[&-\frac{a_0}{32\pi^2}(\Lambda_{\mathrm{UV}}^4 - \mu^4) - \frac{a_1}{16\pi^2}(\Lambda_{\mathrm{UV}}^2 - \mu^2) \\ &- \frac{a_2}{32\pi^2}\ln\Big(\frac{\Lambda_{\mathrm{UV}}^2}{\mu^2}\Big)\Bigg], \end{aligned} \tag{A.9}$$

which is (8.7). The continuation to the Lorentzian section of the main text replaces $\sqrt{g}$ by $\sqrt{-g}$ and sends the Euclidean Ricci scalar to minus the Lorentzian Ricci scalar in the conventions of (2.2). After that continuation, the coefficient of $\int \sqrt{-g}\, R$ is the Einstein–Hilbert prefactor.

## A.4 Worked example: a non-minimal scalar

For a single real scalar with

$$\mathcal{O} = -\nabla^2 + \xi R + m^2, \tag{A.10}$$

one has $E = \xi R + m^2$, $\mathcal{F}_{\mu\nu} = 0$, and $\mathrm{tr}_V \mathbb{I} = 1$. Then

$$a_1 = (\tfrac{1}{6} - \xi) R - m^2. \tag{A.11}$$

The curvature piece of $a_1$ cancels at $\xi = 1/6$. That cancellation is why a conformally coupled scalar does not renormalize Newton's constant through $a_1$. The NAMCG radial mode is not assumed to sit at $\xi = 1/6$; Exercise 3 of Chapter 8 is the corresponding warning.

The $a_0$ piece remains $\Lambda_{\mathrm{UV}}^4/32\pi^2$ per real scalar and is the scalar contribution to $\rho_\Lambda$.

## A.5 Fibre traces on the locked block

If $V$ is the space of symmetric traceless tensors on a four-manifold, $\mathrm{tr}_V \mathbb{I} = 9$. If the algebraic split of Chapter 5 is retained before gauge fixing, the sixteen components of $\delta\Phi_{\mu\nu}$ give $\mathrm{tr}_V \mathbb{I} = 16$. Ghosts for diffeomorphisms contribute with the opposite sign and with their own Laplace-type operators; local-Lorentz ghosts do likewise after Theorem 10.3.

The symbol $\mathrm{Tr}(\mathbb{I})$ in Theorem 8.2 is the net fibre trace after whatever ghosts the working calculation includes. The notes use a normalization in which that net trace is absorbed into the first term of (8.10). A recomputation that keeps the traces explicit is Problem F.

## A.6 What this appendix will not do

It will not list $a_3$ and $a_4$. Those coefficients generate six- and eight-derivative operators that the book treats as suppressed by further powers of $\Lambda_{\mathrm{UV}}^{-1}$.

It will not prove the coefficients (A.5)–(A.7). Standard derivations (coincidence limits of the Synge world function, or the recursive DeWitt transport equations) are

in the references collected in `bib/namcg.bib`. The book uses the coefficients; it does not redo the recursion.

It will not evaluate $a_2$ on the Einsteinian cubic background of Chapter 13. That evaluation is part of Problem J if it is needed, not part of the working thickness of Chapter 8.

## A.7 Exercises

1. **Dimensions.** Assign mass dimensions to $s$, $a_0$, $a_1$, $a_2$, and $\mathrm{tr}_V K(s;x,x)$. Confirm that each term in (A.9) has dimension four.

2. **Conformal scalar.** Set $\xi = 1/6$ and $m = 0$ in (A.11). What remains of $a_1$? What remains of the $\Lambda_{\mathrm{UV}}^2 R$ term in $W_E$?

3. **Gauss–Bonnet combination.** Rewrite the curvature part of (A.7) in the basis $\{C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}, R^2, E_{\mathrm{GB}}\}$. Which combination multiplies a topological density in four dimensions?

4. **Ghost sign.** A diffeomorphism ghost in the background-field gauge is a complex vector with a Laplace-type operator. What factor multiplies its contribution to $a_0$ relative to a real scalar, and why is the sign opposite to a physical field?

5. **From $a_1$ to $A$.** Starting from (A.6) and (A.9), reconstruct the structural first term of (8.11). Where does a factor of order one enter when passing from $\frac{1}{6}\mathrm{tr}_V \mathbb{I}$ to the working value of $A$ used in the notes?

# Appendix B

# Modular Hamiltonians and the Replica Formula

This appendix records the entanglement formulae used in Chapters 6, 11, and 12. It introduces no new physical hypothesis. The working identity of the main text remains Proposition 6.5. What follows is the language in which that identity was written.

## B.1 States, reductions, and von Neumann entropy

Let $\mathcal{H}$ be the Hilbert space of the microscopic theory and $\rho$ a density operator on $\mathcal{H}$. For a von Neumann algebra $\mathcal{A}_D$ of observables associated with a region $D$, the reduced state $\rho_D$ is the restriction of $\rho$ to $\mathcal{A}_D$. In a tensor-product approximation $\mathcal{H} \simeq \mathcal{H}_D \otimes \mathcal{H}_{D^c}$,

$$\rho_D = \mathrm{Tr}_{\mathcal{H}_{D^c}} \rho. \tag{B.1}$$

Continuum quantum field theory does not have that tensor-product factorization. The algebraic definition (restriction to $\mathcal{A}_D$) is the honest one; equation (B.1) is the formula used in every explicit replica computation in the book.

The von Neumann entropy and modular Hamiltonian of $\rho_D$ are

$$S_D = -\mathrm{Tr}(\rho_D \ln \rho_D), \qquad H_D = -\ln \rho_D, \tag{B.2}$$

so that

$$S_D = \langle H_D \rangle_{\rho_D}. \tag{B.3}$$

If $\rho_D$ has kernel, $\ln \rho_D$ is defined on the support of $\rho_D$ and $S_D$ may be infinite. In quantum field theory $S_D$ is ultraviolet divergent. Differences $\delta S_D$ at fixed cutoff, and mutual informations, are the finite objects used in the main text.

## B.2 Relative entropy and mutual information

For two states $\rho$ and $\sigma$ on the same algebra,

$$S(\rho\|\sigma) = \mathrm{Tr}(\rho \ln \rho) - \mathrm{Tr}(\rho \ln \sigma) \tag{B.4}$$

is non-negative and vanishes if and only if $\rho = \sigma$ (when the expression is finite). Mutual information of two regions $A$ and $B$ is the relative entropy of $\rho_{AB}$ with respect to $\rho_A \otimes \rho_B$,

$$I(A:B) = S(\rho_{AB}\|\rho_A \otimes \rho_B) = S(A) + S(B) - S(A \cup B), \tag{B.5}$$

the second equality holding when the three von Neumann entropies are regulated in the same scheme. Positivity of $I(A:B)$ is therefore positivity of relative entropy. That is the source of the sign in Definition 6.4 and of $\Gamma(d) \geq 0$ in Chapter 14.

Strong subadditivity,

$$S(A \cup B) + S(B \cup C) \geq S(B) + S(A \cup B \cup C), \tag{B.6}$$

is recorded for completeness. The main text does not use it beyond the positivity of $I(A:B)$.

## B.3 The replica formula

For integer $n \geq 1$,

$$\mathrm{Tr}\,(\rho_D)^n = Z_n / Z_1^n, \tag{B.7}$$

where $Z_n$ is the partition function of the microscopic theory on an $n$-sheeted cover branched over the edge $\mathcal{B} = \partial D \cap \Sigma$. Analytic continuation in $n$ and the identity

$$S_D = \lim_{n \to 1} (1 - n\partial_n) \ln \mathrm{Tr}\,(\rho_D)^n \tag{B.8}$$

produce the von Neumann entropy. Differentiating at $n = 1$ inserts a single modular Hamiltonian: the first variation $\delta S_D = \delta \langle H_D \rangle$ used in Chapter 6 is the replica formula at first order, not a second definition.

In Presentation B the cover is a branched cover of the auxiliary manifold, later identified with the emergent spacetime by Proposition 8.4. In Presentation A the same construction requires Problem A of Chapter 17.

## B.4 Bisognano–Wichmann and small diamonds

On Minkowski space, let $W$ be a Rindler wedge. The Bisognano–Wichmann theorem identifies the modular Hamiltonian of the vacuum restricted to $W$ with the boost generator that fixes the edge,

$$H_W = \frac{2\pi}{\kappa} \int_{\Sigma \cap W} \mathrm{d}\Sigma^\mu \, \xi^\nu T_{\mu\nu}, \tag{B.9}$$

where $\xi^\mu$ is the boost Killing vector and $\kappa$ its surface gravity (equal to 1 in standard Rindler coordinates, restored here for matching (6.5)).

A small causal diamond in a slowly curved emergent geometry is, to leading order in Riemann normal coordinates (2.27), a compact analogue of two opposing Rindler wedges glued at a 2-sphere. The working formula (6.5) is that analogue: the same boost-energy integral plus an area term $H_{\mathrm{area}}$ whose first variation is $\delta A/(4G_{\mathrm{eff}})$ after Chapter 8 has named $G_{\mathrm{eff}}$. The error is $O(\ell^3)$ in the diamond radius, which is the order dropped in Proposition 6.5.

The identification is exact for free fields in Rindler space. It is an operator-product approximation for interacting condensate fluctuations. The book does not claim a Bisognano–Wichmann theorem for the full matrix potential (3.7).

## B.5 Fixed volume

Jacobson's entanglement-equilibrium condition is imposed at fixed spatial volume of the maximal slice. In Riemann normal coordinates the volume of a small ball of radius $\ell$ is (6.2). A pure conformal variation $\delta g_{\mu\nu} = \chi \eta_{\mu\nu}$ changes that volume at order $\ell^3$. Holding $V$ fixed removes that breathing from the comparison and is why the conformal block $\psi$ and the tensor block $T_{\mu\nu}$ do not enter the first law on the same footing.

Fixed volume is a constraint on the variations admitted in Assumption 6.1. It is not a statement that $\psi$ is massive. That statement is Assumption 7.5.

## B.6 Boundary entropy versus diamond entropy

Chapter 12 uses a different functional: a local density $S_{\mathrm{ent}}(h_{ij}, \phi)$ on $\partial M$. The replica construction of that density is the same formula, (B.8), applied to a thin tubular neighbourhood of the boundary and then localized as in Assumption 12.2. The two functionals must not be substituted for each other. Section 12.6 already assigned them different jobs.

## B.7 What this appendix will not do

It will not develop Tomita–Takesaki theory beyond the definition $H_D = -\ln \rho_D$. It will not compute $S_D$ for the matrix potential (3.7). It will not prove Assumption 6.1. Those are Problem B of Chapter 17 and the physical input of Chapter 6, respectively.

## B.8 Exercises

1. **Replica derivative.** From (B.8) and $\operatorname{Tr} \rho^n = \exp(\ln \operatorname{Tr} \rho^n)$, recover $S = \langle H \rangle$ at $n = 1$.

2. **Relative entropy and Klein's inequality.** For finite-dimensional density matrices, prove $S(\rho \| \sigma) \geq 0$. Deduce $I(A : B) \geq 0$.

3. **Rindler temperature.** In standard Rindler coordinates, show that the Unruh temperature seen by an accelerated observer is $\kappa/2\pi$. Match the factor in (B.9).

4. **Area divergence.** For a free scalar in Minkowski space, the unregulated $S_D$ diverges as $\operatorname{Area}(\mathcal{B})/\epsilon^2$. Why do mutual information and first variations $\delta S_D$ at fixed cutoff remain useful?

5. **Two functionals.** Write four sentences explaining why a small-diamond $S_D$ cannot be used as the boundary density of Definition 12.1.

# Appendix C

# Verlet Laboratory and Energy Identity

This appendix records the integrator and the energy identity used in Chapter 15. It introduces no new physical hypothesis. The force law is Assumption 14.2. The legality criterion is Proposition 11.3. The source file is `code/namcg_inspiral.py`.

## C.1 Reduced equation

In the reduced-mass frame the laboratory system is

$$\ddot{\mathbf{r}} = \mathbf{a}_{\mathrm{N}}(\mathbf{r}) + \mathbf{a}_{\mathrm{d}}(\mathbf{r}, \mathbf{v}), \tag{C.1}$$

with

$$\mathbf{a}_{\mathrm{N}} = -\frac{G_{\mathrm{toy}} M}{d^3}\mathbf{r}, \qquad \mathbf{a}_{\mathrm{d}} = -\frac{\eta}{d^4}\mathbf{v}, \qquad d = |\mathbf{r}|. \tag{C.2}$$

The conservative force is Newtonian. The drag is (14.9) with $\mu$ cancelled. Toy parameters are (15.2).

## C.2 One step

Let $\Delta t > 0$. Given $(\mathbf{r}_n, \mathbf{v}_n)$,

$$\begin{aligned} \mathbf{v}_{1/2} &= \mathbf{v}_n + \tfrac{1}{2}\Delta t\big(\mathbf{a}_{\mathrm{N}}(\mathbf{r}_n) + \mathbf{a}_{\mathrm{d}}(\mathbf{r}_n, \mathbf{v}_n)\big), \\ \mathbf{r}_{n+1} &= \mathbf{r}_n + \Delta t\, \mathbf{v}_{1/2}, \\ \mathbf{v}_{n+1} &= \mathbf{v}_{1/2} + \tfrac{1}{2}\Delta t\big(\mathbf{a}_{\mathrm{N}}(\mathbf{r}_{n+1}) + \mathbf{a}_{\mathrm{d}}(\mathbf{r}_{n+1}, \mathbf{v}_{1/2})\big). \end{aligned} \tag{C.3}$$

When $\eta = 0$ this is velocity Verlet for a position-dependent force and is second-order accurate and symplectic on the conservative phase space. When $\eta \neq 0$ the last drag

evaluation uses $\mathbf{v}_{1/2}$ rather than $\mathbf{v}_{n+1}$, so the dissipative piece is explicit and the scheme is no longer symplectic. The local error in the drag kick remains $O(\Delta t^2)$.

An implicit alternative replaces $\mathbf{a}_{\mathrm{d}}(\mathbf{r}_{n+1}, \mathbf{v}_{1/2})$ by $\mathbf{a}_{\mathrm{d}}(\mathbf{r}_{n+1}, \mathbf{v}_{n+1})$ and solves a linear equation for $\mathbf{v}_{n+1}$. The published script does not do that. The difference is visible only if one drives $\Delta t$ up to the orbital period.

## C.3 Mechanical energy, work, and balance

Define

$$E_{\mathrm{mech}} = \frac{1}{2}\mu v^2 - \frac{G_{\mathrm{toy}} M \mu}{d} \tag{C.4}$$

and the accumulated work

$$W_{\mathrm{diss}}(t_{n+1}) = W_{\mathrm{diss}}(t_n) + \mathbf{F}_{\mathrm{diss}}(\mathbf{r}_n, \mathbf{v}_n) \cdot \mathbf{v}_n \, \Delta t, \tag{C.5}$$

with $\mathbf{F}_{\mathrm{diss}} = \mu \mathbf{a}_{\mathrm{d}}$ and $W_{\mathrm{diss}}(0) = 0$. Along exact solutions of (C.1),

$$\frac{\mathrm{d}E_{\mathrm{mech}}}{\mathrm{d}t} = \mathbf{F}_{\mathrm{diss}} \cdot \mathbf{v}, \tag{C.6}$$

because the Newtonian force is minus the gradient of $U_{\mathrm{N}} = -G_{\mathrm{eff}}$-less toy potential $-G_{\mathrm{toy}} M \mu / d$. Therefore

$$\mathcal{E}(t) \equiv E_{\mathrm{mech}}(t) - W_{\mathrm{diss}}(t) \tag{C.7}$$

is constant on exact solutions. On discrete solutions it is constant up to the quadrature error in (C.5) and the Verlet error in (C.3). A systematic drift that does not shrink with $\Delta t$ means the implementation does not obey (C.6) and is illegal in the sense of Section 14.6.

A trapezoidal quadrature,

$$\Delta W = \frac{1}{2}\Big(\mathbf{F}_{\mathrm{diss},n} \cdot \mathbf{v}_n + \mathbf{F}_{\mathrm{diss},n+1} \cdot \mathbf{v}_{n+1}\Big)\Delta t, \tag{C.8}$$

reduces that drift by one order in $\Delta t$. The published script uses the left-endpoint rule (C.5) because it is enough to make $\mathcal{E}$ look flat on the scale of Figure 15.1.

## C.4 Angular momentum

The Newtonian force is central and exerts no torque. The drag is antiparallel to $\mathbf{v}$, so

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \mathbf{r} \times \mathbf{F}_{\mathrm{diss}} = -\eta\mu \frac{\mathbf{r} \times \mathbf{v}}{d^4}. \tag{C.9}$$

Angular momentum is therefore not conserved when $\eta \neq 0$. That is required: an inspiral that conserved $|\mathbf{L}|$ at fixed mass would be incompatible with a shrinking circular orbit. The identity (C.9) is a useful code check (Exercise 4 of Chapter 15).

## C.5 Initial data and units

The published run uses

$$\mathbf{r}(0) = (1, 0), \qquad \mathbf{v}(0) = (0, \sqrt{G_{\text{toy}} M}), \tag{C.10}$$

with $G_{\text{toy}} = 36$, $\eta = 0.006$, $\Delta t = 10^{-3}$, and $1.2 \times 10^4$ steps. These are code units. They are chosen so that decay is visible, not so that $\eta$ matches a tidal-disruption estimate from Problem I of Chapter 17.

Restoring dimensions, $\eta$ in (C.2) has dimension length$^3$/time. A mutual-information tail (14.7) can supply that combination from $\ell_A^2 \ell_B^2 / t_*$ times a kinematic factor. The appendix does not evaluate the factor.

## C.6 How to run the script

From the project root, after `figs/` exists:

```
python code/namcg_inspiral.py
```

The script writes `figs/namcg_gr_simulation_corrected.pdf`. Point Figure 15.1 at that PDF for the print interior. A legacy PNG from an earlier notebook may be kept for draft viewing.

If `matplotlib` is unavailable, the integrator still runs and can write `traj`, `E_mech`, and `W_diss` to text; the figure is then optional.

## C.7 What this appendix will not do

It will not introduce gravitational waves, horizons, or a relativistic two-body Hamiltonian. It will not treat $G_{\text{toy}}$ as $G_{\text{N}}$. It will not fit $\eta$ to data. Those refusals are the content of Sections 15.6 and 16.7.

## C.8 Exercises

1. **Conservative order.** With $\eta = 0$, measure the secular drift of $E_{\text{mech}}$ over $10^4$ steps at three values of $\Delta t$. Confirm second-order scaling.

2. **Quadrature.** Replace (C.5) by (C.8) and compare the drift of $\mathcal{E}$.

3. **Implicit drag.** Solve the linear equation for $\mathbf{v}_{n+1}$ when the last drag evaluation uses $\mathbf{v}_{n+1}$. Does $\mathcal{E}$ improve enough to justify the extra algebra?

4. **Torque check.** Integrate (C.9) in lockstep with the trajectory and compare with $\mathbf{L}_{n+1} - \mathbf{L}_n$.

5. **Illegal control.** Set $\mathbf{a}_{\mathrm{d}} = -\gamma \mathbf{v}$ with constant $\gamma$ and watch $W_{\mathrm{diss}}$ as $d$ decreases. This is Exercise 3 of Chapter 15, recorded here so the appendix can be used alone as a lab sheet.

# Appendix D

# SPARC Rotation Curves: A Coefficient-Level Consistency Check

## D.1 What is being tested

The function

$$\nu_{\surd}(g_{\rm N}, g_s) = \frac{1}{1 - \exp(-\sqrt{g_{\rm N}/g_s})} \tag{D.1}$$

is the published Radial Acceleration Relation interpolator of McGaugh, Lelli & Schombert (2016). Appendix H records that a derivation of this (or any other) interpolator from the mutual-information functional remains open (Problem H). The present appendix asks only whether the same function, with the single scale fixed at the published value $g_s = g_\dagger = 1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, is compatible with the SPARC rotation-curve sample.

The predicted circular velocity is

$$V = V_{\rm b}\sqrt{\nu_{\surd}(g_{\rm N}, g_s)}, \tag{D.2}$$

where $V_{\rm b}$ is the Newtonian baryonic contribution computed from the SPARC mass models. No additional galactic parameters are introduced.

## D.2 SPARC comparison

The comparison uses the public SPARC rotmod files (175 galaxies, 3391 radial points). Three models are scored by $\chi^2$ and AIC:

- baryons only ($\nu = 1$);

- simple MOND with one free acceleration scale $a_0$;
- $\nu_{\surd}$ with $g_s$ fixed at $g_\dagger$ (zero free parameters) and, for reference, with $g_s$ free.

Table D.1: Global $\chi^2$ and AIC on 175 SPARC galaxies (3391 points). Lower AIC is better.

| Model | Free params | $\chi^2$ | AIC | Scale ($\mathrm{m\,s^{-2}}$) |
|---|---|---|---|---|
| $\nu_{\surd}$ (fixed $g_s = g_\dagger$) | 0 | $1.695 \times 10^5$ | $1.695 \times 10^5$ | $1.20 \times 10^{-10}$ |
| $\nu_{\surd}$ (free $g_s$) | 1 | $1.695 \times 10^5$ | $1.695 \times 10^5$ | $1.20 \times 10^{-10}$ |
| Simple MOND (free $a_0$) | 1 | $1.691 \times 10^5$ | $1.691 \times 10^5$ | $1.16 \times 10^{-10}$ |
| Baryons only | 0 | $1.960 \times 10^6$ | $1.960 \times 10^6$ | — |

The fixed-$g_s$ and free-$g_s$ versions of $\nu_{\surd}$ are statistically indistinguishable; the free fit returns $g_s = 1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Relative to the best one-parameter MOND fit the $\chi^2$ penalty is approximately 0.2%. Both modified-gravity models improve on baryons-only by more than an order of magnitude in $\chi^2$.

## D.3 Numerical implementation

Listing D.1: Core evaluation functions.

```
import numpy as np

KMS2_PER_KPC_TO_SI = 1.0e6 / 3.085677581491367e19
G_DAGGER = 1.20e-10

def gn_si(vb, r):
    r = np.maximum(np.asarray(r, float), 1e-6)
    return (np.asarray(vb, float)**2 / r) * KMS2_PER_KPC_TO_SI

def nu_sqrt(gn, gs):
    gs = max(float(gs), 1e-18)
    x = np.clip(np.asarray(gn, float)/gs, 0.0, 1e12)
    s = np.sqrt(x)
    denom = 1.0 - np.exp(-s)
    nu = np.empty_like(denom)
    tiny = denom < 1e-12
    nu[tiny] = 1.0 / np.maximum(s[tiny], 1e-30)
    nu[~tiny] = 1.0 / denom[~tiny]
    return nu
```

```
def v_namcg(r, vb, gs=G_DAGGER):
    return vb * np.sqrt(np.clip(nu_sqrt(gn_si(vb, r), gs), 1.0, 1e6))
```

## D.4 What the comparison is and is not

At low Newtonian acceleration, $\nu_{\surd}$ approaches the square-root map $g_{\rm obs} \approx \sqrt{g_{\rm N} g_\dagger}$. On the SPARC disks this keeps $V$ from falling as fast as $V_{\rm b}$. The numerical result shows that this particular function, frozen at the published $g_\dagger$, is compatible with the sample at the same level as a one-parameter MOND fit.

The comparison does not:

- derive $\nu_{\surd}$ from $I(A:B)$ or from the modular spectrum (Problem H remains open);
- fix $g_\dagger$ from the condensate;
- constrain $m_T = 0$, the heat-kernel coefficients, or the conservation identity;
- rule out particle dark-matter halos on other scales.

It is a coefficient-level consistency check: the infrared equation of the monograph can wear the published RAR interpolator without contradiction on this data set. That is the strongest statement the SPARC sample supports.

# Appendix E

# A Provisional Bridge from Mutual Information to the RAR

This appendix states the strongest logical contact between NAMCG and the SPARC radial acceleration relation that the book can currently defend. It does not solve Problem H. It does not derive $g_\dagger$ from $\Phi^I{}_A$. It replaces an unspecified $\mu_{\rm ent}$ with a single modular conjecture whose failure would leave Appendix D as a coefficient ranking and nothing more.

## E.1 What the two sides already share

NAMCG, after Theorems 7.3–11.1, is Einstein gravity plus a conserved extra tensor obtained by varying an entanglement functional of baryonic regions. In a static disk that statement has three immediate consequences, none of which uses SPARC.

1. The extra acceleration is slaved to the baryonic Newtonian field. It cannot be an independent halo function of $r$ (Assumption 13.2).

2. At Solar-System and inner-disk accelerations the extra piece must vanish, or Cassini bounds fail: $\nu \to 1$ as $g_{\rm N} \to \infty$.

3. The only scalar available to mark the transition is an acceleration built from the same modular data that defined the diamonds of Chapter 6 and the infrared boundary of Chapter 12.

A fourth consequence is needed once the subsystem is not isolated. The entanglement functional is a functional of the baryonic configuration, not of an excised disk. A slowly varying external Newtonian field $\mathbf{g}_{\rm ext}$ therefore enters the argument of the

boost (Assumption 13.3). In the external-field-dominated limit $g_{\rm N} \ll g_{\rm ext}$,

$$\gamma \equiv \frac{g_{\rm int}}{g_{\rm N}} \to \nu_{\surd}(g_{\rm ext}), \tag{E.1}$$

not $\nu_{\surd}(g_{\rm N})$. At the Sun, $v_\odot^2/R_0 \approx 2.2 \times 10^{-10}\,{\rm m\,s^{-2}} \approx 1.8\,g_\dagger$, so $\nu_{\surd}(g_{\rm gal}) \approx 1.1$. An isolated evaluation of (E.2) at $g_{\rm N} = 10^{-11}\,{\rm m\,s^{-2}}$ wants $\gamma \approx 4$ and is illegal for a solar-neighborhood binary.

The RAR is an empirical map of exactly that type [1, 3]:

$$g_{\rm obs} = \nu_{\rm RAR}(x)\, g_{\rm N}, \qquad x = \frac{g_{\rm N}}{g_\dagger}, \qquad \nu_{\rm RAR}(x) = \frac{1}{1 - e^{-\sqrt{x}}}, \tag{E.2}$$

with $g_\dagger = (1.20 \pm 0.02) \times 10^{-10}\,{\rm m\,s^{-2}}$ and residuals that do not track radius or Hubble type. Appendix D showed that (E.2), frozen at $g_\dagger$, is the disk law NAMCG is presently allowed to wear.

The shared content is therefore the *logical type* of the law, not the algebra of the interpolator and not the numerical value of $g_\dagger$.

## E.2 Why the first NAMCG-shaped interpolator failed

Chapter 6 supplied a conformal large-separation tail

$$I(A:B) \;\sim\; c\,\frac{\ell_A^2 \ell_B^2}{d^{2\Delta}} \tag{E.3}$$

with $\Delta$ of order one [4]. A linear-response force $\partial I/\partial d$, converted carelessly to a local boost and written as a function of $g_{\rm N}$, produced the working family

$$\nu_{\rm exp} = \frac{1}{1 - \alpha \exp(-g_{\rm N}/g_s)}. \tag{E.4}$$

On the same 175 SPARC files that family lost to simple MOND by a wide AIC gap. The transition was too sharp in $g_{\rm N}$ and needed a second amplitude $\alpha$ that disks do not want.

The failure is useful. It shows that "mutual information implies a boost" is not enough. The *argument of the exponential* matters. SPARC selects $\sqrt{g_{\rm N}}$ in that argument, not $g_{\rm N}$.

## E.3 An algebraic identity, not a derivation

Write $y = \sqrt{x} = \sqrt{g_{\rm N}/g_\dagger}$. Then

$$\nu_{\rm RAR}(x) = 1 + \frac{1}{e^y - 1} = 1 + n_{\rm BE}(y), \tag{E.5}$$

where $n_{\rm BE}$ is the Bose–Einstein occupation of a single mode with dimensionless energy $y$. This is an identity. It does not, by itself, mean that galactic baryons are a Bose gas. It does mean that any microscopic argument that produces a thermal factor $e^{-\sqrt{g_{\rm N}/g_s}}$ in a screen occupation, rather than $e^{-g_{\rm N}/g_s}$, lands on the published interpolator automatically.

The linear-exponential (E.4) is the Boltzmann factor of a scale set by $g_{\rm N}$ itself. The RAR is the Bose factor of a scale set by $\sqrt{g_{\rm N}}$. That is the whole algebraic distinction.

## E.4 A modular conjecture

Chapter 6 already assigns a diamond an Unruh temperature $T_{\rm Unruh} = \kappa/2\pi$ and a modular Hamiltonian $H_D$. Chapter 12 already assigns the cosmology an infrared boundary scale through $\Lambda_{\rm eff}$. Milgrom's coincidence [5],

$$2\pi g_\dagger \;\sim\; cH_0 \;\sim\; c^2\sqrt{\Lambda/3}, \tag{E.6}$$

says that the SPARC scale sits, to a factor of order $2\pi$, on the same infrared accelerations NAMCG already possesses as diamond and boundary data.

**Assumption E.1** (Square-root modular response)**.** For a slowly varying disk of condensed parcels, the leading reversible variation of $U_{\rm ent}$ with respect to a local Newtonian acceleration is thermal in the variable

$$y = \sqrt{\frac{g_{\rm N}}{g_s}}, \qquad g_s \sim \frac{\kappa_{\rm IR}}{2\pi}, \tag{E.7}$$

where $\kappa_{\rm IR}$ is the surface gravity of the infrared boundary used in Chapter 12, not a new galactic parameter. The occupation that weights the extra stress is $n_{\rm BE}(y)$. Then $\nu = 1 + n_{\rm BE}(y)$ is (E.2), and $g_s$ is identified with $g_\dagger$ rather than fitted to SPARC.

Assumption E.1 is the missing sentence of Problem H. It is not implied by Theorem 7.3. It is not implied by the conformal tail (E.3). It is the simplest contact that turns the book's modular objects into the interpolator SPARC already selected.

Why a square root rather than a linear Unruh factor $T \propto g_{\rm N}$ is not derived here. A linear factor reproduces (E.4) and has already failed. A justification would have to come from the spectrum of $H_D$ on a diamond whose size is set by the local baryonic

curvature radius, or from a two-horizon sharing between the local diamond and the HBEV boundary. That calculation is the content of Problem H, restated.

## E.5 What this bridge excludes

Verlinde's 2016 emergent-gravity formula for finite disks depends explicitly on radius as well as on $g_{\mathrm{N}}$. Lelli, McGaugh and Schombert found that those residuals correlate with radius and that acceptable fits require stellar mass-to-light ratios in tension with population synthesis [6]. NAMCG cannot take that route. Assumption 13.2 plus the RAR residuals already require a map of $g_{\mathrm{N}}$ only.

A cosmological running $g_s(z) \propto H(z)$ is likewise not licensed by this appendix. Equation (E.6) is a present-day coincidence used to name $g_s$. Promoting it to $g_s(z)$ is extra physics and would have to survive high-redshift rotation curves before it is written as a NAMCG result.

Isolated application of (E.2) to wide binaries is likewise not licensed. Assumption 13.3 saturates the boost at $\nu_{\surd}(g_{\mathrm{ext}})$. A Gaia sample consistent with Newton constrains the isolated reading, not Theorem 7.3. A reported $\gamma \approx 1.4$ that survives quality cuts is not explained by $\nu_{\surd}(g_{\mathrm{gal}}) \approx 1.1$; that residual, if real, is still open.

## E.6 How the bridge would be confirmed or killed

**Confirmation.** The published diamond first law recovers Einstein gravity at first order in the metric, hence $\nu \to 1$. An interpolator requires the modular spectrum of $K_D$ in the two-scale geometry of Chapters 6 and 12 (local diamond plus infrared surface gravity $\kappa_{\mathrm{IR}}$). Problem H is solved if the leading eigenvalue for a weak-field source is $\lambda_0 \propto \sqrt{g_{\mathrm{N}}/\kappa_{\mathrm{IR}}}$ and the occupation is bosonic. Then $g_s = \kappa_{\mathrm{IR}}$ is not a SPARC fit, and Appendix D becomes a consistency check. A linear eigenvalue $\lambda_0 \propto g_{\mathrm{N}}/\kappa_{\mathrm{IR}}$ would reproduce the retired $\nu_{\mathrm{exp}}$ and kill Assumption E.1.

The same computation must grow an external-field argument before wide binaries are a prediction. Until it does, Assumption 13.3 is a coefficient in the same sense as Assumption 13.2: it is the QUMOND external-field limit written with $\nu_{\surd}$, not a lemma from $I(A:B)$.

The explicit modular Hamiltonian of de Sitter diamonds [7] is the operator to use for the eigenvalue test (17.13). The Euclidean Schwarzschild–de Sitter action is minus the sum of the two horizon entropies, not a geometric mean of surface gravities; that instanton does not by itself produce $\sqrt{g_{\mathrm{N}}/\kappa_{\mathrm{IR}}}$. The numerical coincidence $cH_0/2\pi \sim g_\dagger$ may be recorded after $\lambda_0$ is computed, not used to fix $g_s$.

**Failure.** The same computation produces (E.4), a radius-dependent kernel, or a light $\psi$. Then disks may still follow the RAR, but not as a NAMCG prediction.

Assumption 13.2 remains a coefficient; Problem H remains open; Appendix D is unchanged.

## E.7 One paragraph for Chapter 13

The working disk law is (E.2). The reason NAMCG is allowed to use it on an isolated disk is Assumption 13.2. The reason a subsystem embedded in the Galaxy must not use the isolated evaluation is Assumption 13.3. The reason the book might eventually own either sentence is Assumption E.1. Until the last is a lemma, the first two are borrowed, and the galactic sector is the RAR with the external-field effect.

## E.8 Exercises

1. Prove (E.5) in three lines.

2. Expand $\nu_{\mathrm{RAR}}$ and $\nu_{\exp}$ for $x \ll 1$ and $x \gg 1$. Where do they first differ by order one?

3. Using $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, compute $cH_0$ and $cH_0/2\pi$ in $\mathrm{m\,s^{-2}}$. Compare with $g_\dagger$. This is a coincidence check, not a derivation.

4. Why does an explicit $r$ in $\nu$, as in Verlinde's finite-disk formula, threaten the residual structure reported by [3]?

5. **External field.** Take $g_{\mathrm{ext}} = 2.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $g_{\mathrm{N}} = 10^{-11}\,\mathrm{m\,s^{-2}}$. Compute $\nu_{\surd}(g_{\mathrm{N}})$ and $\nu_{\surd}(g_{\mathrm{ext}})$. Which one is the legal boost under Assumption 13.3, and which published wide-binary claim does each contact?

# Appendix F

# Einstein Gravity Plus a Conserved Extra Tensor

This appendix writes the infrared field equation of Theorems 7.3–11.1 next to Einstein's equation. It introduces no new hypothesis.

## F.1 Einstein

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T^{\text{matter}}_{\mu\nu}. \tag{F.1}$$

Bianchi identities force $\nabla^{\mu} T^{\text{matter}}_{\mu\nu} = 0$ on shell.

## F.2 NAMCG infrared equation

After the composite metric, $m_T = 0$, the heat-kernel map, Lorentz locking, and conservation are granted,

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G_{\text{eff}} (T^{\text{matter}}_{\mu\nu} + T^{\text{tensor}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu}). \tag{F.2}$$

$G_{\mu\nu}$ is the Einstein tensor of the composite metric $g_{\mu\nu} = \eta_{AB} e^{A}{}_{\mu} e^{B}{}_{\nu}$. $G_{\text{eff}}$ and $\Lambda_{\text{eff}}$ are the coefficients of Theorems 8.2 and 12.4. The extra source is

$$T^{\text{ent}}_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta U_{\text{ent}}[I(A:B)]}{\delta g^{\mu\nu}} = T^{\text{ent,rev}}_{\mu\nu} + T^{\text{ent,irrev}}_{\mu\nu}. \tag{F.3}$$

Chapter 11 requires

$$\nabla^{\mu} (T^{\text{matter}}_{\mu\nu} + T^{\text{tensor}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu}) = 0. \tag{F.4}$$

In the infrared, $T^{\text{tensor}}_{\mu\nu}$ is absorbed into the Einstein tensor of $g_{\mu\nu}$. What can appear as extra gravity in static disks is $T^{\text{ent,rev}}_{\mu\nu}$.

## F.3 Weak field

Newtonian gauge, static source:

$$\nabla^2\Phi = 4\pi G_{\rm eff}(\rho_{\rm b} + \rho_{\rm ent}), \qquad \rho_{\rm ent} = T_{00}^{\rm ent,rev}. \tag{F.5}$$

Assumption 13.2 slaved $\rho_{\rm ent}$ to $\rho_{\rm b}$. The interpolator $\nu_{\surd}$ of Appendix D is a coefficient choice for that assumption, not a consequence of (F.2).

## F.4 What this does and does not change

High acceleration: $T_{\mu\nu}^{\rm ent,rev} \to 0$, so (F.2) returns to (F.1) with $(G_{\rm eff}, \Lambda_{\rm eff})$ in place of $(G, \Lambda)$. Low acceleration: $\rho_{\rm ent}$ may dominate $\rho_{\rm b}$ without a particle halo, if and only if Assumption 13.2 holds. Clusters, the CMB, and colliding systems test that assumption, not $G_{\mu\nu}$. Gravitational waves test $m_T = 0$, not (F.3). The irreversible drag of Chapter 14 is a different tensor structure and is not a static extra potential.

## F.5 One-line summary

Einstein equates curvature to ordinary stress. NAMCG equates the same curvature to ordinary stress plus a divergenceless entanglement stress of baryonic regions.

# Appendix G

# Heat-Kernel Area Term on a Causal Diamond

This appendix records the standard heat-kernel contribution to the entanglement entropy of a causal diamond. It does not derive the equilibrium condition $\delta S_D = \delta\langle H_D\rangle$ and does not recover the Einstein equation from the matrix sector alone.

## G.1 Replica formula and cone expansion

The von Neumann entropy of the reduced state on a diamond $D$ is obtained from the replica trick,

$$S_D = (1 - n\partial_n) \ln Z(n)\Big|_{n=1}, \tag{G.1}$$

where $Z(n)$ is the partition function on the $n$-sheeted cover branched along the entangling surface $\Sigma$. For a Laplace-type fluctuation operator the heat-kernel expansion on the cone produces the area term

$$S_D = \frac{\mathrm{Area}(\Sigma)}{4G_{\mathrm{eff}}} + S_{\mathrm{pot}}, \tag{G.2}$$

with the prefactor identified from the Seeley–DeWitt coefficient that multiplies the deficit. This is the standard result used in Chapter 6 and Appendix B. The numerical value of $G_{\mathrm{eff}}$ remains the structural heat-kernel map of Theorem 8.2; the coefficients $A, B, C$ are not fixed here (Problem F).

## G.2 What is not derived

- The equilibrium condition $\delta S_D = \delta\langle H_D\rangle$ under fixed volume. It is the hypothesis of Chapters 6–7 that forces $m_T = 0$. It is not obtained as a path-integral Ward

identity from the replica variation alone.

- An evaluation of the fluctuation operator of the locked NAMCG condensate (including diffeomorphism and local-Lorentz ghosts) that would determine the precise endomorphism contribution to the area coefficient.
- Recovery of the infrared equation. Once the area law and the equilibrium condition are both assumed, the Einstein equation with a matrix source follows by the usual Jacobson argument; that assumption is not removed.

## G.3 Status

The area term in $S_D$ is the standard heat-kernel result. The appendix adds no new derivation of the Ward identity, no scheme-fixed coefficient, and no alteration to the open-problem ledger of Appendix H.

# Appendix H

# Status of the Open Problems

This appendix records the precise status of Problems A–J as of the present text. It collects standard supporting calculations that are already rigorous, states the exact missing step for each problem, and does not convert any open problem into a lemma. The research agenda of Chapter 17 remains the authoritative list; this appendix is a working ledger.

## H.1 Problem A: dictionary

The Weyl–Wigner expansion

$$[\hat{X}_\mu, \hat{\Phi}] = i\Theta_{\mu\nu} D^\nu \Phi + \frac{1}{24}\Theta_{\mu\alpha}\Theta_{\beta\gamma} D^\alpha D^\beta D^\gamma \Phi + \mathcal{O}(\Theta^3) \tag{H.1}$$

is a standard result of the Moyal product (see, e.g., the calculation in Section **??** of the removed material). It is not the dictionary required by the programme.

**Missing step.** An error-controlled map from the matrix order parameter (or from $[\hat{X}, \Phi]$) into the modular Hamiltonian $H_D$ of a causal diamond, or into the mutual-information functional $U_{\text{ent}}[I(A:B)]$, that holds before the composite metric is defined. Until that map is a lemma, Chapter 8 computes a heat kernel in a theory whose relation to Chapter 3 is still Assumption 3.5.

## H.2 Problem B: replica Ward identity

The heat-kernel expansion on an $n$-sheeted branched cover and the resulting area term in $S_D$ are standard. The fixed-volume condition $\delta V = 0$ isolates the symmetric traceless sector in the usual way.

**Missing step.** A derivation of the Ward identity $\delta S_D = \delta\langle H_D\rangle$ that does not already presuppose the composite metric or the split of $\delta\Phi$ performed in Chapter 5. The present argument remains the one given in Chapters 6–7.

## H.3 Problem C: conformal mode and Cassini

On the locked vacuum the radial singlet $\sigma$ of the polar decomposition (4.4) has tree-level mass squared $\lambda v^2$. The second variation of the potential term in (3.6), taken with $\Lambda$ held fixed at the locked vacuum, is the ordinary massive Higgs quadratic form

$$\delta^2_\sigma S_V = -\int d^4x\sqrt{-g^{(0)}}\,\frac{1}{2}m^2_\sigma\sigma^2, \qquad m^2_\sigma = \lambda v^2. \tag{H.2}$$

By Definition 4.1 the official vierbein is evaluated at $\sigma = 0$, so a pure $\sigma$ fluctuation does not enter the composite metric at linear order. For any $\lambda$ not tuned below $(10^{-18}\,\mathrm{eV}/v)^2$ the pole is short-range on an astronomical unit and invisible to the Cassini link.

The conformal mode $\psi$ is the residual trace of $\delta\Phi_{\mu\nu}$ orthogonal to this pure-$\sigma$ direction. The mixed second variation of the kinetic term in (3.6) that would couple $\psi$ to $\sigma$ (after the scaffold is identified with the composite metric and the projection orthogonal to $\sigma$ is performed) is not present in the text.

**Missing step.** Explicit evaluation of that mixed quadratic form and diagonalization of the resulting mass matrix. If the light eigenvalue is a positive multiple of $\lambda v^2$, the infrared theory is general relativity and Cassini is satisfied for free. If a massless eigenmode survives, the infrared theory is scalar–tensor and the bound applies to that mode. No galaxy observable enters the fork.

## H.4 Problem D: local Lorentz

The counting of generators under the locked vacuum is given in Chapter 10. The zero-torsion condition that maps the internal connection onto the spin connection is imposed, not derived.

**Missing step.** Either a derivation of $\mathrm{SO}(3,1)$ from the matrix algebra alone, or an explicit statement that Presentation A is not ontologically prior. The present text does the latter.

## H.5 Problem E: signature

Euclidean positivity of the matrix action and the formal Wick rotation of one coordinate are recorded. They do not select Lorentzian signature without an additional hypothesis.

**Status.** Remains Assumption 4.6.

## H.6 Problem F: coefficients $A, B, C$

The structural map

$$\frac{1}{G_{\text{eff}}} = A\,\mathrm{Tr}(\mathbb{I})\Lambda_{\text{UV}}^2 + Bv^2\ln(\Lambda_{\text{UV}}/\mu) + C\mu^2 \tag{H.3}$$

is Theorem 8.2. The numerical values of $A, B, C$ are not scheme-fixed.

**Missing step.** Recomputation of the Seeley–DeWitt coefficient $a_1$ (and the relevant part of $a_2$) on the locked $4 \times 4$ block, including diffeomorphism and local-Lorentz ghosts. Fibre-trace tables that assign $N_{\text{eff}} = +2$ without this computation do not close the problem.

## H.7 Problem G: HBEV

The boundary functional of Chapter 12 regulates $\Lambda_{\text{eff}}$ once an infrared entropy density is supplied. The working value $\Lambda_{\text{eff}} = 3/L_{\text{IR}}^2$ is a choice consistent with that functional, not a derived necessity.

**Status.** Closed as a consistent regulation mechanism under the stated boundary entropy; not closed as a derivation of the entropy density from the matrix theory.

## H.8 Problem H: interpolating function

Appendix D shows that the function

$$\nu_{\surd}(g_{\text{N}}) = \frac{1}{1 - \exp(-\sqrt{g_{\text{N}}/g_s})} \tag{H.4}$$

with $g_s = g_\dagger$ is compatible with the SPARC sample at the coefficient level (approximately 0.2% behind the best one-parameter MOND fit). This is not a derivation.

**Missing step.** Extraction of the leading modular eigenvalue $\lambda_0(g_N, \kappa_{IR})$ from the spectrum of $H_D$ on a weak-field diamond. Success requires $\lambda_0 \propto \sqrt{g_N/\kappa_{IR}}$ with bosonic occupation and $g_s = \kappa_{IR}$ fixed by the infrared surface gravity, not by a fit. A linear eigenvalue would reproduce the retired exponential and kill the target. Refitting parameters to rotmod files does not solve the problem.

## H.9 Problem I: drag exponent

The fall-off $I(A : B) \propto d^{-4}$ and the linear-response integral that produces $\mathbf{a}_{irrev} \propto -\mathbf{v}/d^4$ are the phenomenological inputs already used in Chapter 14. They are consistent with the conservation identity of Theorem 11.1 once the stress is divergenceless, but they are not derived from the modular data.

**Status.** Coefficient, not load-bearing. A derivation from $I(A : B)$ would be desirable; its absence does not kill the framework.

## H.10 Problem J: regular cores

Under the Einsteinian cubic truncation the static spherical reduction admits the approximate regular-core profile of Chapter 13. The profile is not exact, and the truncation itself is not derived from the matrix theory.

**Status.** Closed as an approximate solution inside the cubic theory; not closed as a resolution of singularities from the microscopic action.

## H.11 What this appendix does not do

It does not produce an unsuppressed laboratory coupling between a controllable external field and $T^{ent}_{\mu\nu}$ or $G_{eff}$. The observational window therefore remains empty. It does not alter the chain of Chapter 16 or the agenda of Chapter 17.

# Appendix I

# Gaia DR3 Wide Binaries: A Coefficient-Level Consistency Check

This appendix records a comparison of Gaia DR3 wide-binary kinematics with the external-field-corrected form of the coefficient-level interpolator already examined on SPARC in Appendix D. It is not a test of Theorems 7.3–12.4, not a derivation of $\nu_{\surd}$, and not evidence that Problem H is solved.

## I.1 What is being compared

The relevant acceleration scale for a solar-neighbourhood binary is not the internal Newtonian acceleration alone. The Galactic external field at the Sun satisfies

$$g_{\rm gal} \approx (1.5\text{–}2.2) \times 10^{-10}\,\mathrm{m\,s^{-2}} > g_\dagger .$$

Under the external-field saturation already noted for the coefficient layer (Assumption 13.3), the isolated deep-regime boost is suppressed. The working prediction used here is therefore the residual enhancement obtained by evaluating $\nu_{\surd}$ on the quadrature sum of the internal and external fields (or the equivalent AQUAL/QUMOND external-field prescription). Asymptotically this residual is of order 20%, not the factor $\sim 4$ of the isolated map.

The comparison is performed on the public El-Badry et al. Gaia DR3 wide-binary catalogue after the standard quality cuts (RUWE$<$ 1.3, fractional parallax error $< 0.06$, projected separations 2–30 kAU). Projection, unresolved companions, and chance alignments are modelled by the same mixture techniques employed in the controlled literature analyses; they are not optional.

## I.2 What the data show

Controlled analyses that include the Galactic external field, realistic triple contamination, and the quality cuts above find relative velocities consistent with Newtonian gravity. The residual MOND/RAR-like enhancement is excluded at high significance (Banik et al. 2024; Pittordis, Sutherland & Shepherd 2025; quality-framework study 2026). Apparent boosts reported in less filtered samples diminish once these systematics are treated.

Consequently the external-field-corrected $\nu_{\surd}$ (the coefficient-level map of Appendix D) is also consistent with the data only to the extent that its residual boost remains inside the observational uncertainties. It does not outperform Newtonian gravity on this sample, nor does it produce a statistically preferred rising velocity ratio of the size generated by the isolated interpolator.

## I.3 What this is not

- It is not a derivation of $\nu_{\surd}$ from $I(A : B)$ or from the modular spectrum. Problem H remains open.
- It is not a constraint on $m_T = 0$, on the heat-kernel map for $G_{\text{eff}}$, or on the conservation identity. A null result under the external-field-corrected prediction is the outcome expected once Assumption 13.3 is respected.
- It is not evidence that an isolated reading of the RAR interpolator applies to solar-neighbourhood binaries. That reading is already ruled out by $g_{\text{gal}} > g_{\dagger}$.
- It is not a laboratory handle on the entanglement stress.

## I.4 Status

The Gaia DR3 wide-binary sample, analysed with the external field included, is compatible with Newtonian gravity and with the modest residual enhancement allowed by the coefficient-level NAMCG/RAR map under external-field saturation. It supplies no new support for the framework beyond the SPARC consistency check already recorded, and it does not alter the open-problem ledger of Appendix H.

# Appendix J

# Infrared Limits That Follow from the Proved Chain

This appendix collects the infrared limits that follow directly from the theorems and structural results already established in Chapters 3–12. It does not treat open problems as closed, and it does not claim additional recovery of general relativity beyond those limits.

## J.1 Limits that are obtained

Under the named assumptions of the monograph the following hold.

1. **High-acceleration Einstein equation.** When the reversible entanglement stress is negligible compared with ordinary matter stress, the infrared equation reduces to
$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G_{\text{eff}} T^{\text{matter}}_{\mu\nu}. \tag{J.1}$$
This is the content of the high-acceleration limit of the equation already written after Theorem 11.1.

2. **Massless tensor polarizations.** The entanglement-equilibrium condition of Chapters 6–7 forces $m_T = 0$, so the infrared spectrum contains the two transverse-traceless modes of a massless spin-2 field. This is Theorem 7.3; it is not re-derived here.

3. **Conservation.** The total stress (matter + tensor + entanglement) is divergenceless,
$$\nabla^\mu (T^{\text{matter}}_{\mu\nu} + T^{\text{tensor}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu}) = 0, \tag{J.2}$$
by Theorem 11.1. Any reduced drag used in later chapters is legitimate only because this identity has already been established.

4. **Structural form of $G_{\text{eff}}$.** The prefactor of the Einstein–Hilbert term is given by the heat-kernel map of Theorem 8.2. The numerical values of the coefficients $A, B, C$ remain unfixed (Problem F).

5. **Regulation of the quartic term.** The holographic boundary-entropy vacuum of Chapter 12 supplies a consistent mechanism that can cancel the leading $\Lambda_{\text{UV}}^4$ contribution against a boundary term. The specific working value $\Lambda_{\text{eff}} = 3/L_{\text{IR}}^2$ is a choice compatible with that mechanism, not a derived necessity.

## J.2 Limits that are not obtained

The following are frequently listed as recoveries but do not follow from the proved chain.

- Exact $\gamma - 1 = 0$ in the solar system. The conformal mode $\psi$ is not identical to the radial Higgs; its mass matrix has not been diagonalized (Problem C). Cassini therefore remains a constraint on any light eigenmode that may survive.

- A first-principles drag exponent or numerical rate. The $d^{-4}$ form and the estimate $\gamma_{\text{drag}} \sim 10^{-15}\,\text{s}^{-1}$ are the phenomenological inputs of Chapter 14 (Problem I). They are consistent with conservation once the stress is divergenceless; they are not derived from the modular spectrum.

- Singularity resolution from the matrix theory. The regular-core profile is an approximate solution inside the Einsteinian cubic truncation (Chapter 13). It is not obtained from the microscopic action (Problem J).

- Derivation of local Lorentz invariance or of Lorentzian signature from the matrix algebra alone (Problems D and E). Both remain under the assumptions stated in Chapters 3 and 10.

- An interpolating function for galactic or binary kinematics derived from $I(A:B)$ (Problem H). Coefficient-level compatibility with the published RAR map is recorded in Appendix D; it is not a derivation.

## J.3 Status

The infrared theory obtained from the proved chain is Einstein gravity with a structural $G_{\text{eff}}$, a massless spin-2 sector, a divergenceless entanglement stress, and a consistent regulation of the leading vacuum-energy term. Every stronger recovery claim requires the closure of at least one open problem listed in Appendix H. This appendix adds no further recovery.

# Appendix K

# Heat-Kernel Response to an External Field Strength

This appendix records the leading response of the Seeley–DeWitt coefficient $a_1$ to a laboratory field strength and the precision constraints that any residual conformal mixing must satisfy. It does not recompute $a_1$ on the locked block, does not derive a kinetic mixing from the entanglement functional, and does not open an observationally allowed laboratory window.

## K.1 Vanishing of the linear term in $a_1$

On a Laplace-type operator $\Delta = -D^\mu D_\mu + E$ the first Seeley–DeWitt coefficient is

$$a_1 = \mathrm{tr}(E) - \frac{1}{6} R \,\dim\,. \tag{K.1}$$

If a controllable field strength enters the endomorphism only through an antisymmetric generator,

$$E = E_0 + \xi F_{\alpha\beta} S^{\alpha\beta}, \tag{K.2}$$

the trace of $S^{\alpha\beta}$ vanishes on every representation that appears in the locked $4 \times 4$ block and in the diffeomorphism and local-Lorentz ghost sectors. Consequently

$$\delta a_1 = 0. \tag{K.3}$$

This is a standard algebraic fact. It is not a recomputation of $a_1$ on the NAMCG fluctuation operator (Problem F remains open), nor does it cancel the higher-order suppression by $\Lambda_{\mathrm{UV}}$ already identified for laboratory fields.

## K.2 Quadratic appearance in $a_2$

The first non-vanishing correction linear in the laboratory field appears in $a_2$ and is proportional to $F_{\mu\nu}F^{\mu\nu}$. Variation of the resulting effective action with respect to the metric produces a contribution proportional to $g_{\mu\nu}$. The coupling is therefore confined to the conformal mode. No transverse-traceless response is generated at this order.

## K.3 Precision constraints on conformal mixing

Any additional kinetic mixing between the conformal mode $\psi$ and a laboratory scalar that might be introduced by hand is constrained by existing data:

- Cassini: $|\gamma - 1| \lesssim 2 \times 10^{-5}$;
- torsion-balance equivalence-principle tests: $\sin^2\theta \lesssim 10^{-13}$.

A mixing angle large enough to produce a macroscopic laboratory acceleration violates these bounds. No calculation within the present framework cancels the parametric suppression $F/\Lambda_{\rm UV}^2$ while leaving a residual that survives the bounds. The observational window therefore remains empty.

## K.4 What is not shown

- A completed evaluation of $a_1$ (or the relevant part of $a_2$) on the locked $4 \times 4$ block with ghosts.
- A derivation of a kinetic mixing term from $U_{\rm ent}[I(A:B)]$ or from the modular Hamiltonian.
- A dynamical cancellation of quadratic ultraviolet divergences that leaves a finite infrared residual.
- Any projection of galactic rotation-curve discrepancies onto solar-system scales, or any fit to pulsar-timing-array correlation data. Those exercises do not follow from the heat-kernel expansion and are omitted.

## K.5 Status

The linear coupling of a laboratory field strength to $a_1$ vanishes for antisymmetric generators, and any residual conformal mixing is limited by existing precision tests to a level far below laboratory interest. This appendix adds no new handle on the entanglement stress or on $G_{\rm eff}$ and does not alter the open-problem ledger of Appendix H.

# Appendix L

# A Phenomenological Redshift-Dependent Multiplier (Outside the Proved Chain)

This appendix records a purely phenomenological modification that lies outside the NAMCG derivation. It is not obtained from the entanglement stress, the heat-kernel map, or any theorem of Chapters 3–12, and it does not close any open problem.

## L.1 The inserted multiplier

One may choose, by hand, a redshift-dependent factor

$$\beta(z) = \beta_{\mathrm{ref}}\Big(\frac{1+z_{\mathrm{ref}}}{1+z}\Big)^{\gamma} \tag{L.1}$$

and insert it into the standard linear growth factor according to

$$D(z) = D_{\Lambda\mathrm{CDM}}(z)\big[1+\beta(z)\ln(1+z)\big]. \tag{L.2}$$

The parameters $\beta_{\mathrm{ref}}$, $z_{\mathrm{ref}}$ and $\gamma$ are free. No equation in the monograph fixes them.

## L.2 Illustrative high-redshift comparison

With the particular choice $\beta_{\mathrm{ref}} = 0.5$, $z_{\mathrm{ref}} = 11$, $\gamma = 0.8$, the modified growth can be fed into a schematic Sheth–Tormen mass function and a fixed mass-to-light conversion. The resulting curves can be plotted against a small set of representative high-redshift UV luminosity points. Agreement or disagreement with those points is adjustable by changing the three free parameters and does not test the infrared equation, the

conservation identity, or the value of $G_{\text{eff}}$.

## L.3 What the illustration is not

- It is not a derivation of an epoch-dependent coupling from the matrix condensate or from mutual information.
- It is not a solution of the continuity or Euler equation sourced by $T^{\text{ent}}_{\mu\nu}$.
- It is not a statistical fit to the public JADES or CEERS catalogs.
- It does not constrain, or supply evidence for, any load-bearing theorem.
- It does not open a laboratory handle on the entanglement stress.

## L.4 Status

The construction is an external multiplier that can be tuned to high-redshift galaxy counts. It stands outside the proved chain and is recorded only as an illustration of how additional free functions may be added once one leaves the framework. It is omitted from the claim list of Chapter 16 and from the open-problem ledger of Appendix H.

# Conclusion

The monograph constructs a single derivation chain from a non-Abelian matrix order parameter to an infrared equation that is Einstein gravity plus a divergenceless entanglement stress, with a structural heat-kernel map for $G_{\rm eff}$ and a massless spin-2 sector forced by entanglement equilibrium. That chain is complete only under the named assumptions of Chapters 3–12. Several load-bearing steps remain open.

## What has been obtained

Under those assumptions one has:

- a composite metric defined from the condensate,
- $m_T = 0$ from the diamond Ward identity,
- the structural form of $1/G_{\rm eff}$,
- covariant conservation of the total stress including $T^{\rm ent}_{\mu\nu}$,
- a consistent regulation of the leading vacuum-energy term via the holographic boundary functional.

Coefficient-level compatibility of the published RAR interpolator with the SPARC sample is recorded in Appendix D. It is not a derivation of that interpolator.

## What has not been obtained

- a derivation of $\nu_{\surd}$ (or any other interpolator) from mutual information or the modular spectrum (Problem H);
- scheme-fixed values of the heat-kernel coefficients $A, B, C$ (Problem F);
- diagonalization of the conformal-mode mass matrix (Problem C);
- the dictionary map into the modular Hamiltonian (Problem A);

- an unsuppressed, observationally allowed laboratory coupling to the entanglement stress;
- empirical support from Gaia DR3 wide binaries once the Galactic external field is included (Appendix I);
- any redshift-dependent growth multiplier; such constructions lie outside the proved chain.

The framework therefore does not yet resolve the dark-matter problem, nor does it supersede existing phenomenological fits on galactic or binary scales.

## Next steps

Progress requires completion of at least one load-bearing calculation:

1. extraction of the leading modular eigenvalue on a weak-field diamond (Problem H);
2. recomputation of $a_1$ on the locked $4 \times 4$ block with ghosts (Problem F);
3. diagonalization of the $\psi$–$\sigma$ mass matrix (Problem C);
4. an error-controlled dictionary map (Problem A).

Until one of these is performed and shown to survive existing precision bounds, the infrared equation remains exactly as written in the main text and no stronger claim is licensed. The open-problem ledger of Appendix H is the working list.

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